Solar-Powered Rover

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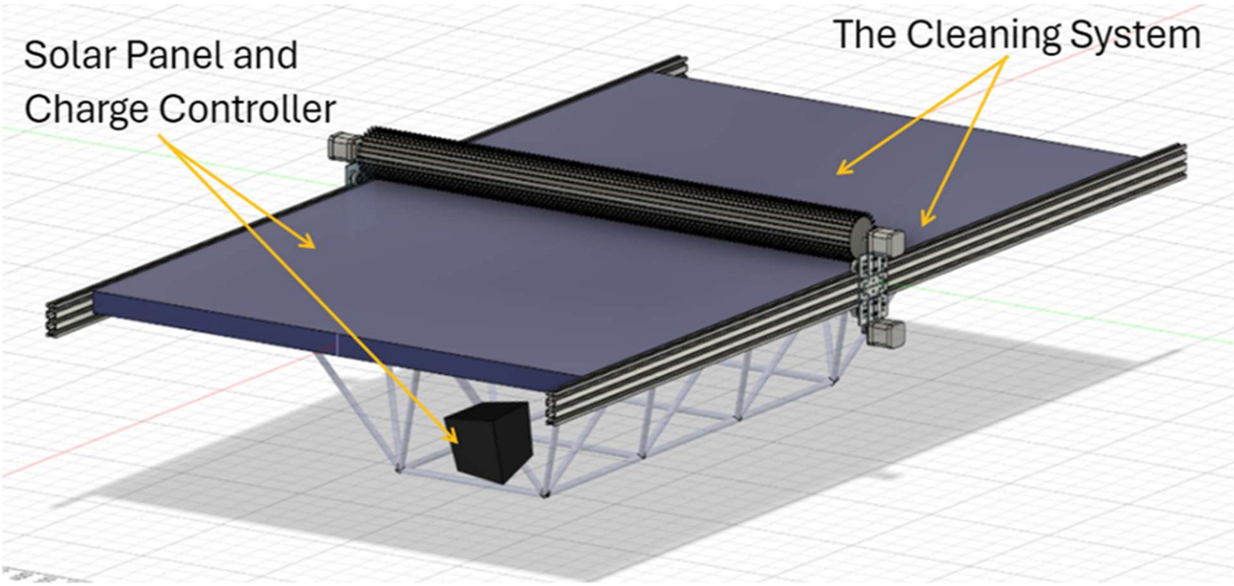

I. Abstract

Expanding on a previous senior design project that had built a rover for educational purposes, this project intended to enhance LOUIS, the rover, by integrating a solar panel to the rover,

creating autonomous switching between solar and battery power, and developing a cleaning system for the solar panel. With a $4,000 budget, the project had to consider the limited leftover chassis space and the maximum additional load capacity of 36.3kg. The cleaning system had to be gentle on the solar panel, so as not to scratch or damage it, thus maximizing the energy output. The various components were to withstand the different weather conditions in Louisiana, such as wind, precipitation, and heat. The project aimed to have a system that switches to battery power if the solar panel is not receiving adequate sunlight due to clouds or nighttime conditions. Additionally, the group needed to make sure to maintain the current functions, despite the additions. As a result of the team's engineering, the battery life expectancy improved by 206.8%, and the cleaning system restores 77.8% of the clean panel's original power generation. We maintained 92.4% of the rover's original speed, and we added 36.2kg, 0.1kg below the maximum load. We left plenty of room in the chassis, only taking up 1% of its space. The additions were weatherproof, due to the design of the equipment or the hydrophobic spray, and we included instructions on how to disassemble LOUIS on our website to maintain portability. Lastly, we remained $475 below the budget.

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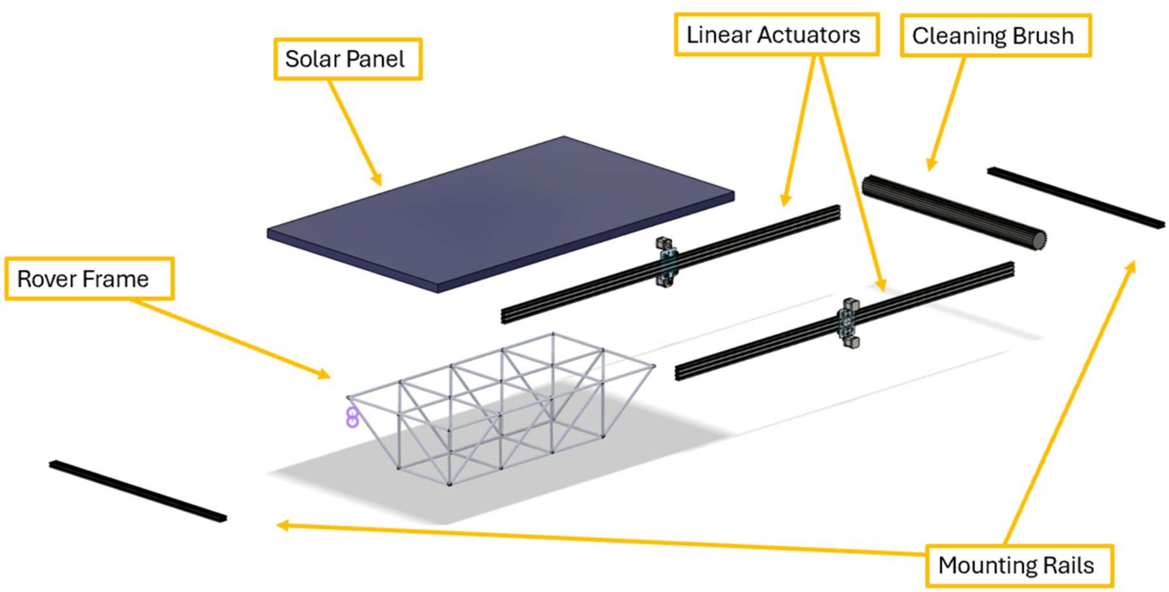
II. Executive Summary

Objectives and Problem Definition: The main objective of this project is to enhance the previously built rover, which is battery powered, by integrating solar power, autonomous power switching between solar and battery power, and an independent cleaning system for the solar panel.

Key Engineering Specifications: The rover must convert solar power into useable energy for the rover, autonomously switch between solar and battery power, and clean the solar panel.

Product/System Description: In this system, the solar panel absorbs solar energy that is then converted to useable power. The charge controller, which is attached to the solar panel, battery, and rover, then regulates the voltage to be sent at a constant level to the rover. When not needing power, the charge controller sends the extra voltage to charge the battery. If the solar panel is too dirty or sunlight conditions are poor, the charge controller can detect the power output loss. Through this detection, the charge controller can switch to battery power only. A dual-function activation system then initiates the cleaning system, where a brush rolls across the panel. Our cleaning system can be both manually activated and autonomously activated using signals from the relay driver and pyranometer.



Exploded View Solid Model and Physical Prototype Picture

Engineering Analysis Result Summary: The project had many considerations such as the current draw of all new subsystems being added. The team also had to consider the power supplied from the addition of the solar panel. Lastly, the team had to analyze how the battery life would be improved by the new system.

Manufacturing/Assembly Summary: This project has more assembly than manufacturing, most of the parts were sourced from vendors and just needed to be assembled together using bolts and screws to secure the design to the rover frame. Some components also needed to be cut to length, like the mounting rails and wiring, while other parts, like the flange couplings and gantry plates, needed to be custom manufactured by us.

Safety Considerations: During our Senior Design process, safety has always been the highest priority for our team. Without any experience in the machine shop our team made sure to take the appropriate time to learn the machinery needed to accomplish our project. This resulted in safe operation of machinery without any incidents. We also made sure to practice electrical safety while working on our rover.

Design-Phase Testing Results: Team 66 did not conduct any testing during the design phase.

Validation Test Plan & Results Summary: Testing was conducted on the power production of the solar power system, as well as the autonomous cleaning system. This project was successful in greatly increasing the battery life, and the cleaning system proved to be very effective and passed all activation testing. Testing was also done pertaining to the portability and speed of the rover after the additions made for this project, finding that the rover retained as much portability as possible, and the speed of the rover was not significantly affected by the additional weight.

Statement of Prototype Performance: The rover has satisfied every test apart from some needed design enhancements. This includes battery life extension of up to 206%. The cleaning system of the rover effectively cleans dust and debris off the panel at an acceptable rate. The rover has been driven in light rain with no problems. The additions made to the rover are all quickly detachable. The added solar panel takes a variable current load off of the battery depending on the weather.

III. Engineering Specification

III.A. Objective Statement

The main objective of this project is to enhance the previously built rover, which is battery powered, by integrating solar power, autonomous power switching between solar and battery power, and an independent cleaning system for the solar panel.

III.B. Introduction

III.B.1. Background Information

The original inspiration behind the creation of LOUIS was to expose young engineers to the development and design of rovers, as it can easily be seen that space exploration will continue to be a prominent area of work for engineers in the future. We intended to build upon that idea by incorporating solar power into the design of LOUIS. As educational proof of concept, we also designed a power switching system to autonomously switch between battery and solar power, depending on the environment. As many NASA rovers also use solar power, this addition pushed the rover one step closer to the design of rovers currently being used in space exploration missions. This made LOUIS an increasingly relevant tool for future students who are interested in working in this field.

Picture of LOUIS:

III.B.2. Problem Description and Motivation for the Project

The motivation for our project is to make power improvements to a space rover engineered by a previous group of Senior Design students. We hope that our solar-oriented improvements will enable enhancement opportunities for future students. Overall, our teams' motivation is to improve upon a platform that can be used as a vessel for teaching students about space rovers and solar power systems.

III.B.3. Existing/Competing Technologies

- SolarEdge Monitoring Portal: SolarEdge offers a cloud-based monitoring platform that provides real-time data on solar system performance, energy consumption, and grid interaction. The team is planning on doing something similar with a charge controller just on a smaller scale. Instead of grid integration we would be looking at solar panel voltage and current output and calculating the load needed to operate the rover.

- Nanosolar Coating: This advanced nanotechnology-based coating repels water and dirt, ensuring that rainwater washes away debris without leaving residues. This is a viable way to have a passive cleaning system for solar panels.
- Ecoppia E4: This autonomous robot uses soft brushes to clean panels without water. It can navigate arrays independently and is designed for large-scale solar farms. This robot uses a similar waterless brush system that the team is currently considering using to clean the solar panel. The main difference is that it is designed for a solar farm.
- NASA's Spirit and Opportunity Rovers: These are rovers launched by NASA in 2003. They both used solar panels to generate power. The panels collected sunlight and charged the rovers' batteries. This will be similar to what the team plans to do by adding solar panels to the existing rover. However, the team's solar panels will be able to both charge the existing battery as well as take on the electrical load of the rover.

III.B.4. Potential Customers

Primary: The LSU Electrical Engineering Department can utilize LOUIS for research, development, and educational purposes for their students.
Secondary: Future senior design students can enhance LOUIS's capabilities to learn more about the design and building process. Additionally, companies can use and adapt this technology for their own purposes, like energy companies adapting the technology to have access to unreachable areas.

III.C. Functional Requirements

Table III-1: Required Functions

#	Function	Explanation	Type(s) of Validation Test(s)
F1	F1	Convert solar energy to power the rover.	An ideal (sunny), non-ideal (cloudy), and an indoor power production test to assess how much power is being generated in various environmental conditions.
F2	F2	Autonomously switch power between power sources.	A battery charging test to see if the solar power is sufficient to charge the battery when LOUIS is stationary. A battery operation test to see if the battery life is extended during normal operation due to the addition of the solar panel.
F3	F3	Clean the solar panel.	An effectiveness test to see how power generation is affected before and after cleaning, and an activation test to see if the cleaning system is activated only when it must be.

The above functions were chosen based on the demands of the sponsor.

III.D. Qualitative Constraints

Table III-2: Required Qualitative Constraints

#	Qualitative Constraint	Explanation	Type(s) of Validation Test(s)
Q1	Compatibility with Existing Design	Make sure that the new implementations to the previous rover are compliant with each other. The additions should merge with the existing design seamlessly.	Speed Test on an incline hill to see how the existing weight added affects the speed of the rover. These results will be compared to the previous teams' incline speed test with 36kg of added weight
Q2	Weatherproof Design	The solar panel system should be built to withstand the elements of Louisiana such as wind, precipitation, and heat.	The group will be coating the electronics in a water-proof spray that we are safely assuming will protect the components rather than risk the components by testing them.
Q3	Portability	Make sure that the system doesn't hinder the transportation of the rover. It should be convenient to maneuver without disassembly.	The portability test was conducted by last year's team, showing that it could be carried by 2 people. To retain this portability, the group plans to time, the assembly and disassembly of the added components.

The above qualitative constraints were chosen based on optimal functionality of the rover, given its potential vulnerability to the weather, necessity for seamlessness between the new and old design, and previous constraints from the last project.

III.E. Measurable Engineering Specifications

Table III-3: Required Measurable Engineering Specifications

#	Name	Symbol	Units	Value(s)	Explanation	Type(s) of Validation Test(s)
M1	Budget	B	$	4000	A budget of $4000 was decided upon by the sponsor.	Running total of expenses
M2	Interior Space	S	M^3	0.977	This is the amount of available space inside the rover's chassis to install new electronic components.	Measurements of added components
M3	Weight Limit	W	Kg	<36.3	The max additional weight load was measured at 36.3kg. The team would like to stay under this variable.	Weighing of added components

The above measurable engineering specifications were chosen based on the financial constraint given by our sponsor and the limitations of the already-built rover.

III.F. Deliverables

Upon completion of our project, we will be delivering a functional solar power system which will be added onto the existing planetary exploration rover LOUIS. This solar power system will have autonomous power switching between battery and solar power, depending on available sunlight and user preference. We will also have an autonomous cleaning system incorporated into our design to clean the solar panel of dust or debris intermittently. Our team will ensure that the additions we make will only be improvements to the rover, without losing any of the general functionality existing prior to our additions. Other deliverables include comprehensive reports and presentations, as well as complete engineering analyses detailing the technical aspects of the project.

IV. Embodiment

IV.A. Functional Breakdown - Objective Tree(s)

One of our three main functions is to be able to clean our solar panel. To achieve this, our brush cleaning system is using a dual-function system where we will be able to manually activate as well as autonomously activate our brush cleaning system. The manual activation uses a push button beside our Arduino to activate the cleaning system, and the autonomous activation uses a signal from our relay driver along with a signal from our pyranometer to activate the cleaning system. These signals must be held for 1 minute before activation takes place, this was designed this way to avoid false positives with our cleaning system. Both signals activate the brush cleaning system which clears debris from the surface of the solar panel. This function affects the other two functions since debris greatly reduces the efficiency of power generation of solar panels, and since debris build up results in power loss it can lead to a switch between power sources. The second function is to be able to generate solar power utilizing the solar panel that is going to be attached to the rover. This solar panel will absorb sun rays and convert them to electricity which will then be used to power the rover and the battery. These affect the two other functions because the power from the solar panel will be used to power the two other functions. The third function is to be able to autonomously switch power sources. Using a charge controller, the rover should be able to detect power consumption and execute a pre-programmed action to be able to prioritize a more suitable power source. This affects the solar power generation function because it may result in a change in power source priority when the solar panel is not adequately supplying enough power due to lack of sunlight.

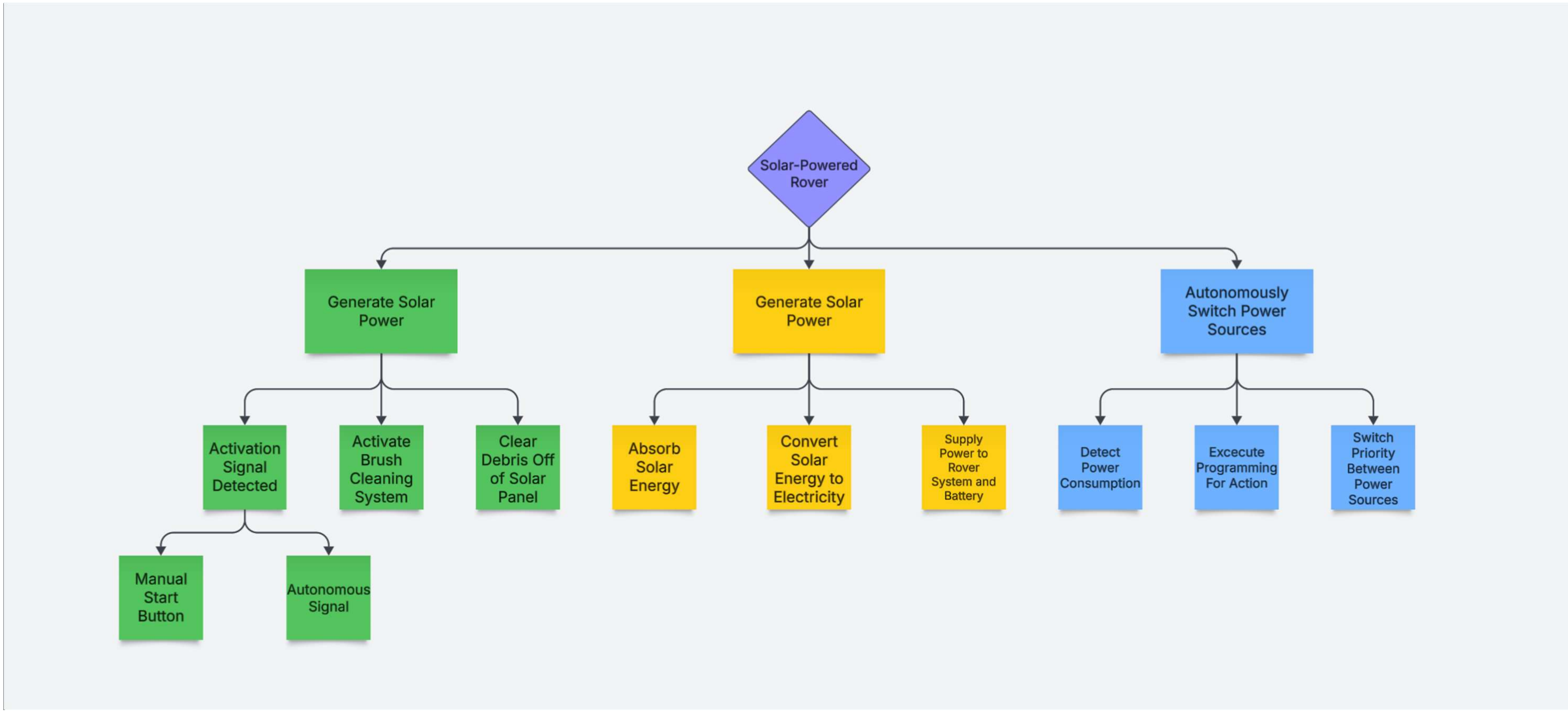


Figure IV-1: Project Objective Tree

IV.B.Concept/Solution Generation, Evaluation and Selection

IV.B.1. Concept/Solution Generation Method(s) Used

In general, each of the team members researched different pieces of the project and presented the material to one another to discuss the best options when proceeding forward. From here, we made sure to have a person or two to challenge the consensus by offering potential difficulties with each of our decisions to see if there was a potential alternative we should investigate that offered the desired solutions. We also drew inspiration from existing technologies, like the Lotus P4000 and Geva Bot, which are shown below.

Image 1: GEVA Bot

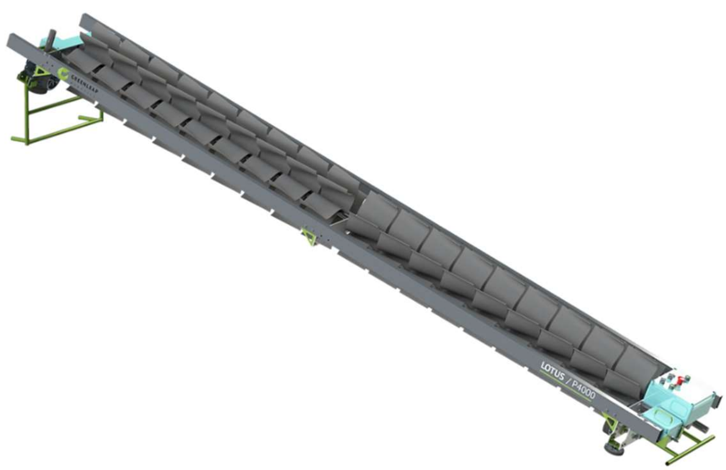


Image 2: LOTUS P4000

IV.B.2. Concept/Solution Evaluation Method(s) Used

We originally drew up a pros and cons list for each of our functions: the solar power, the autonomous power switching, and the cleaning system. After this, a decision matrix was made for each function to ensure our decisions were properly made, rather than by bias. The decision matrices matched what we had thought was the best option for each.

IV.B.3. Concept/Solution for Function F#1 – Convert Solar Energy for Power

IV.B.3.a. Original Concept/Solution Selected for Function F#1

The team proceeded with a design incorporating the monocrystalline solar panel. This solar panel has PV cells made from a single silicon crystal. This increases efficiency because the single silicon crystal used allows electrons to move more freely, generating more electricity. The lower temperature coefficient of a monocrystalline panel makes it more suited for the Louisiana environment, allowing it a longer life span. As for the cons, this type of solar panel is the most expensive. It is also heavier than some of the other options.

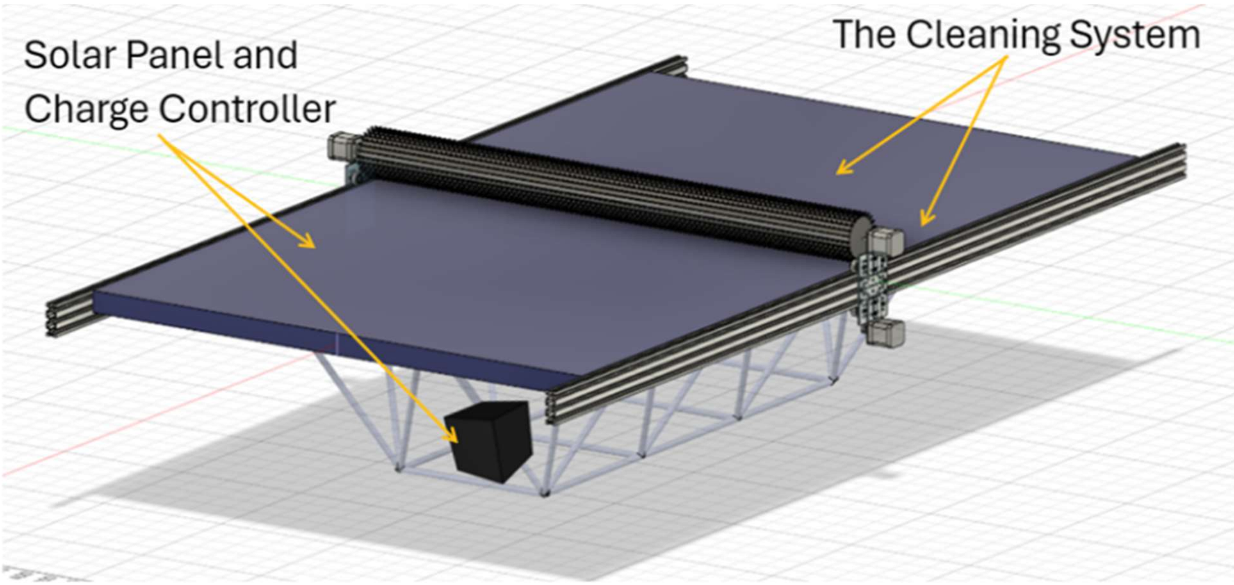


Figure IV-2: Illustration of Original Concept for Function F#1

IV.B.3.b. Re-design Concept/Solution for Function F#1

There was no redesign required for solution for function# 1

IV.B.4. Concept/Solution for Function F#2 – Autonomously Switch Power Between Power Sources

IV.B.4.a. Original Concept/Solution Selected for Function F#2

The concept selected to autonomously switch between power sources is the solar charge controller. As shown in the picture below the charge controller is able to control how much power the battery receives depending on the current draw of the load. This protects the battery from being overcharged, so if the battery is getting close to 100% it will send more of the array power to the load rather than charging the battery, the opposite goes for is the battery is low. We decided specifically on a max power point tracking (MPPT) charge controller over a pulse width modulation

(PWM) charge controller because the MPPT has a higher power conversion rate of about 99% compared to PWMs 75%.

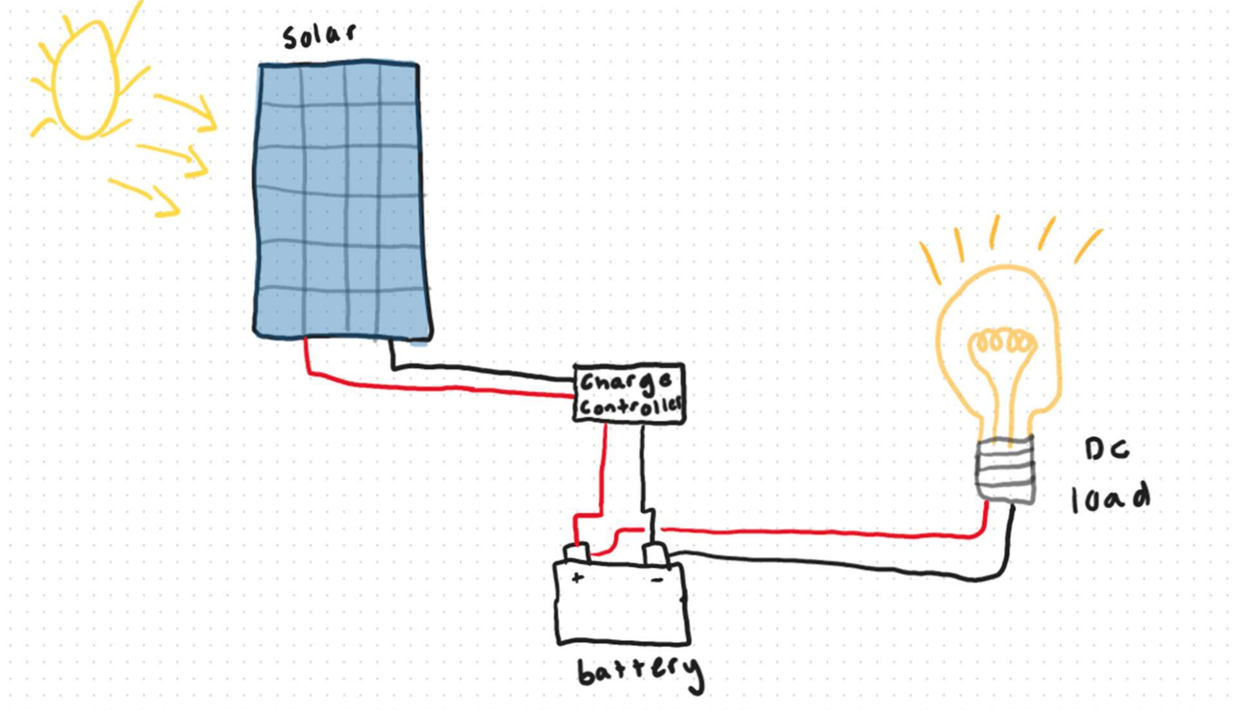


Figure IV-3: Illustration of Original Concept for Function F#2

IV.B.4.b. Re-design Concept/Solution for Function F#2

No redesign was needed for this concept

IV.B.5. Concept/Solution for Function F#3 – Clean the Solar Panel

IV.B.5.a. Original Concept/Solution Selected for Function F#3

The concept we originally selected for the cleaning system was the brush method. As shown in the image below, the purple rectangle represents the rotating brush, and the orange rectangles represent rails that the brush rolls across. We drew inspiration for this idea from the Lotus P4000 and the Geva bot shown on part IV.B.1. This was the most commercialized idea we saw that is popularly employed on solar farms, utilizing wet or dry-cleaning methods. This was the most feasible method we generated due to low cost, relative effectiveness, and ease of maintenance.

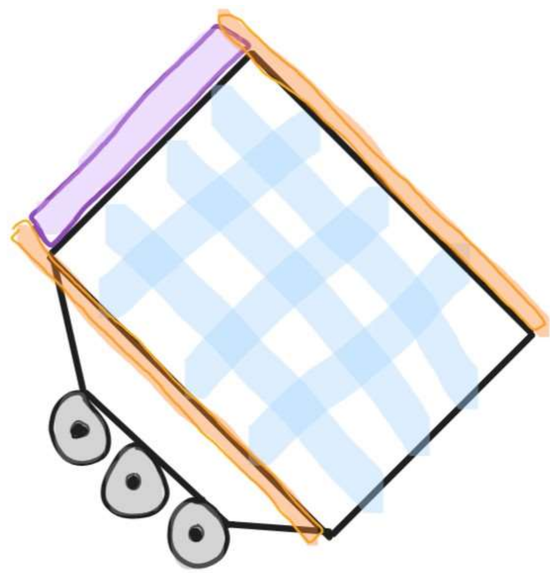


Figure IV-4: Illustration of Original Concept for Function F#3

IV.B.5.b. Re-design Concept/Solution for Function F#3

No re-design was necessary for this function.

IV.C. System Description/Product Architecture

Solar energy is converted into usable power for the rover by the solar panel as a new primary power source. A programmable solar charge controller sends power to the load or the battery, based on power consumption of the rover and power available from the solar panel. If solar power is insufficient for the rover to function, the rover will rely on battery power to supply the remaining electrical load. The solar panel is cleaned using a rotating brush. The brush is moved along the length of the solar panel by two linear actuators driven by DC stepper motors. Another DC motor rotates the brush as it moves across the panel. The activation of the cleaning system relies on two variables: solar panel current and solar radiation. A relay driver that is in constant communication with the solar charge controller monitors solar panel current, sending a signal to the Arduino when the current drops below a set threshold. A pyranometer is used to measure solar radiation, sending that measurement to the Arduino as well. If the solar panel current is below the threshold, but the pyranometer is detecting that the rover is outdoors and in sufficient sunlight, this signals that the solar panel is dirty and not producing adequate power, activating the cleaning system. Inputs will be sent to the three DC motor controllers which will drive the DC motors to clean the solar panel. However, if the pyranometer measurement is below a set threshold, indicating that the rover is indoors or not in bright or direct sunlight, the cleaning system does not activate. This eliminates the possibility of having false positives when the rover is taken indoors or in the shade.

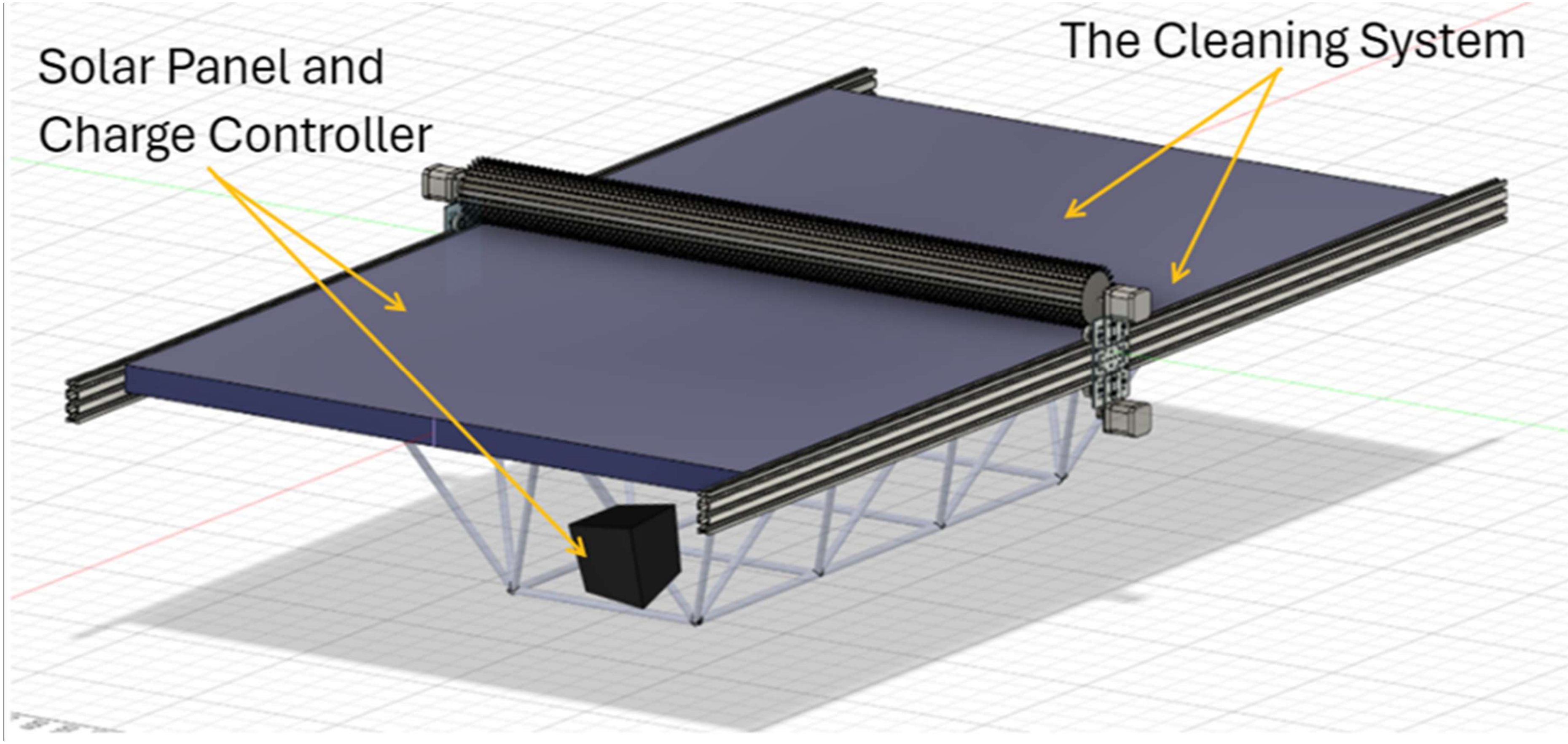


Figure IV-5: Assembled Prototype Drawing

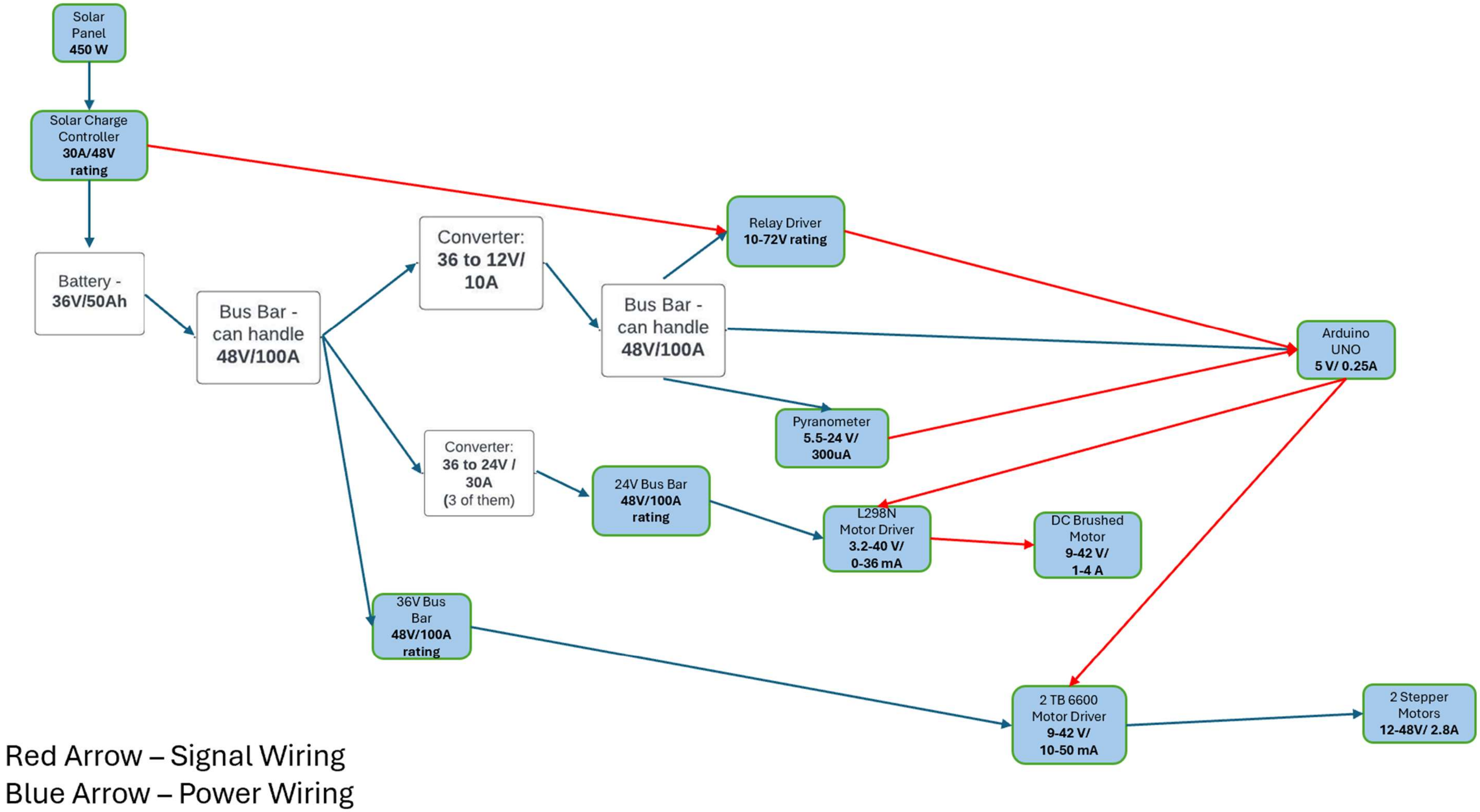


Figure IV-6: Signal Wiring Diagram

Table IV-1: List of Sub-Systems

#	Name	Brief Description of Sub-System Functionality
SS1	Solar Panel and Charge Controller	Provide solar power, voltage regulation, and power distribution to the rover.
SS2	The Cleaning System	The linear actuators run alongside the solar panel to cause the brush to move across the panel. There is also a gear motor that will cause the brush to rotate.

IV.C.1. Description of Sub-System SS#1 – Solar Panel and Charge Controller

This subsystem of the project is the newly added solar power system. The solar panel is used to generate power for the system. It does this by absorbing solar radiation and then converting those solar rays to electricity. This is used to power the rover and charge the battery. For the goal of autonomous power flow, a solar charge controller is used. This controller is programmed by the team specifically for the battery and solar panel on-board the rover. The power being sent by the charge controller is capable of carrying more or less of the electrical load, depending on the environment. Any additional power generated by the panel that is not consumed by the rover is sent to charge the battery. In the picture shown below, I1 is the current produced by the panel while I2 is the current the battery is providing to supply the rest of the electrical load, however I2 will switch direction when the rover is at standstill, recharging the battery.

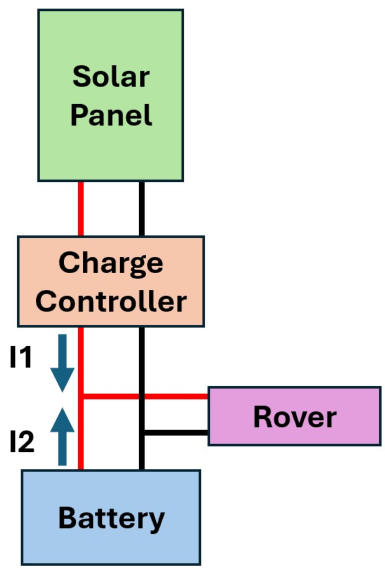


Figure IV-7: Drawing of Sub-System SS#1

IV.C.2. Description of Sub-System SS#2 – The Cleaning System

This newly added subsystem is the Brush Cleaning System of our rover. The overall goal of this subsystem is to detect and clean any form of dust or debris on top of our solar panel to maximize the efficiency of power generation. This subsystem is made up of linear actuators, a motor driven brush, and various electronics for added intelligence. The linear actuators are composed of the linear actuator rails and stepper motors to control the overall movement of the brush cleaning system. The motor driven brush is then connected to this linear actuator system to perform the cleaning movement. The brush itself is a solar panel safe brush connected to DC gear motor on one end and a bearing on the other end to allow for a spinning motion which is designed to help with cleaning the solar panel. This whole cleaning system is controlled by our Arduino Uno. The Arduino Uno has an embedded code that controls the movement of the entire cleaning system. Aside from movement the Arduino Uno is in control of activation of the cleaning system. This is in the form of a manual activation or an automatic activation using our relay driver and our pyranometer. The Arduino Uno also controls the safety features of the cleaning system. We designed the cleaning system to have a cancel button along with a buzzer to alert operators nearby that a cleaning cycle is about to begin. This subsystem will be effective in maximizing power production while being safe and smart.

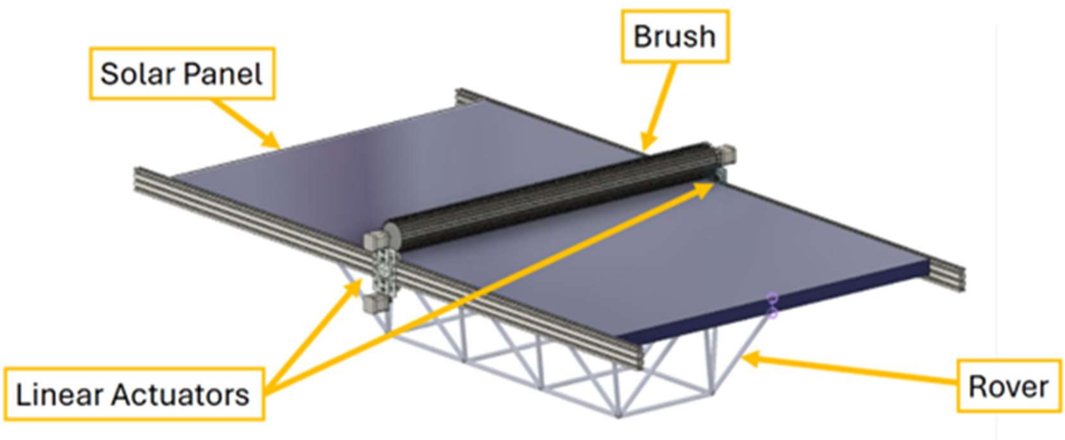


Figure IV-8: Drawing of Sub-System SS#2

V. Engineering Analysis and Materials Selection

V.A. Engineering Analysis for SS#1 – Solar Panel and Charge Controller

V.A.1. Types of Eng. Analysis Conducted:

- MATLAB Simulink
  - State of Charge
  - Temperature vs Efficiency
  - Solar Radiation vs Power Generation

V.A.2. Eng. Analysis & Materials Selection for SS#1-P#1&2 – Solar Panel and Charge Controller

Bowen Williamson and William Mancuso

While working on the analysis, we discovered that charge controllers that use PWM or pulse width modulation, have a power conversion rate of about 75% as opposed to the charge controllers with MPPT or max power point tracking technology, which have a conversion rate of about 99%. This meant much more efficiency, which made that the obvious choice since one of the main constraints was increasing the battery life of the rover. This analysis was conducted using a MATLAB Simulink model of a MPPT solar charge controller system.
Here are some of the results:

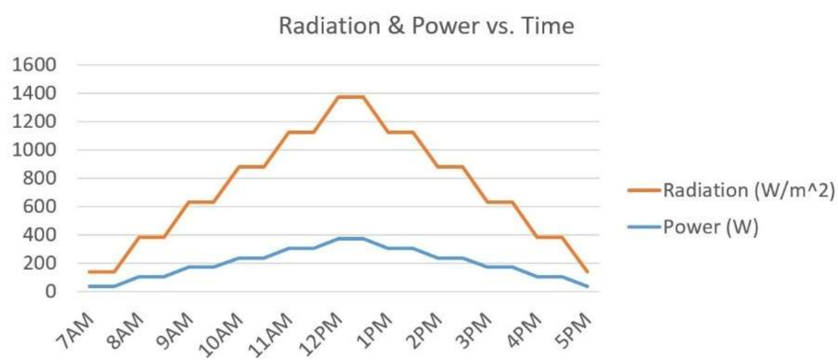


Figure V-1: Solar Radiation & Power vs. Time Graph

This graph shows the relationship of solar radiation and power generation. With varying solar radiation shown in orange, simulating different times of day. The blue line shows the corresponding increase and decrease in power produced by the panel, with our peak power output occurring between 12 and 1 o'clock, hitting almost 400W, which was consistent with actual values that we measured during our testing.

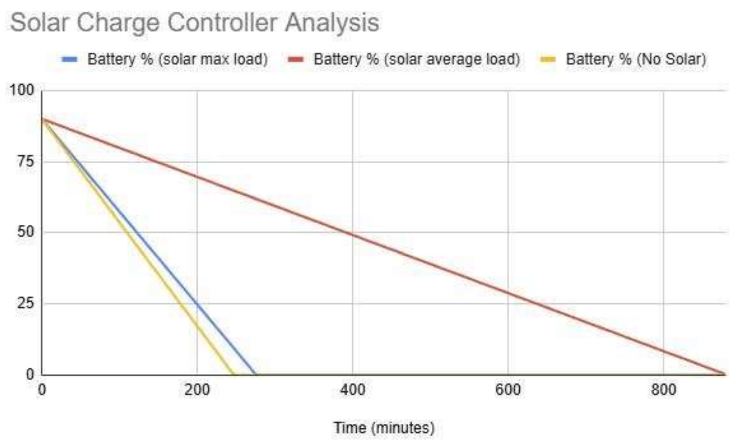


Figure V-2: Battery State of Charge vs. Time Graph

This graph shows the battery's state of charge over its entire operating life. Here we're comparing the performance of the battery before and after the addition of solar power. The yellow line represents battery discharge without any solar input. It's based on the rover's average current draw of about 8 amps, which gives us a runtime of approximately 4.5 hours. The blue and red lines show simulations with solar power connected. The red line shown on the graph is from a simulation run with the average 8-amp load again, but with solar power factored in. This shows a significant improvement in the life of our battery, giving us a runtime of about 14 hours. The blue line shows the battery lifetime if the rover was continuously drawing its maximum current—which is around 11 amps—and this still results in a slight improvement in battery life.

V.A.3. **Eng. Analysis & Materials Selection for SS#1– Overall circuit**

Kenny Bui

The overall goal of this analysis is to see how the existing battery's state of charge would respond with the addition of our new load that we have implemented to our rover. To perform this analysis, I reused a load analysis performed by the previous senior design group and added it to our own load analysis on the new components that we plan on adding to our rover. I used these calculations and implemented them into a Simulink model that depicts our overall circuit. After running the simulation, we noticed an expected decrease in state of charge, however, we confirmed that the battery could handle our additional load. The images below show the Simulink model along with the results. After an elapsed time of one hour, the original load is at 60.9% state of charge and the new load is at 38.9% state of charge. Since our battery can handle the new load, this analysis justifies the additions to our overall circuit.

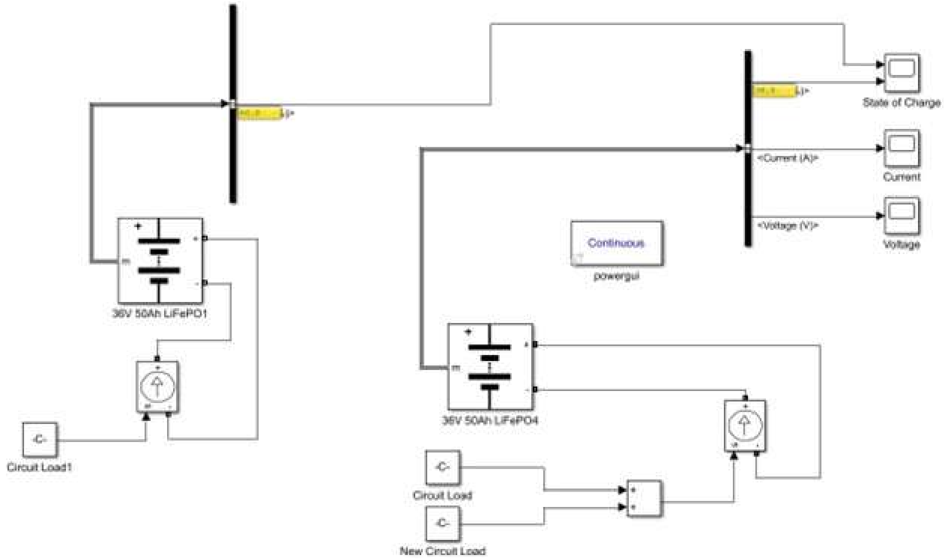
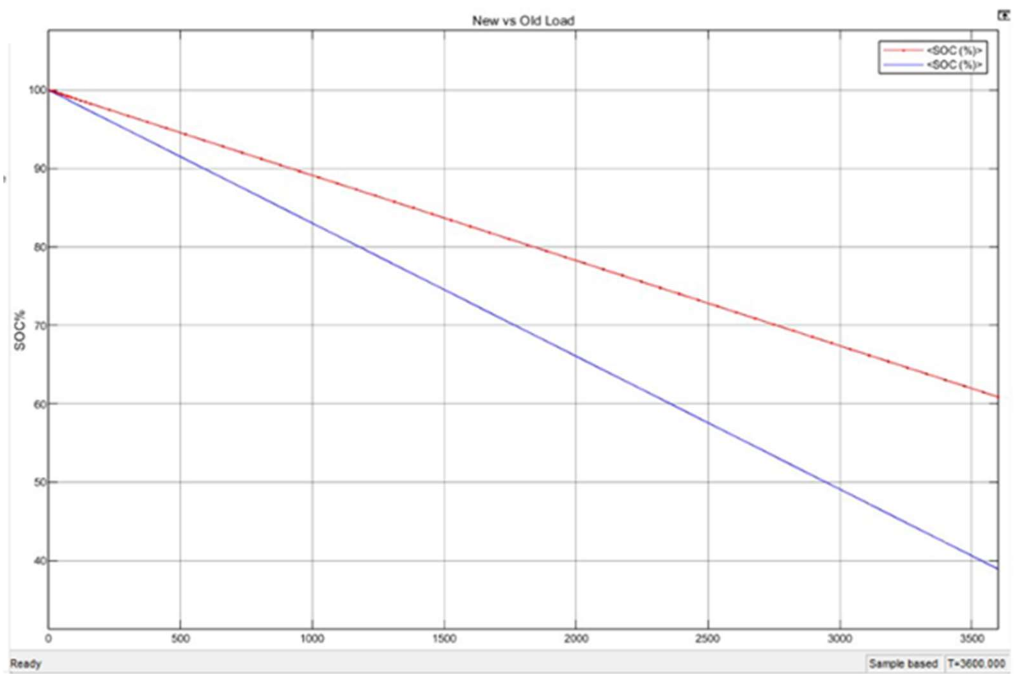


Figure V-3: Complete Circuit Simulink Model



Red: Original Load (60.9%) Blue: New Load (38.9%)

Figure V-4: State of Charge vs. Time Graph; Original and New Load

V.B. Engineering Analysis for SS#2 – The Cleaning System

V.B.1. Types of Eng. Analysis Conducted:

- The current draw of the stepper motor was measured to determine the load requirements and ensure cohesion with our solar panel system. Additionally, this is to see how the stepper motor driver should be configured to match the motor's requirements, optimize performance, and prevent overheating/inefficiencies.
- The current draw of the geared DC motor was measured to determine over all cohesion with the existing power system along with the addition of the solar panel. The torque to shaft speed along with the amount of current draw to achieve the desired speed to extend the motor life and achieve cleaning.

V.B.2. Eng. Analysis & Materials Selection for SS#2-P#6 – Arduino Uno

Kenny Bui

For the material selection of our electronic intelligence, we did thorough research to determine which type of intelligence would be best suited for our needs. When determining the right intelligence for our rover, we did research into integrated circuits, microcontrollers, and microcomputers. Integrated circuits are a cheaper, more simply approach, but would require a lot of parts and could be affected by the interference around the chassis. Microcontrollers are a little more complex and expensive but provide an adequate amount of electronic control without overcomplicating the objective. Microcomputers are the most expensive and complex option. These can certainly get the job done but are suitable for more complex and complicated objectives. We chose the Arduino Uno because it is readily available, at a reasonable price, and is appropriate for this kind of objective. The Arduino Uno has input and output locations that make our activation design more accomplishable. This device also carries the appropriate amount of memory making our system reusable and operates smoothly. The Arduino Uno microcontroller is overall the best selection for our application.

V.B.3. Eng. Analysis & Materials Selection for SS#2-P#1 – NEMA 23 Stepper Motor

Margaret Burkes

<u>Calculations:</u>
Mass of stepper motors: 0.75kg
Mass of gearmotor: 0.5kg
Mass of brush: 2.7kg

Total mass: 3.95kg

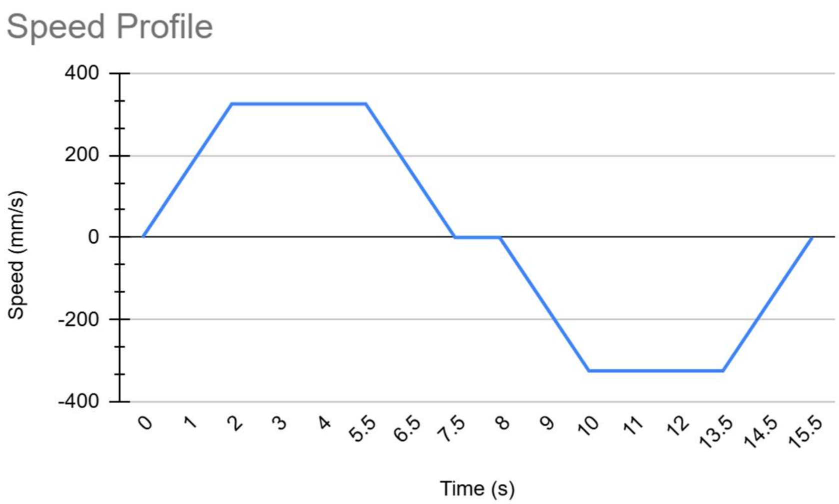


Figure V-5: Speed Profile for Stepper Motor

$$r = \frac{(\#\, of\ teeth)(pitch)}{2\pi} = \frac{20 \times 3mm}{2\pi} = 9.5mm$$
$$\omega_1 = \frac{v}{r} = \frac{150\ mm/s}{9.5mm} = 15.71\ rad/s$$
$$\omega_2 = \frac{v}{r} = \frac{300\ mm/s}{9.5mm} = 31.42\ rad/s$$
$$\frac{d\omega}{dt} = \frac{\Delta\omega}{\Delta t} = \frac{31.42 - 15.71}{2 - 1} = 15.71\ rad/s$$
$$T_{em} = \frac{1}{2}Mr^2\frac{d\omega}{dt} + r^2M\frac{d\omega}{dt} + rf_l = 0.38Nm$$

<u>Selected Motor Torque Curve:</u>

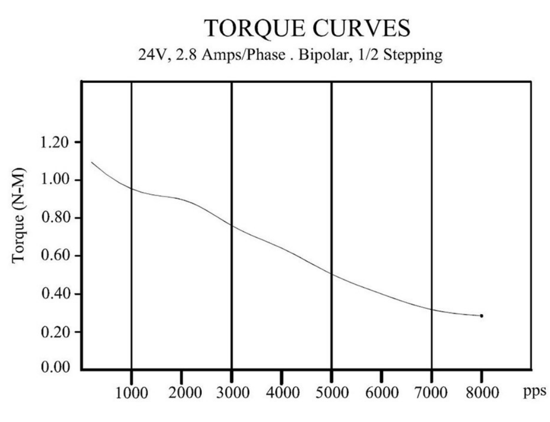


Figure V-6: Torque Curve for Stepper Motor

After running a MATLAB simulation, the team measured the maximum current draw to be 2.5A. For more information on this simulation, refer to the appendix.

V.B.4. Eng. Analysis & Materials Selection for SS#2-P#7 – TB6600 Stepper Motor Driver

Margaret Burkes

TB6600 Stepper Motor Driver Specs:

Operating Voltage	9-42V
Input Current	1-4A
Max Output Current	3.5A
Input Control Voltage	5-24V
Max Subdivision	32 Microsteps

The above chart shows how the operating voltage was within the operating voltage of the NEMA 23 stepper motor. The max output current is a little high for the NEMA 23 stepper motor, however, there are settings on the driver that allow us to be able to control the output current. The dip switches were set to quarter-stepping mode with a maximum current of 2.8A.

V.B.5. Eng. Analysis & Materials Selection for SS#2-P#9– Inline gearmotor (model SEI 37JB270G/32ZYT30-2468)

Keldon Ngo

The 24V DC gearmotor is electrically optimized for low-speed, high-torque applications. In this project, it is used to rotate a 6-pound cylindrical brush is rotated by a ball bearing at the opposite end. This setup introduces rotational inertia and requires steady torque delivery at low speed for consistent operation. Electrically, the motor is powered by a 24V DC supply and draws approximately 60 mA at no load. The gearmotor's internal gear reduction (270:1) enables it to produce significant output torque—estimated around 8 in-lbs.—despite its compact size and modest current draw. This reduction also means the motor's electrical efficiency is maximized at low speeds, making it ideal for rotating heavy, slow-moving components like the brush.

V.B.6. Eng. Analysis & Materials Selection for SS#2-P#8– L298N Dual H-Bridge

Keldon Ngo

The L298N dual H-bridge motor driver serves as a critical component in controlling the motion of two DC motors or a single stepper motor, offering full bidirectional control and current amplification. Its primary functional requirements include the ability to switch motor direction, handle logic-level signals from a microcontroller, and enable precise speed regulation via PWM inputs. Each channel of L298N is rated for up to 2A continuous current, with a maximum peak of 3A under short-duration loads. Mechanically, the L298N module includes mounting holes to secure it in robotic or mechatronic assemblies where vibration may be present such as driving the rover over rough terrain. The L298N supports a wide input voltage range from 5V to 46V for the motors, while the logic voltage is maintained at 5V.

V.B.7. **Eng. Analysis & Materials Selection for SS#2-P#4– Pyranometer**

Bowen Williamson

The pyranometer plays a key role in the rover's autonomous cleaning system. Its primary function is to measure solar irradiance in real time and help determine whether the rover is operating indoors or outdoors. This information is used in conjunction with the solar panel's current output to assess whether the panel's performance is being hindered by surface contamination. When the pyranometer detects high irradiance levels but the panel output is low, the onboard control logic infers that the panel may be dirty, triggering the cleaning mechanism.

Given its placement on the front of the rover, the pyranometer is fully exposed to outdoor environmental conditions, including direct sunlight, dust, and potential moisture. While dust accumulation is a concern, the sensor's hemispherical geometry helps mitigate this by reducing flat surfaces where particles could settle and block light. This self-cleaning aspect by design made additional protective enclosures unnecessary, balancing simplicity with adequate environmental resistance.

For mechanical integration, the pyranometer is mounted using a 3D-printed support bracket. This bracket was designed to slide into an aluminum T-slot rail that forms part of the custom frame supporting the solar panel. This modular approach allows for adjustability in sensor placement and easy removal for maintenance.

From an electrical perspective, the selected pyranometer outputs an analog voltage in the 0–5 V range, making it directly compatible with the Arduino's analog input pins without the need for level shifting or signal conditioning. This simplifies the electrical integration and reduced the component count. Additionally, the sensor's wide operating voltage range of 5.5–24 V allowed it to be powered directly from an available 12 V bus bar on the rover, further streamlining the power distribution design.

VI. Manufacturing & Assembly

VI.A. Manufacturing

For the manufacturing of this project, gear for welding was used to weld to the frame of the solar panel to install rails used for mounting the system, this includes all of the necessary safety equipment, like welding hoods, gloves, and jackets. These rails also needed to be cut to length so since they are made of aluminum, the group used a bandsaw, particularly one that is meant specifically for cutting aluminum. We also used a drill to tap a threaded hole through the rails of the cleaning system and the welded t track rails. To do this the group first drilled a hole and then used a M5 x 0.8 tapping bit to thread the hole in the rails. The installed screws of the newly tapped holes needed to be flat head screws to be sure they do not inhibit the movement of the gantry plate sliding back and forth on the cleaning system.

To make the flange couplings, an aluminum rod was used for its relatively light weight. The flange couplings were slowly turned down to the measurements taken of the brush core and the motor shaft, however as those measurements got close, we began checking the fit, because we wanted this to fit very snug around the shaft and in the core, even with the set screws holding them. Once the flange couplings were machined to the correct size, the one used for connecting the brush to the gear motor needed to accommodate an M5 set screw to secure the coupling to the D-cut motor shaft. This was done on the tap table in the machine shop. The hole was drilled with a size 20 drill bit, and then tapped with an M5 tap.

For the manufacturing of the gantry plates, we first measured the holes on the base plate to be 20mm apart, with that, we 3D printed a gantry plate with 5mm of space between them as a stencil. After drawing where the motor and ball bearing would be best fit on the stencil we then used a drill press to create holes large enough for the motor shaft and the attaching screws. We then secured the motor to the stencil and placed the entire assembly on the base plate. After attaching the assembly, we realized we needed the slots to be about 11mm longer to have enough adjustable room for the brush to sit comfortably on the panel. We then took an aluminum plate and cut the appropriate-sized piece from it, sanding the material to make it uniform and remove jagged edges. After manufacturing the plate, we milled in the appropriate slots, leaving a 5mm gap between the next slot. We also extended the existing 25mm slots to 36mm. After ensuring the plate fit we then used the previously created stencil to drill the appropriate holes for the bearing and motor. During assembly of the entire system, we realized that the previously used T-track rails were too small for the newly milled slots. We then took a smaller aluminum plate and cut it into a rectangle of 25mm and tapped it to created a mounting bar for the screws to secure the gantry plate to the base plate. After mounting the plates, we made sure the flanges are lined up and at the appropriate height to ensure they sit evenly on the panel. This concludes my manufacturing and assembly of the custom gantry plates. Lastly, we created a website to include reference information and instructions that can be found here: https://megburkes1.github.io/LOUIS/index.html.

VI.B.Assembly

The assembly of the new additions to the rover will start with the solar panel. The two 20x20 rails are welded to the bottom of the solar panel frame. These rails have tracks that will fit t-track nuts. These nuts secure a two-hole electrical strap around the frame using screws on the other end. This secures the solar panel to the frame of the rover. Next to be added are the cleaning system rails. Those are secured to the sides of the solar panel and screwed into the rails that were welded underneath. The rails were tapped through to add threads for the screws. The screws are flat head to make sure they are flush for the gantry plate to roll over, and not impede the movement, they can be removed for disassembly.

For electrical wiring the plan was to add on to the existing design. Shown below is the picture of the add-ons with the old wiring document. The white boxes are the old components, the blue boxes are new components, blue arrows are new electrical wiring. Also included in the appendix is a more in-depth look at the wiring. This included the pin slots and fuses added to the connections. The solar panel is connected to the charge controller with the included wiring, however the IP68 connectors were swapped out for a waterproof switch. The same will happen for the battery; we added a switch that can disengage the power to the charge controller at any moment. The stepper motor drivers are powered with 36 V and that busbar was connected to the drivers via 16 AWG wiring. Those motor controllers connect the motors with the wiring the motors came with. The rotational motor is powered with 24V and both the busbar connection to the driver and the driver to the motor, is wired with 14 AWG. The relay driver is fed with 24 V, and 14 AWG, and the signal wiring goes to the Arduino with 22 AWG. Finally, the Arduino sends a signal to the controllers telling them how to operate.

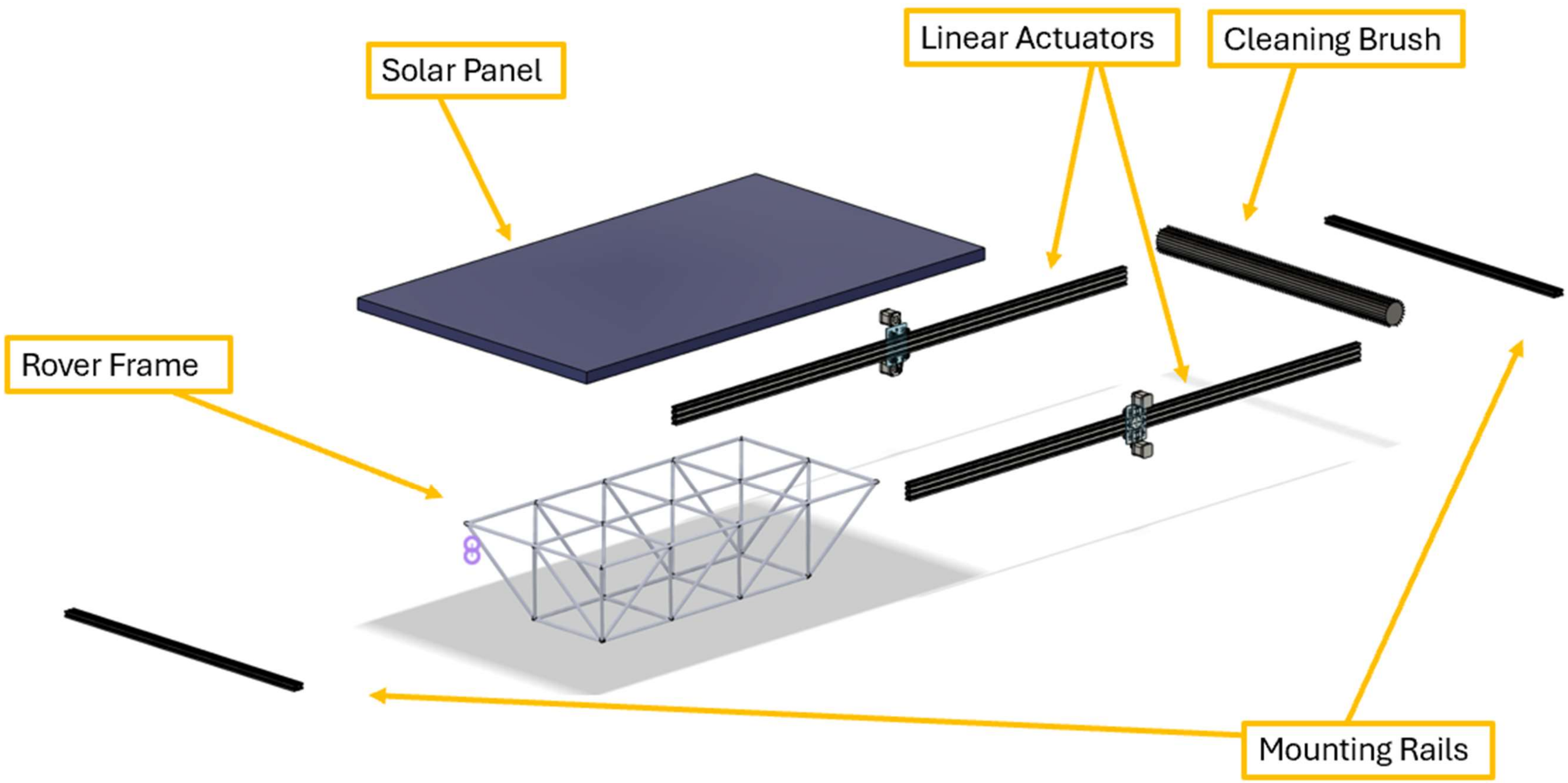


Figure VI-1: Exploded View Assembly Drawing of Prototype – Sub-System Level

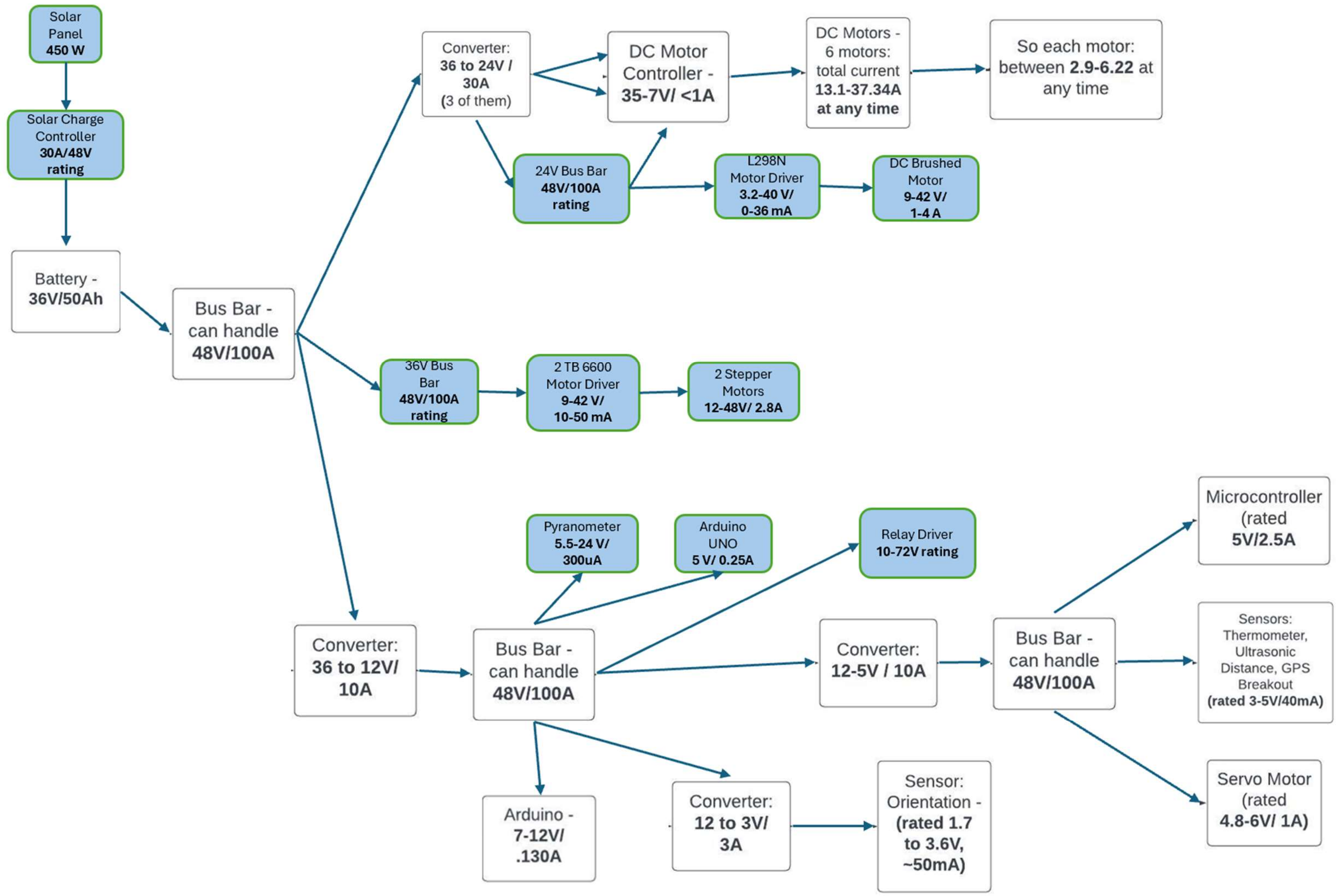


Figure VI-2: Electrical Diagram – Sub-System Level

VII. Safety Considerations

VII.A. Operational Safety

During our project process, our group thought about various safety concerns during operation that we have provided solutions to. The first concern is electrical shock and arc flash hazard. To prevent this hazard, we have made sure that all our electrical connections are grounded properly and that any bus bars are insulated to prevent a short across the terminals. We also implemented properly sized fuse connections to important electrical connections to prevent overcurrent damage. We were also concerned with the moving parts of the brush cleaning system causing a hazard, so we added a cancel button and a buzzer to create a system to stop the process of the cleaning system and add an auditory warning to those nearby. Lastly, in the case of emergencies, we have kill-switches on the battery terminals to completely de-energize the rover. Bus bars and important buttons have been labeled as well for ease of vision. These factors were thought out to ensure operational safety for our rover and systems.

VII.B. Safety During Manufacturing, Assembly and Testing

For safety during manufacturing, assembly, and testing, we utilized all of the safety considerations from operating the rover, but we also had additional safety considerations. For manufacturing and assembly, our group has spent a significant amount of time in the workshop, so making sure that we acknowledged proper workshop safety was essential. For example, proper PPE for our tasks as well as making sure to be aware of our surroundings are essential safety considerations that we had. For testing, our group ensured that the tests being run were setup correctly to prevent any unexpected results that may be deemed dangerous. With these safety considerations in mind, our safety during manufacturing, assembly, and testing was fulfilled.

VIII. Environmental Impact Assessment

VIII.A. Manufacturing Related Environmental Impact

The manufacturing plan includes many components however all the components are recyclable. The largest environmental impact coming from manufacturing is the welding of some components for mounting purposes. This includes air pollution such as metal fumes and gaseous emissions.

Welding also generates waste such as electrodes and filler metals. These can be mitigated with proper ventilation and eco-friendly materials.

VIII.B. Operation/Usage Related Environmental Impact

It is confirmed that the addition of solar power to the rover will increase the intervals between charging the onboard battery, decreasing the total energy required from a utility company for charging the battery. The main advertised advantage for using solar power is that you will not have to consume as much power from the grid. The rover can be charged while idle with solar power engaged. There should not be any negative environmental impacts from the operation of the solar power system.

VIII.C. End-Of-Life (Disposal) Related Environmental Impact

The motors and electronics used for the self-cleaning system should be recycled at their end-of-life, as they can be reused in the manufacturing of new devices. The solar panel should also be recycled once it becomes obsolete or no longer produces sufficient power. Solar panels contain heavy metals such as cadmium and lead, which will become hazardous materials if a solar panel is simply thrown away in a landfill. Recycling can also repurpose the rare elements used in photovoltaic cells such as gallium and indium, which could be used in the production of new solar panels.

IX. Design-Phase Testing

IX.A. Design-Phase Testing Objectives, Rationale & Brief Description

Team 66 did not conduct design-phase testing.

IX.B. Design-Phase Testing Outcomes & Conclusions

Team 66 did not conduct design-phase testing.

X. Testing, Validation, and/or Implementation

X.A. Testing & Validation of Function F#1 – Solar Panel Power Output

Keldon Ngo

The following tests were performed: Sunny conditions test, partial cloud cover test, heavy cloud cover test, and indoor test.

X.A.1. Objective, Rationale & Brief Description - F1

Sunny Conditions test

Objective: To test the panel voltage, current, and power output under sunny conditions with no cloud cover.

Test Description: The panel was placed flat on the ground with no shadows being cast onto the panel to maximize power generation. The voltage and current were then measured using a special multimeter.

Partial cloud cover test

Objective: To test the panel voltage, current, and power output under partly cloudy weather conditions.

Test Description: The panel was placed in the same spot as the sunny conditions test to maximize consistency. There were no additional shadows cast onto the panel, the only difference being the partly cloudy conditions. The voltage and current were then measured using a special multimeter.

Heavy cloud cover test

Objective: To test the panel voltage, current, and power output under cloudy weather conditions.

Test description: The panel was placed in the same spot as all previous tests to maximize consistency, the only factor being there was extremely heavy cloud cover. The voltage and current were then measured using a special multimeter.

Indoor Test:

Objective: to determine if the panel would generate power indoors at all and to see if the battery would charge while being in storage.

Test description: The panel was placed on top of the rover in which it was assumed it would be stored. The room used had no windows to minimize any false power generation. The voltage and current were then measured using a special multimeter.

X.A.2. Testing & Validation Summary of Results & Conclusions – F1

Table X-1: Sunny test conditions results

Test #	Date & Time	Voltage	Current	Power

Test 1	2/25 11:33 AM	50.1V	7.0A	350.7W
Test 2	2/25 11:39AM	50.3V	7.1A	357.1W
Test 3	4/25 12:15PM	54.9V	8A	439W

Figure X-1: Sunny Conditions test conducted in the front lawn of PFT

Table X2: Partly Cloudy Conditions Test

Test #	Date & Time	Voltage	Current	Power
Test 1	3/13 10:58AM	51.5V	8.3A	427.5W
Test 2	3/13 11:06AM	48.6V	7.8A	379.1W

Table X3: Cloudy Conditions Test

Test #	Date & Time	Voltage	Current	Power
Test 1	3/19 3:25PM	50.7V	8.3A	420.8W
Test 2	3/19 3:40PM	45.7V	7.3A	333.6W

Table X4: Indoor test

Test #	Date & Time	Voltage	Current	Power
Test 1	3/18 11:18AM	47.6V	6.45A	307.0W
Test 2	3/18 11:35AM	46.9V	6.23A	292.2W

X.B. Testing & Validation of Function F#2 – Autonomously Switch Power Between Power Sources

Bowen Williamson

X.B.1. Objective, Rationale & Brief Description - F#2

Per the requirements of the project, the sponsor Dr. Meng has requested that the rover have autonomous power switching, between solar and battery power. Even though the largest reasonable panel was purchased for the project, the electrical demands of the rover are too much to be supplied by the solar panel alone. This test was conducted to prove autonomous power flow was taking place, meaning the solar panel supplies more or less of the load as the environment varies, while the battery acts as a secondary source of power to make up the difference.

X.B.2. Testing & Validation Summary of Results & Conclusions – F#2

Testing was conducted with the motors of the rover driven at full speed. All measurements of battery current were done by monitoring the mobile app that connects to the battery, and solar power current was measured using a clamp-style multimeter. A baseline measurement of current draw was taken with no solar power connected. This time with solar power turned on, measurements of both solar panel current and battery current were obtained on both a cloudy day and a sunny day, and the corresponding values are shown in the data table below. This test confirmed goal of autonomous power flow was achieved, as the solar panel now acts as the main power source and battery will provide any additional power needed to supply the electrical load. The results of this test also help explain the significant increase in battery life with the solar power addition, when looking at the difference in the electrical load the battery has to sustain.

Table X-2: Test Result Data Table

	No Solar	Cloudy	Full Sun
PV Current		2.5A	7.7A
Battery Current	11.0A	8.8A	3.4A

X.C. Testing & Validation of Function F#3 – Clean the Solar Panel

Margaret Burkes and Kenny Bui

X.C.1. Objective, Rationale & Brief Description - F#3

The goals were to ensure that the cleaning system: activates when triggered by a dirty solar panel and effectively removes debris from the solar panel with the goal of restoring the optimal power output given the weather conditions. To attain this goal, we conducted the following tests:

- Test 1: Activation Test

The objective of this test was to see if the activation portion of the brush cleaning system would work. We wanted to see if the conditions that we have set within the Arduino code would work given the real-world scenarios. To execute this test, we uploaded the updated Arduino code to the Arduino Uno. Then we placed jackets on top of our solar panel to simulate dust and debris. After we would stand by to see if the Arduino Uno would recognize the "dirtiness" of the panel and execute an activation of the brush cleaning system.

- Test 2: Effectiveness Test

The procedure is as follows: using a multimeter, measure the power output of a clean solar panel. This will be the base measurement for comparison purposes. Apply a measured amount of flour evenly across the panel and record power generation at varying levels of obstruction. Activate the cleaning system and re-measure power generation. Compare the power generation from before to after cleaning to assess effectiveness.

X.C.2. Testing & Validation Summary of Results & Conclusions – F#3

Test 1: Activation Test

While no exact numerical results were produced from this test, we do have video recordings of the activation taking place. From these videos, we can confirm that the activation test of our brush cleaning system was a success. This test proved to our group that the code is functional and allows fine-tuning to make the cleaning system optimal. Adjustments to the pyranometer reading output along with adjustments to the activation signal hold will not only reduce false activations but also make the cleaning system cycles more efficient.

Test 2: Effectiveness Test

Table X-3: Test Result Data Table

	Current (A)	Power (W)
Before	8.00 A	450 W
During	2.25 A	125 W
After	6.5 A	350 W

Using these results, we used the following equations to determine its effectiveness:

$$\eta_{cleaning} = \frac{P_{after} - P_{during}}{P_{during}} \times 100\% = \frac{350 - 125}{125} = 180\%$$

$$Restoration = \frac{P_{after}}{P_{before}} = \frac{350W}{450W} = 77.8\%$$

In summary, the cleaning system improved the power generation by 180% from when it was dirty to when it got cleaned, restoring it to 77.8% of its original power generation. This number is likely even higher due to variable cloud coverage during the day of testing.

Figure X-2: Test Results

X.D. Testing & Validation of Qualitative Constraint Q#1 – Compatibility with Existing Design

Margaret Burkes

X.D.1. Objective, Rationale & Brief Description - Q#1

The objective of this test is to conduct a maximum speed test to see how the rover's previous capabilities were affected by the additions. To conduct this, the team went to a parking lot and marked out an allotted length, measured the time ran using a stopwatch, and ran LOUIS at full speed. Using distance divided by time, the team was able to assess the new top speed of LOUIS with the additions.

X.D.2. Testing & Validation Summary of Results & Conclusions – Q#1

Table X-4: Test Result Data Table

Trial Number	Distance	Time	Speed	Average:
1	15 meters	12.14 seconds	1.24 m/s	1.22 m/s
2	15 meters	12.21 seconds	1.23 m/s	
3	15 meters	12.60 seconds	1.19 m/s	

In conclusion, as a result of our additions, LOUIS can now move up to 1.22 m/s, 92.4% of its original top speed.

Figure X-3: Test Results

X.E. Testing & Validation of Qualitative Constraint Q#2 – Weatherproof Design

Keldon Ngo

X.E.1. Objective, Rationale & Brief Description - Q#2

The rover is built for Louisiana's weather conditions, so it must be able to withstand the rain, heat, and humidity of the environment. Electronics that are not shielded by the panel will have to be coated in a waterproof coating. All additions made to the rover have also been made of aluminum to mitigate rusting and corrosion of key components.

X.E.2. Testing & Validation Summary of Results & Conclusions – Q#2

Table X-5: Test Result Data Table

Condition tested	Results
Temp: 85 F Humidity:70% Rain: none	Louis was still completely functional
Temp: 86 F Humidity:77%	Louis was still functioning and was able to withstand additional testing later in the day

Rain: light-moderate	

Figure X-4: The Rover getting rained on during weight testing

X.F. Testing & Validation of Qualitative Constraint Q#3 - Portability

William Mancuso

X.F.1. Objective, Rationale & Brief Description - Q#3

One of the qualitative constraints for this project was portability, before the additions were made, the rover could be carried by 2 people if needed. After these additions, even with portability being accounted for, it is expectedly more difficult to transport. To account for this, the design was made to be completely disassembled with relative ease. To prove this, we tested the assembly and disassembly time for the new solar power and cleaning system, and it's previous battery powered system.

X.F.2. Testing & Validation Summary of Results & Conclusions – Q#3

Table X-6: Test Result Data Table

	Time 1	Time 2
Disassembly	6 minutes 39 seconds	6 minutes 24 seconds

Assembly	10 minutes 8 seconds	10 minutes 27 seconds

As the table shows, the new additions can be disassembled returning the rover to its battery powered state in about 6.5 minutes, and can be reassembled, in about 10 minutes. The instructions for this can be found on the website.

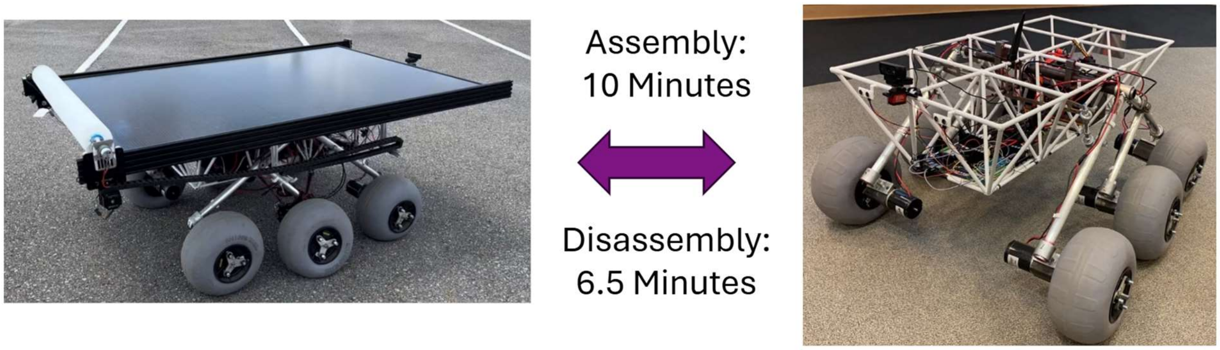


Figure X-5: Test Results

X.G. Testing & Validation of Quantitative Constraint M#1 - Budget

Bowen Williamson

X.G.1. Objective, Rationale & Brief Description - M#1

Because the project has been given a strict budget, the team had to ensure the desired results could be delivered within the budget given by our sponsor. The team was given the opportunity to increase the budget a few months ago, however it was deemed unnecessary, and the decision was made to move forward with the original budget. $4000 was allotted for the completion of the project. This includes any and all purchases for the project, such as components and any tools that we do not already have access to that we need for manufacturing the project. It was important to accurately keep track of the amount of money spent so as not to run out of funding before the project reached completion. A master list of all purchases made by the team was compiled and consistently updated with each purchase or return. This spreadsheet included the cost of items, quantity, delivery dates, and links to each individual item online.

X.G.2. Testing & Validation Summary of Results & Conclusions – M#1

At the completion of the project, the total spending was tallied up to be $3525, leaving almost $500 remaining. Thus, the project was finished comfortably under budget, satisfying this constraint. More information can be found in XI.B. Budget and XIII.H.2. Budget Details.

Table X-7: Test Result Data Table

Mounting Hardware	$333.76

Electronics/Wiring	$921.86
iPad	$1387.61
Mechanical Cleaning System Parts	$657.52
Testing	$79.11
Misc.	$145.52
Total	**$3525.38**

X.H. Testing & Validation of Quantitative Constraint M#2 – Interior Space

Kenny Bui

X.H.1. Objective, Rationale & Brief Description - M#2

The objective of this test is to determine the amount of usable space within the chassis of our rover after our additions. Our team wanted to leave an adequate amount of interior space for further additions in the future. To perform this test, we used the interior space measurement from the previous team and subtracted the volume of our new additions from the existing interior space.

X.H.2. Testing & Validation Summary of Results & Conclusions – M#2

After performing this test, our team concluded that there is still an adequate amount of interior space for future additions. The previous team had an interior space of 0.977 cubic meters and after our additions our new additions the new interior space measurement is 0.971 cubic meters. This means that our group only consumed about 0.006 cubic meters, leaving a reasonable amount for future projects.

Components	Volume (M^3)		
L298N	0.000049851		
TB6600 (2)	0.000354816		
Arduino Uno	0.000054909		
TriStar	0.005378		
RD-1	0.0004346		
Bus Bar (3)	0.0000648		
	0.006336976		
Original Volume	Occupied Space		
0.977	0.006336976	0.970663024	

Figure X-5: Test Results

X.I. Testing & Validation of Quantitative Constraint M#3 - Weight

William Mancuso

X.I.1. Objective, Rationale & Brief Description - M#3

The goal of this constraint was to not exceed the previous test 36.3 kg load that was done by last years team. Although this wasn't considered to be the max load over the rover, it was the max tested load, and our goal was to not exceed it to protect the lifespan of the rover.

X.I.2. Testing & Validation Summary of Results & Conclusions – M#3

Table X-8: Test Result Data Table

Previous weight of the rover	Current weight of the rover	Difference
77.25 kg	113.45	36.2 kg

The difference between the previous weight and the current weight show that the entire weight of the components added come out to around 36.2 kg just below the limit set of 36.2 kg.

Figure X-6: Test Results -Weigh Station

XI. Project Management

XI.A. Schedule and Milestones

Our project starts with our Project Preview milestone. During this period, our group did a lot of preparation and introduction for our project. This involved meeting with our sponsor and choosing a faculty advisor we thought would have the most expertise in our area. We also were able to do an interpretation of our project and assign each team member with their own role that would be beneficial to the entire team. We then move on to the Planning and Design portion of our project. During this time, we begin researching the main functions of our project. This includes solar panels, a solar panel cleaning system, and autonomous power switching system. From our research we were able to turn our ideas into concept generations. Using these concept generations, our group was able to make selections on the concepts that we deemed would be most suitable and beneficial to our rover design. To visualize these, we created 3D rendering in Fusion. Next, our group will move into the Analysis phase of our project. During this phase, our group will run simulations and experiments to ensure that our functions and subfunctions will in fact work when we assemble and run them. This is also the time to make sure that our functions will be compatible with the previous rover's system, and to make any necessary changes to our functions. This is identified as our critical failure point and critical path. Without proper analysis our ideas could fail if they prove not to be feasible. We must ensure that proper analyses are done to prove that our system does work. Once we have verified that our functions are feasible, we can now investigate ordering the components needed to manufacture our function systems. That leads us to the next semester, where we will begin assembling our function systems once the essential parts arrive during the Manufacturing and Testing stage. Once we have assembled each function system, our group will make sure to test that specific function to make sure that it works properly and how we planned it to. Once we have thoroughly built and tested each function, we will implement all our systems onto the pre-built rover. From then we will run another series of tests to make sure that our system is working properly with the original system. This testing allows us to fine tune our systems to ensure that they are working the best that they can be. Once fine tuning and one final round of testing is completed, our team will be ready to give our final presentation. After completing the analysis phase of our project, we have discovered that our battery is capable of supporting the additional load that we will add, our solar panel will generate the adequate power to aid in the longevity of the rover run time, the proper sizing of stepper motors and DC motors to execute a brush cleaning system, and constructed a code to activate the brush cleaning system. Using simulations and calculations, we confirmed that our functions are feasible. We will now move onto our manufacturing phase, where the proper assembly of our functions is a new critical path and critical failure point. If our functions are not assembled properly, our whole project scope and functions will fail causing a long delay or possibly failure. That is why it is critical that we stay on the right path for assembly and ensure that our functions are suitable for the testing phase. Up to this point our team has finished both the Manufacturing and Testing phases of our senior design process. Our team was able to finish assembling within a time efficient manner and begin testing on each of our individual functions. Finishing the Manufacturing phase was crucial to our project because we had to begin testing our individual functions to determine if they truly worked and satisfied our desires. Even though proper assembly was the previous critical path and failure, we believe that the testing phase has become the new critical path and failure point. Even though our functions are assembled, our team needed to make sure that they were thoroughly tested and worked. Without proper testing our team was at risk of not having any functional deliverables which would not satisfy the needs of our project, rendering it a failure. During this senior design process, our team learned that it is important to achieve target dates. Staying on course allowed our team to not only have adequate time for other objectives, but it allowed our team to have extra time for things that may have surprisingly come up and become a realization. We determined that it is important to finish objectives in a time efficient manner and provide extra time for unwarranted obstacles.

Project Preview			
Met Sponsor/Decide Faculty Advisor	100%	9/5/24	9/6/24
Project Interpretation and Team Roles	100%	9/7/24	9/10/24
Planning and Design			
Research Previous Rover Project	100%	9/11/24	9/12/24
Research Solar Panel Cleaning	100%	9/13/24	9/16/24
Research Methods to Autonomously Switch	100%	9/14/24	9/17/24
Research Solar Panels & Connections	100%	9/15/24	9/18/24
Create Concept Generations for Each Functio	100%	9/19/24	9/24/24
Decide/Select Ideal Concept Generations	100%	9/25/24	10/2/24
Create 3D Models in Fusion for Each Functio	100%	10/3/24	10/20/24

Analysis		
Simulations to See Power Capabilities of Solar Panel	10/21/24	10/28/24
Solar Panel with Rover System as Load Test	10/21/24	10/28/24
Run Analysis on Charge Controller for Power Switching	10/29/24	11/4/24
Analysis on DC Brush Motor for Cleaning Brush	10/29/24	11/4/24
MATLAB Stepper Motor Analysis	10/29/24	11/4/24
Construct Code and Analyze For Cleaning System Activatio	11/5/24	11/11/24
Experiment with Physical Model with Real Debris	11/5/24	11/11/24
Load Analysis for New Electrical Components	11/12/24	11/18/24
Evaluate Results and Adjust Concepts Accordingly	11/12/24	11/18/24
Verify Compatibility With Previous Rover System	11/19/24	11/25/24
Order Necessary Parts For Manufacturing	11/26/24	12/7/24

Manufacturing		
William Will Fix Rover Breaker	1/21/25	1/27/25
Kenny Will Be Coding Arduino	1/21/25	2/3/25
Keldon Will Design Gantry Plate	1/21/25	2/10/25
Margaret Will Make/Configure Website	1/21/25	2/24/25
Margaret In Charge Of Stepper Driver Configuration	1/28/25	2/3/25
William Will Fix and Adjust Camera	1/28/25	2/3/25
Keldon In Charge Of Gear Driver Configuration	1/28/25	2/3/25
Kenny Will Construct Linear Actuator	1/28/25	2/10/25
Kenny Will Incorporate Limit Switches	1/28/25	2/10/25
Margaret Is In Charge Stepper Motor Wiring	2/4/25	2/10/25
Keldon Is In Charge Of Gear Motor Wiring	2/4/25	2/10/25
William and Bowen In Charge Of Pyranometer Platform	2/4/25	2/10/25
William and Bowen In Charge Of Program Charge Controller	2/11/25	2/17/25
William and Bowen In Charge Of Pyranometer Wiring	2/11/25	2/17/25
William and Bowen In Charge Of Connect Charge Controller	2/18/25	2/24/25
William Will Assemble Solar Panel and Frame	2/18/25	3/2/25
Bowen Will Assemble Flange and Brush System	2/25/25	3/2/25
Testing		
Keldon Will Solo Test Brush Cleaning System	3/3/25	3/16/25
Kenny Will Ensure Motors Actuate Properly	3/3/25	3/16/25
Keldon Will Solo Test Solar Panel Power Generation	3/17/25	3/30/25
Keldon Will Test Production In Different Scenarios	3/17/25	3/30/25
William and Bowen Solo Test Autonomous Power Switching	3/31/25	4/13/25
William and Bowen Test Power Switching with Different SOC	3/31/25	4/13/25
The Group Will Test Rover With Total Integration	4/14/25	4/20/25
The Group Will Make Adjustments as Needed	4/14/25	4/20/25
The Group Will Conduct a Series of Final Tests	4/21/25	4/27/25

Figure XI-1: Gantt Chart with Project Time-Line

XI.B. Budget

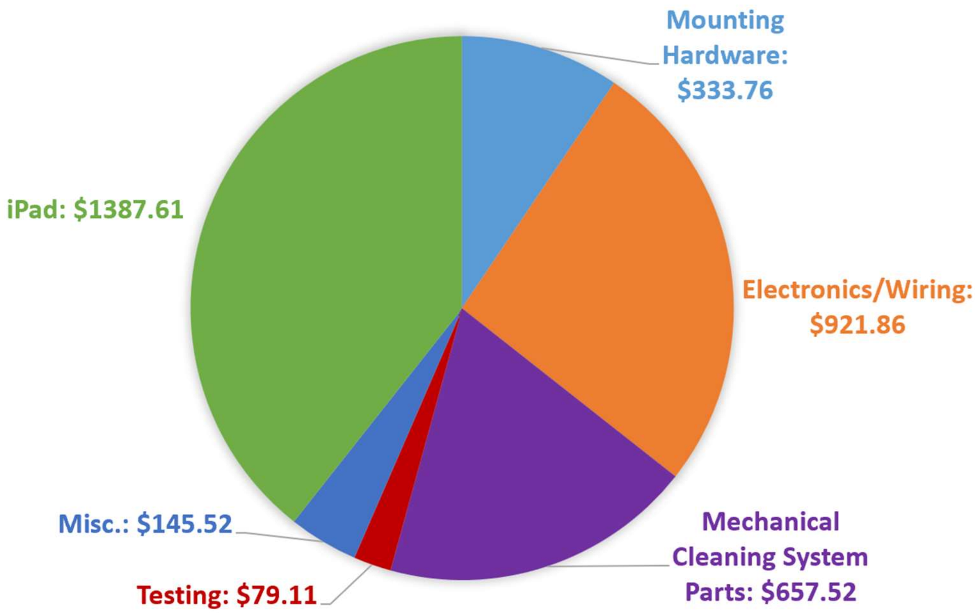


Figure XI-2: Budget Pie Chart

The breakdown of the budget is shown above, separated into major categories. More information can be found in XIII.H.2. Budget Details.

XII. Summary and Conclusions

Our project aimed to enhance LOUIS's battery life and stay safely under the budget ($4,000), which was accomplished. We integrated a solar panel as the primary power source, implemented autonomous power switching to optimize power input, and developed a cleaning system to maximize solar panel efficiency. The solar panel produces an average of 382.3W on sunny days, 390.3W on cloudy days, and 220.2W indoors. Autonomous power switching is overseen by the MorningStar TriStar-MPPT-30 charge controller, which utilizes maximum power point tracking to alternate between solar and battery power. The cleaning system consists of linear actuators with a stepper motor for linear motion and a gearmotor for rotational motion of the brush. As a result of the team's engineering, the battery life expectancy improved by 206.8% from the previous team's design, and the cleaning system improves the power generation by 180% from dirty to clean. We maintained 92.4% of the rover's original speed, and we added 36.2kg, 0.1kg below the maximum load. We left plenty of room in the chassis, only taking up 1% of its space. The additions were weatherproof, due to the design of the equipment or the hydrophobic spray, and we included instructions on how to disassemble LOUIS on our website to maintain portability.

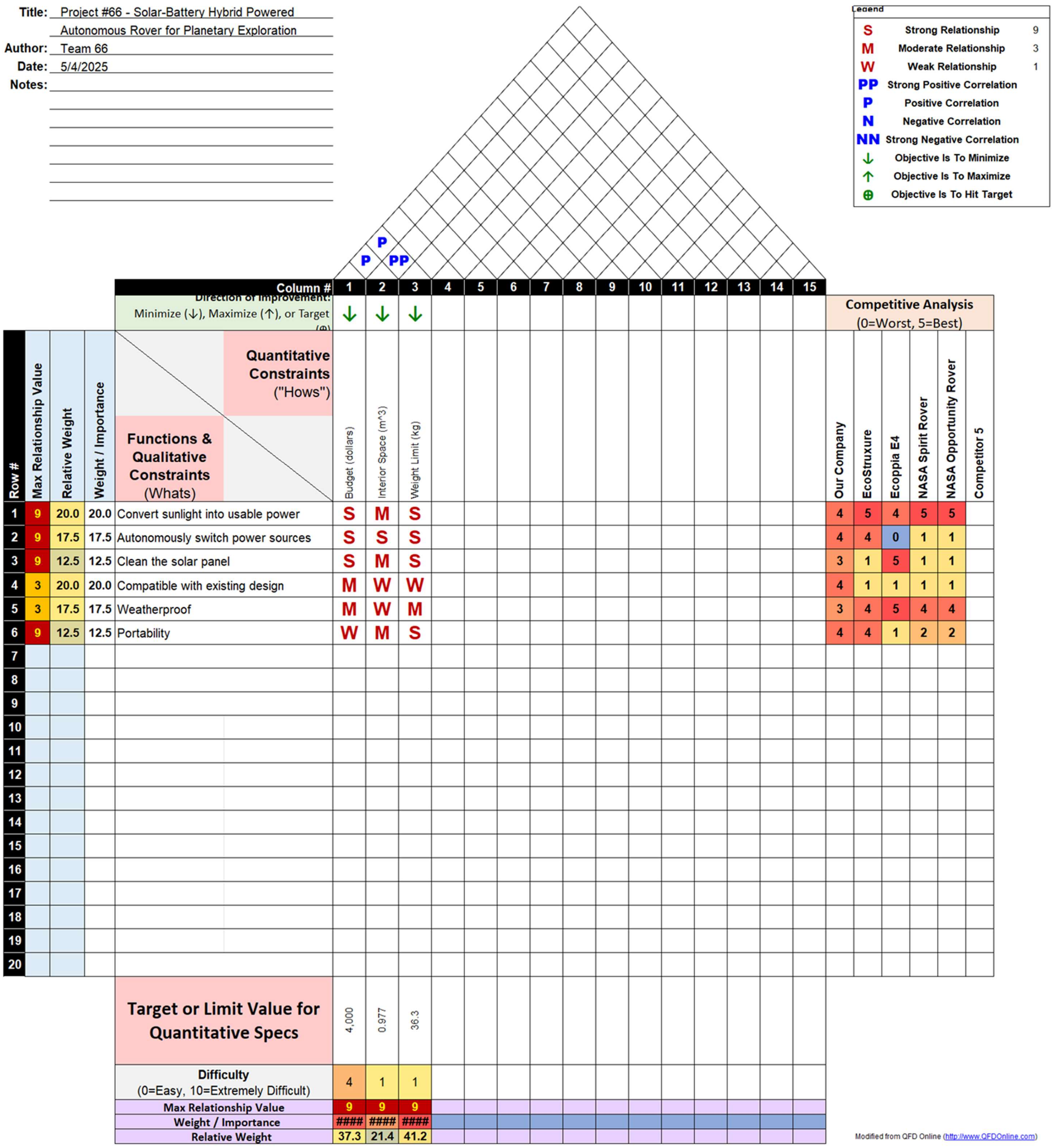
XIII. Appendix

XIII.A. Quality Function Deployment (QFD) - HoQ

The House of Quality pictured below demonstrates the relationship between the functions of the project and constraints that must be met to accomplish these functions. Weights were assigned in ranking of importance from the perspective of the sponsor's desired results. Values of 1, 3, or 9 were assigned to the relationships between functions and constraints to show these relationships as weak, moderate, or strong. The relative weights for the functions and qualitative constraints in the House of Quality were determined by taking the relative weights of each function and quantitative constraint from the tables in the Engineering Specification and dividing them by 2, since they are grouped together in the House of Quality but are two separate tables in the Engineering Specification.

Quantitative constraints are those that can be assigned specific values for physical limitations on the project. As this project was desired by the sponsor to be more of a proof of concept, the team was given very few quantitative constraints, giving us great flexibility in our design. The only sponsor-assigned quantitative constraint was that the project should remain within a budget of $4,000. Upon visual inspection and measurements of the rover, it was determined that any additional electronics or components had to fit within a $0.977m^3$ space to ensure that the electronics are contained within the chassis of the rover. The tested additional load of the rover was 36.3kg, so the team could not exceed this weight when adding the solar power and cleaning systems.

The more components required to accomplish our goals with this project, the higher our costs and the more weight capacity and space would be taken up. This results in positive correlations between cost and available space/weight. There is a strong positive correlation between space being used and added weight to the rover, as taking up more of the available space in the chassis with electronic components increases the weight of the rover as well.



Title: Project #66 - Solar-Battery Hybrid Powered Autonomous Rover for Planetary Exploration
Author: Team 66
Date: 5/4/2025
Notes:

Legend		
S	Strong Relationship	9
M	Moderate Relationship	3
W	Weak Relationship	1
PP	Strong Positive Correlation	
P	Positive Correlation	
N	Negative Correlation	
NN	Strong Negative Correlation	
↓	Objective Is To Minimize	
↑	Objective Is To Maximize	
⊕	Objective Is To Hit Target	

Roof correlations: P, P, PP

Column #					1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	Competitive Analysis (0=Worst, 5=Best)					
				Direction of Improvement: Minimize (↓), Maximize (↑), or Target (⊕)	↓	↓	↓																		
Row #	Max Relationship Value	Relative Weight	Weight / Importance	Quantitative Constraints ("Hows") / Functions & Qualitative Constraints (Whats)	Budget (dollars)	Interior Space (m^3)	Weight Limit (kg)													Our Company	EcoStruxure	Ecoppia E4	NASA Spirit Rover	NASA Opportunity Rover	Competitor 5
1	9	20.0	20.0	Convert sunlight into usable power	S	M	S													4	5	4	5	5	
2	9	17.5	17.5	Autonomously switch power sources	S	S	S													4	4	0	1	1	
3	9	12.5	12.5	Clean the solar panel	S	M	S													3	1	5	1	1	
4	3	20.0	20.0	Compatible with existing design	M	W	W													4	1	1	1	1	
5	3	17.5	17.5	Weatherproof	M	W	M													3	4	5	4	4	
6	9	12.5	12.5	Portability	W	M	S													4	4	1	2	2	
7																									
8																									
9																									
10																									
11																									
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14																									
15																									
16																									
17																									
18																									
19																									
20																									
				Target or Limit Value for Quantitative Specs	4,000	0.977	36.3																		
				Difficulty (0=Easy, 10=Extremely Difficult)	4	1	1																		
				Max Relationship Value	9	9	9																		
				Weight / Importance	####	####	####																		
				Relative Weight	37.3	21.4	41.2																		

Modified from QFD Online (http://www.QFDOnline.com)

Figure XIII-1: The House of Quality for the project.

XIII.B. Concept/Solution Evaluation and Selection Supplement

XIII.B.1. Concept/Solution for Function F#1 - Details

XIII.B.1.a. Original Concept Selected – Details

The monocrystalline solar panel was selected after completing a pros and cons chart along with a decision matrix. The monocrystalline solar panel was the most efficient and is much higher than both of the other options. This allows for a smaller panel, saving space. Although it is the highest cost, this is weighed by its efficiency. The weight was similar to a polycrystalline panel.

XIII.B.1.b. Re-design Concept Selected – Details

No re-design was needed for this concept.

XIII.B.1.c. Original and Re-designed Concepts Considered but not Selected.

		Concepts		
Criterion	Weights	Thin Film	Polycrystalline	Monocrystalline
Efficiency	40	-1	0	1
Cost	25	1	0	-1
Weight	20	1	-1	-1
Life-span	15	-1	1	1
Weighted Total	100	-.1	-.05	0.1

Pros and Cons of solar panel types.

	Monocrystalline	**Polycrystalline**	**Thin-Film**
Pros	• High Efficiency: 20-23% • Space-Saving • Longer Lifespan	• Good Efficiency • Cost-Effective • Decent Space Utilization	• Lightweight and Flexible • Performance in High Temperatures • Lower Cost
Cons	• Expensive • Production	• Space • Shorter Life-span • Less efficient	• Low Efficiency: 13-15% • Consume lots of space • Not Durable

Above is both our decision matrix and pros/cons chart for function 1, the evaluation and selection of solar panel types. We were deciding between thin-film, polycrystalline and monocrystalline panels. We weighed efficiency the highest at 40 percent because having a higher efficiency solar panel will also cut down on the size of solar panel needed to provide appropriate power to the rover, and we don't want the rover to lose any capability due to size of panel. Cost is weighed 25 percent as this is going to be the most important component because it is the basis of our project. We placed weight at 20 percent because the rover does have a maximum load weight of about

36 kilograms. The panel will be the heaviest component added. The lifespan of the solar panel is rated at 15 percent, as this will allow future teams to work with the panel. This led us to the decision of a monocrystalline solar panel because it was the only panel that had high enough efficiency to power the rover and charge the battery.

XIII.B.2. Concept/Solution for Function F#2 - Details

XIII.B.2.a. Original Concept Selected – Details

The charge controller was selected after completing the decision matrix. The charge controller method was much more accurate, efficient, and user-friendly, than the voltage regulator or DC/DC converter. The price was a concern for the charge controller because it was the most expensive option, however with our budget of $4,000 the group felt confident it was the right purchase. This was further validated when Morningstar donated the charge controller to us. The charge controller's ease of use and ability to hook right up to the system made it the perfect choice for our application.

XIII.B.2.b. Re-design Concept Selected – Details

No redesign was necessary for this concept.

XIII.B.2.c. Original and Re-designed Concepts Considered but not Selected.

		Concepts		
Criterion	Weights	Voltage Regulator with Arduino	DC/DC Converter with Arduino	Solar Charge Controller
Functional Accuracy	45	-1	0	1
Simplicity	25	0	-1	1
Programmability	20	-1	0	1
Cost	10	1	0	-1
Weighted Total	100	-0.55	-0.25	0.8

Above is the decision matrix for function 2. For the decision matrix, the most important criteria we decided on was functional accuracy. This was weighed 40% because the main goal of this concept is that it can accurately swap between solar and battery power, because of this it was weighed 40%. Simplicity was weighed at 25% because it was important that the switching be easy to operate and understand. Following that was programmability, which as stated before is sort of a subcategory of simplicity. We weighed this 20% because we wanted to make sure that the power switching was able to be set to parameters of our choosing, while also being easy to do. Lastly, the cost was decided to be weighed at 10% because it was decided that of course the $4000 budget cannot be exceeded, and there is enough in the budget to ensure that the best quality item is chosen. For the calculations, 0 meant the "standard" compared to the other methods, -1 meant "poor", and 1 meant "good" and the weight of the criteria was multiplied across and then each concept was added up accordingly.

XIII.B.3. Concept/Solution for Function F#3 - Details

XIII.B.3.a. Original Concept Selected – Details

The brush method was selected after completing a pros and cons chart and a decision matrix. The brush method was far cheaper than the electrostatic method, and cleaned relatively effectively, despite concerns over potential scratching. The solar panel we selected should be able to continually withstand that level of friction. The brush was going to be a lighter system, and given our weight constraint, this was a necessity. It is a potentially abrasive system, as stated earlier, but the risk is low due to the use of a soft bristle brushes and a monocrystalline panel. Maintenance wise, the brush will likely have to be replaced every year or so, depending on how often the cleaning system is used.

XIII.B.3.b. Re-design Concept Selected – Details

No re-design was necessary for our project.

XIII.B.3.c. Original and Re-designed Concepts Considered but not Selected.

Pros and Cons for F3:

	Air Compressor	**Electrostatic Method**	**Brush**
Pros	- Less likely to scratch	- Requires humidity - No scratch - 90% effective	- Easily obtainable parts - Within budget - Low power consumption
Cons	- Weight distribution - Large power consumption - Potentially less effective due to spacing between holes	- Too expensive - May not clean heavier debris	- May scratch - May trap debris

Decision Matrix for F3:

		Concepts		
Criterion	Weights	Air Compressor	Electrostatic Method	Brush Method
Cost	30	0	-1	1
Cleans Effectively	25	-1	1	0
Mass	20	-1	0	1
Non-abrasive	20	0	1	-1
Maintenance	5	-1	1	0
Weighted Total	100	-0.5	0.2	0.3

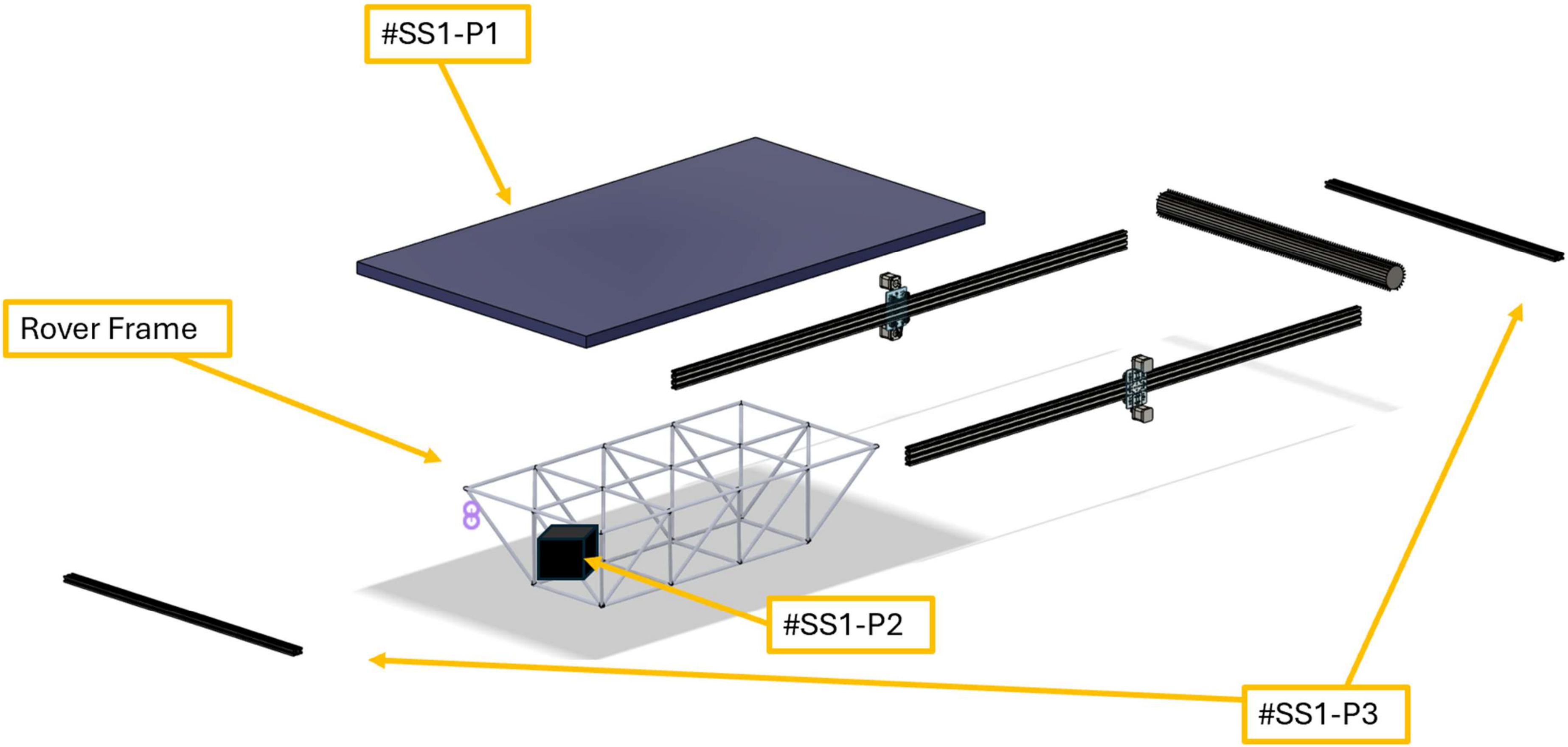
For the pros and cons chart, the air compressor would not scratch the panel because there would be no physical contact. However, it would have a high density and might disrupt the weight balance. Additionally, it would consume larger amounts of power to complete the same task. Lastly, the effectiveness of cleaning is in question due to the spacing between the holes of the perforated pipe. With regards to the electrostatic method, it works most effectively in humid environments due to its reliance upon electrical conductivity in the air. It wouldn't scratch since there would be no physical contact, and MIT researchers have reported that it is 90% effective. However, the indium tin oxide (ITO) alone was around $4000, which would be too much for our budget. We might would have needed to also supply a higher voltage to pick up the heavier debris, which might lead to more power loss than necessary. For the brush method, the parts are much more obtainable than some of the more niche components and lie within budget. This method also did not require lots of power, but it might trap debris within its bristles and scratch the panel.

For the decision matrix, we weighed three different options: an air compressor system, an electrostatic method, and a brush method. In the air compressor system, air would blow through a perforated pipe that would move across the panel. This idea had been generated by the team since we wanted to lower the risk of scratching the solar panel, which would reduce the efficiency of the panel. For the electrostatic method, we discovered this idea through an MIT research article. Through the utilization of electrostatic energy to pick up debris, ITO or indium tin oxide glass would sit on top of the solar panel, and an oppositely charged electrode would pass over it. This creates a field that then would pick up the debris. For the brush method, we saw this idea online as the commercialized option. A rotating brush moves across a solar panel through a dry-cleaning system. For the decision matrix, the main criteria we selected was the cost to stay under the $4000 budget. This was weighed 30% due to the quantitative constraint. An effective cleaning system was weighed 25% to ensure maximum power absorption. A lower mass was 20% to stay under the 36.3kg quantitative constraint. Non-abrasiveness was 20% to again maximize that power absorption. This is a little lower than the effective cleaning since the solar panel we selected is relatively durable. Lastly, ease of maintenance to have minimal repair issues. This was 5% since the rover will be for educational purposes, rather than for space. For the calculations wise, 0 meant the “standard” compared to the other methods, -1 meant “poor”, and 1 meant “good”. To summarize, the air compressor was going to cause uneven weight distribution and might leave dirty spots due to the spacing between the holes. The electrostatic method was way over budget due to cost of materials, so this left us with the brush method, which can be seen below.

XIII.C. System Description/Product Architecture Supplement

XIII.C.1. Sub-System SS#1 – Solar Panel and Charge Controller

The most critical component for this project, the 450W solar panel will absorb solar energy and act as the main power source for the rover when sunlight is available. The chosen solar panel measures 47 inches wide, 68 inches long, and 1.2 inches thick, and weighs 50 pounds. The panel is composed of 88 heterojunction cells with gapless technology. An MPPT solar charge controller will be used to regulate the power output of the solar panel while drawing the appropriate current to extract maximum power from the panel. The charge controller will be sending the power from the panel to the existing electrical system of the rover, including the battery. This charge controller will be programmed maximize battery life and efficiently work in conjunction with the 36V battery currently onboard the rover, charging the battery when solar power produced exceeds that which the rover needs to function at that time. The rover itself will play an important role in the design of the project. This is because every component added must be mounted or fastened securely to the rover. The frame, which consists of a series of skeletal struts made of small cylindrical steel pipe, will provide mounting locations for all of those added components. The rails supporting the solar panel will be mounted to the frame by two-hole electrical straps. The straps will wrap around the circumference of the cylindrical frame struts. The rails have slots that will accommodate T-track nuts, which will be used to screw the ends of the electrical straps to the rails.


#SS1-P1
Rover Frame
#SS1-P2
#SS1-P3

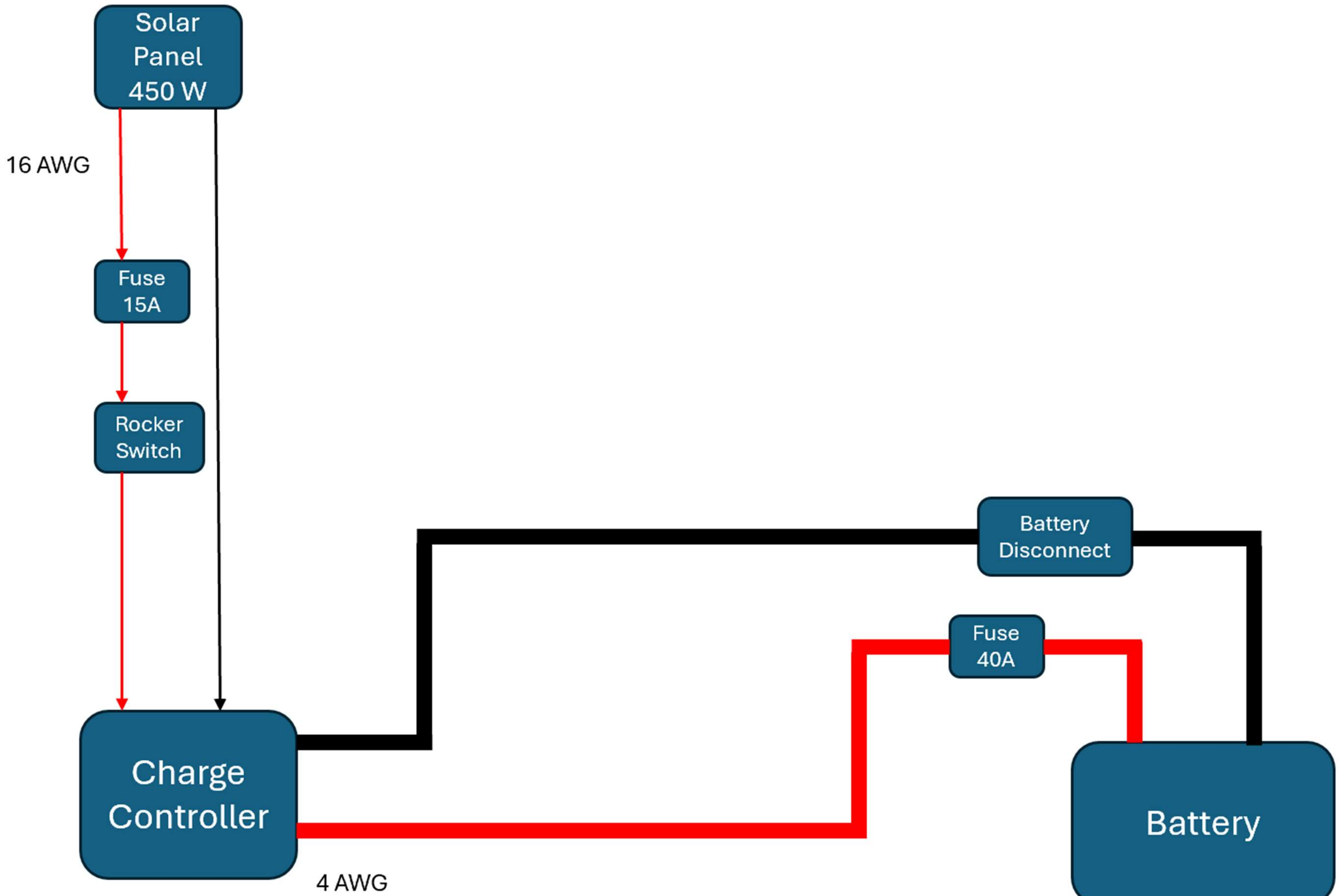

Solar
Panel
450 W
16 AWG
Fuse
15A
Rocker
Switch
Battery
Disconnect
Fuse
40A
Charge
Controller
Battery
4 AWG

Figure XIII-2: Exploded View Assembly Drawing of Sub-System SS#1 - Solar Panel and Charge Controller

XIII.C.1.a. Comprehensive Parts List for SS#1 – Solar Panel and Charge Controller

Table XIII-1: List of Parts for Sub-System SS#1

Part #	Quantity	Material	Name
SS#1-P1	1	N/A	Solar Panel
SS#1-P2	1	N/A	Solar Charge Controller
SS#1-P3	2	Aluminum	T Track Rails

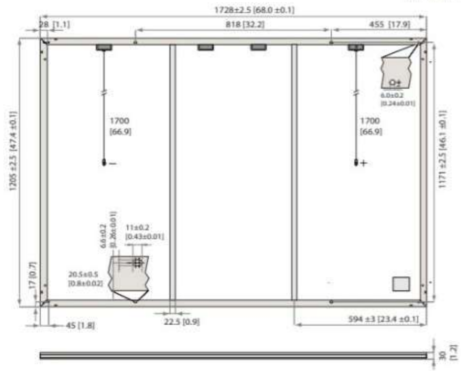
REC ALPHA® PURE-RX SERIES
DATASHEET

REC
SOLAR'S MOST TRUSTED

GENERAL DATA

Cell Type	88 half-cut bifacial REC heterojunction cells, with gapless technology
Glass	0.13 in solar glass with anti-reflective surface treatment in accordance with EN12150
Backsheet	Highly resistant polymer (Black)
Frame	Anodized aluminum (Black)
Junction Box	4-part, 4 bypass diodes, IP68 rated, in accordance with IEC 62790:2020
Connectors	Stäubli MC4 PV-KBT4/KST4 (12 AWG) in accordance with IEC 62852:2014, IP68 only when connected
Cable	12 AWG solar cable, 66.9 in (1.70 m) + 66.9 in (1.70 m) in accordance with EN50618:2014
Dimensions	68.0 x 47.4 x 1.2 in (22.4 ft²) / 1728 x 1205 x 30 mm (2.08 m²)
Weight	50.0 lb / 22.7 kg
Origin	Made in Singapore

Measurements in [inches] and mm



Specifications subject to change without notice.

ELECTRICAL DATA

PRODUCT CODE*: RECXXXAA PURE-RX

STC	Power Output - P_{MAX} (WP)	450	460	470
	Watt Class Sorting - (W)	0/+10	0/+10	0/+10
	Nominal Power Voltage - V_{MPP} (V)	54.3	54.9	55.4
	Nominal Power Current - I_{MPP} (A)	8.29	8.38	8.49
	Open Circuit Voltage - V_{OC} (V)	65.6	65.8	65.9
	Short Circuit Current - I_{SC} (A)	8.81	8.88	8.95
	Power Density (W/ft²)	20.1	20.5	21.0
	Panel Efficiency (%)	21.6	22.1	22.6
NMOT	Power Output - P_{MAX} (W_P)	343	350	358
	Nominal Power Voltage - V_{MPP} (V)	51.2	51.7	52.2
	Nominal Power Current - I_{MPP} (A)	6.70	6.77	6.86
	Open Circuit Voltage - V_{OC} (V)	61.8	62.0	62.1

Values at standard test conditions (STC: air mass AM 1.5, irradiance 1000 W/m², temperature 25°C), based on a production spread with a tolerance of P_{MAX}, V_{OC} & I_{SC} ±3% within one watt class. Nominal module operating temperature (NMOT: air mass AM 1.5, irradiance 800 W/m², temperature 20°C, windspeed 1 m/s). *Where xxx indicates the nominal power class (P_{MAX}) at STC above.

CERTIFICATIONS

ISO 14001; ISO9001; IEC45001; IEC62941	
IEC 61215:2021;IEC 61730:2023;UL 61730	
ISO 11925-2	Ignitability (EN 13501-1 Class E)
IEC 62716	Ammonia Resistance
IEC 61701	Salt Mist (SM6)
IEC 61215:2016	Hailstone (35mm)
UL 61730	Fire Type 2

Intertek

MODULE RATINGS

Module Operating Temperature [T98]§	158°F (70°C)
Min. Environmental Temperature	-40°F (-40°C)
System Voltage	1000 V
Maximum Test Load (4 Point Mounting, Front)*	+7000 Pa (1.02 lbs/in²)
Maximum Test Load (4 Point Mounting, Rear)*	-4000 Pa (0.58 lbs/in²)
Maximum Test Load (6 Point Mounting, Front)**	+8000 Pa (1.16 lbs/in²)
Maximum Test Load (6 Point Mounting, Rear)**	-6000 Pa (0.87 lbs/in²)
Max Series Fuse Rating	25 A
Max Reverse Current	25 A

Design load = Test load / 1.5 (safety factor)
§ 98th percentile operating temperature
*IEC61730/UL61730 certified. Refer to installation manual.
**Internal testing. Refer to installation manual.

TEMPERATURE RATINGS*

Nominal Module Operating Temperature	44 ± 2°C
Temperature coefficient of P_{MAX}	-0.24%/K
Temperature coefficient of V_{OC}	-0.24%/K
Temperature coefficient of I_{SC}	0.04%/K

*The temperature coefficients stated are linear values

DELIVERY INFORMATION

Panels per Pallet	33
Panels per 40 ft GP/high cube container	594 (18 Pallets)
Panels per 53 ft truck	792 (24 Pallets)

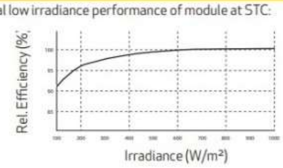
WARRANTY

	Standard	REC ProTrust	
Installed by an REC Certified Professional	No	Yes	Yes
System Size	All	<25 kW	25-500 kW
Product Warranty (yrs)	20	25	25
Power Warranty (yrs)	25	25	25
Labor Warranty (yrs)	0	25	10
Power in Year 1	98%	98%	98%
Annual Degradation	0.25%	0.25%	0.25%
Power in Year 25	92%	92%	92%

REC ProTrust Warranty applies only for i) REC panels installed by an REC Certified Solar Professional, and ii) panels have been registered by the installer with REC. Subject to System Size and further conditions. See www.recgroup.com for details.

LOW LIGHT BEHAVIOR

Typical low irradiance performance of module at STC:



Available from:

Founded in 1996, REC Group is an international pioneering solar energy company dedicated to empowering consumers with clean, affordable solar power. As Solar's Most Trusted, REC is committed to high quality, innovation, and a low carbon footprint in the solar materials and solar panels it manufactures. Headquartered in Norway with operational headquarters in Singapore, REC also has regional hubs in North America, Europe, and Asia-Pacific.

REC Solar PTE. LTD.
20 Tuas South Ave. 14
Singapore 637312
post@recgroup.com
www.recgroup.com

REC

REF: PM-DS-12-06-REV-4.3/BULEN12.2024

Figure XIII-3: Specifications for part SS#1-P#1 – Solar Panel

MORNINGSTAR
PROFESSIONAL SERIES
World's Leading Solar Controllers & Inverters

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sales@morningstarcorp.com

30, 45 or 60 amps at up to 150 volts open circuit.

Shown with optional meter

TriStar MPPT™

SOLAR CONTROLLER WITH MAXIMUM POWER POINT TRACKING

- Maximizes Energy Harvest
- Extremely High Reliability
- Very High Efficiency
- Extensive Networking

Morningstar's TriStar MPPT solar controller with TrakStar Technology™ is an advanced maximum power point tracking (MPPT) battery charger for off-grid photovoltaic (PV) systems with PV array max power (Pmp) up to 4.2 kW. The controller provides the industry's highest peak efficiency of 99% and significantly less power loss compared to other MPPT controllers. Detailed battery programming options allow for advanced battery support for the latest Lithium, Nickel Cadmium, and Lead Acid battery types.

The TriStar MPPT features a smart tracking algorithm that maximizes the energy harvest from the PV by rapidly finding the solar array peak power point with extremely fast sweeping of the entire I-V curve. This product is the first PV controller to include on-board Ethernet for a fully web-enabled interface and includes up to 200 days of data logging.

KEY FEATURES AND BENEFITS

Maximizes Energy Harvest

Our TrakStar MPPT Technology features:

- Better peak power point tracking than other MPPT controllers
- Very fast sweeping of the entire I-V curve
- Recognition of multiple power points during shading or mixed PV arrays
- Excellent performance at sunrise and low solar insolation levels

Extremely High Reliability

- Robust thermal design and no cooling fans
- Parallel circuit design provides less stress and longer life for electronic components
- No mechanical relays
- Extensive electronic protections including PV short circuit protection
- Epoxy encapsulated inductors and conformally coated printed circuit boards

Very High Efficiency

- Peak efficiency of 99%
- Proprietary tracking algorithm minimizes power losses
- Low self-consumption
- Continuous operation at full power to 45°C without need to de-rate
- Selected electronic devices with higher ratings to minimize losses from heating

Extensive Networking and Communications Capabilities

Enables system monitoring, data logging and adjustability. Uses open standard MODBUS™ protocol and Morningstar's MS View software.

- Meterbus: communications between compatible Morningstar products
- Serial RS-232: connection to a personal computer
- EIA-485: communications between multiple devices on a bus
- Ethernet: fully web-enabled interface to a local network or internet; view from a web browser or send email/text messages
- EMC-1: IP based network and internet connectivity (including SNMP)

SNMP

Metering and Data Logging

- TriStar meter and remote meter provides detailed operating data, alarms and faults
- Three LEDs display system status
- Up to 200 days of data logging via meters or communications ports

System Status:

53.60V	28C	54.2A
2867W		MPPT

Data Logging:

Today 46.4 Vmin Batt	Day:-1 47.2 Vmin Batt
Today 58.9 Amax Solar	Day:-1 56.8 Amax Solar
Today 107.2 Vmax Solar	Day:-1 105.5 Vmax Solar

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TRAKSTAR

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MORNINGSTAR
PROFESSIONAL SERIES
World's Leading Solar Controllers & Inverters

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sales@morningstarcorp.com

Technical Specifications

Versions		TS-MPPT-30	TS-MPPT-45	TS-MPPT-60	TS-MPPT-60M
Meter					
TS-M2		Optional	Optional	Optional	Included
TS-RM2		Optional	Optional	Optional	Optional
Electrical					
Maximum Battery Current		30 amps	45 amps	60 amps	
Nominal Maximum Output Power*				Max Output	Max PV Input*
	12 Volt	400 Watts	600 Watts	800 Watts	1100 Watts
	24 Volt	800 Watts	1200 Watts	1600 Watts	2100 Watts
	48 Volt	1600 Watts	2400 Watts	3200 Watts	4200 Watts
Max Recommended Solar PV Input*		~ 130% of Nominal Max Ouput Power (60 Amp models shown above)			
Peak Efficiency		99%			
Nominal System Voltage		12, 24, or 48 volts DC			
Maximum PV Open Circuit Voltage**		150 volts DC (without damage to unit)			
Battery Operating Voltage Range		8-72 volts DC			
Voltage Accuracy		12 / 24 V: ≤ 0.1% ± 50 mV 48 V: ≤ 0.1% ± 100 mV			
Maximum Self-consumption		2.7 Watts			
Transient Surge Protection		4500 Watts/port			
Battery Charging					
Charging Algorithm		4-stage			
Charging Stages		Bulk, Absorption, Float, Equalize			
Temperature Compensation:	Coefficent Settings Range	–5mV/°C/cell (25° ref); –30°C to +80°C Absorption, Float, Equalize, HVD			
Remote Temperature Sensor (RTS)		Absorption, Float, Equalize, HVD			

Certifications:

- CE and RoHS Compliant
- ETL Listed (UL1741)
- cETL (CSA C22.2 No. 107.1-01)
- FCC Class B Part 15 Compliant
- Manufactured in a certified ISO 9001 facility
- IEC 62109-1 (UL/CSA/IEC requires ambient temperature limited to 45ºC)

Options:

- TriStar Meter-2 (TS-M-2)
- TriStar Remote Meter-2 (TS-RM-2)
- Meter Hub (HUB-1)
- Relay Driver (RD-1)
- EMC-1

Notes:

*The PV array power rating may exceed the controller's Max Nominal Output Power specification. The controller will limit battery current and prevent damage. Array oversizing should be considered on a case by case basis. See our array string sizer tool and related tech documentation. https://www.morningstarcorp.com/array-oversizing

**PV Voltage must be greater than Vbattery + 1 Volt to start charging

*** Assumes 75Vmp, unvented enclosure. See operating manual for further performance characteristic data.

Warranty:

Five year warranty period. Contact Morningstar or your authorized distributor for complete terms.

Communication Ports	TS-MPPT-30	TS-MPPT-45	TS-MPPT-60	TS-MPPT-60M
MeterBus	Yes	Yes	Yes	Yes
RS-232	Yes	Yes	Yes	Yes
EIA-485	No	No	Yes	Yes
Ethernet	No	No	Yes	Yes
EMC-1	Yes	Yes	Yes	Yes

Electronic Protections	
Solar	Overload, Short Circuit, High Voltage
Battery	High Voltage
High Temperature	
Lightning & Transient Surges	
Reverse Current at Night	

Environmental	
Ambient Operating Temperature Range	-40ºC to 60ºC
May derate above the following temperature***	TS-MPPT-60 = 45ºC TS-MPPT-45 = 50ºC TS-MPPT-30 = 55ºC
Storage Temperature	-55ºC to +85ºC
Humidity	100% non-condensing
Tropicalization	Epoxy encapsulation, Conformal coating, Marine rated terminals

Mechanical	
Dimensions	29.1 x 13.0 x 14.2 cm 11.4 x 5.1 x 5.6 in
Weight	4.2 kg / 9.2 lbs
Maximum Wire Size	35 mm² / 2 AWG
Conduit Knockouts	M20; ½, 1, 1 ¼ in
Enclosure	Type 1 (indoor and vented) IP 20

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Due to Morningstar's policy of continuous improvement, product availability, features and specifications are subject to change without notice.

Control # MS-001901 Revision: 2/2025

Figure XIII-4: Manufacturing Drawing of part SS#1-P#2 – Charge Controller

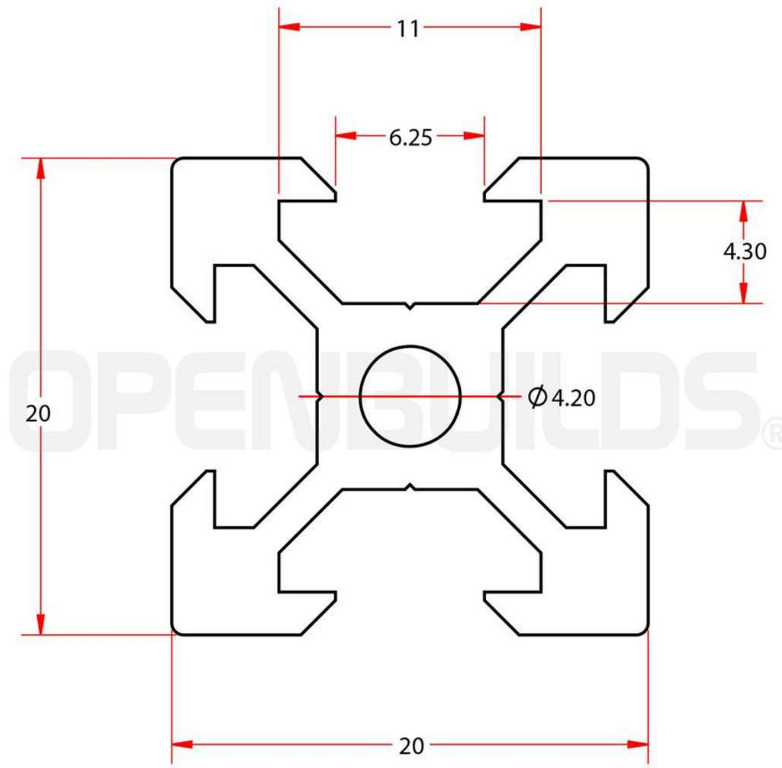


Figure XIII-5: Manufacturing Drawing of part SS#1-P#3– T Track Rails

XIII.C.2. Sub-System SS#2 – The Cleaning System

The brush will move across the length of the panel via linear actuators, while simultaneously rotating to sweep debris off the panel. This rotation will be accomplished using a DC brushed motor mounted on the gantry cart of one of the linear actuators at end of the cleaning brush. The other end of the brush will be supported by a bearing in a bearing housing mounted to the other gantry cart, so this end of the brush will be free spinning. The cleaning brush will ride on linear actuators to move back and forth along the length of the solar panel. These linear actuators will be mounted on the sides of the solar panel. The linear movement will be accomplished by stepper motors on a belt-and-pinion system. The stepper motors are mounted on gantry carts that roll along linear rails. The pinion on the stepper motor shaft moves along a stationary synchronous belt as the motor shaft spins. This belt has teeth that engage with the teeth of the pinion, allowing the motor to drive the gantry cart along the belt. An Arduino microcontroller will be used to control the movement of the stepper motors. In the spring we will determine how to program the cleaning system to autonomously activate when power production of the solar panel drops below a certain threshold.

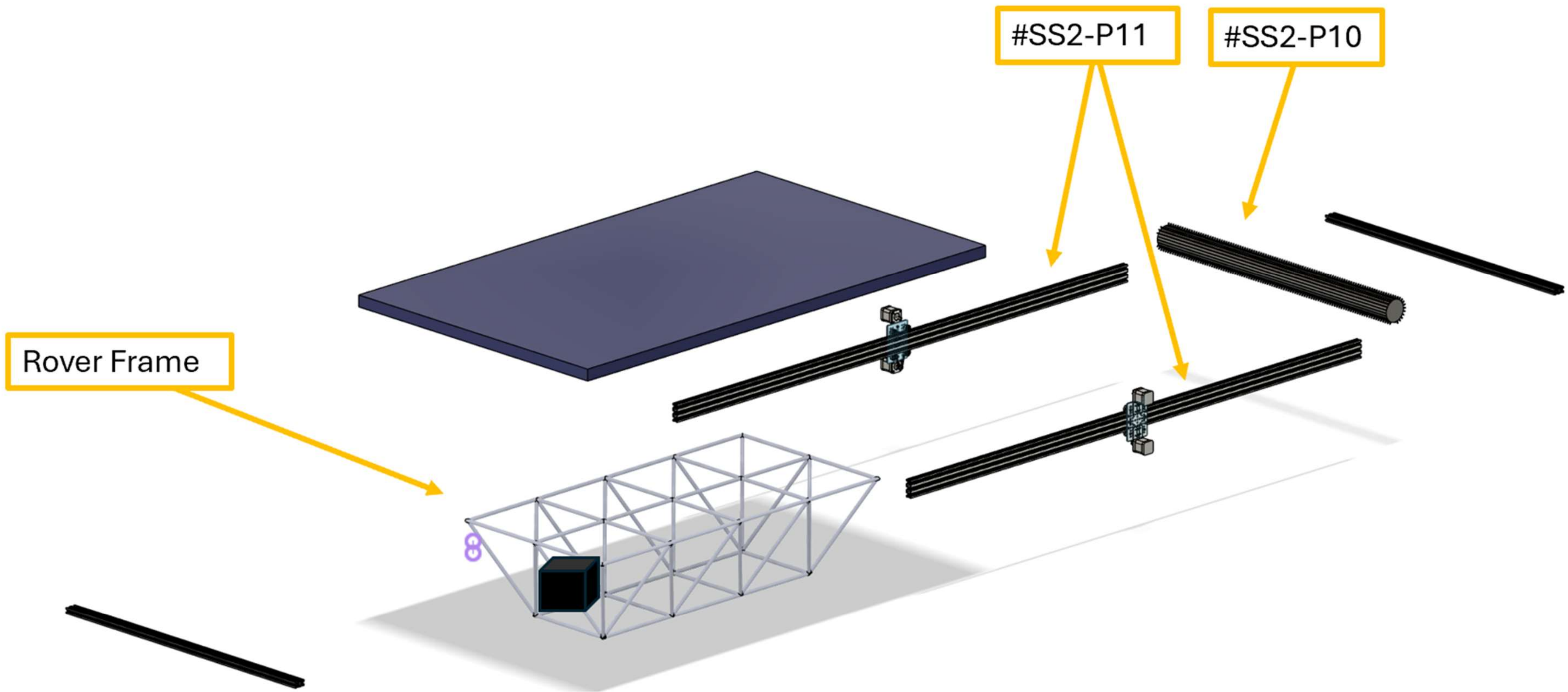


Figure XIII-5: Exploded View Assembly Drawing of Sub-System SS#2 – Cleaning System

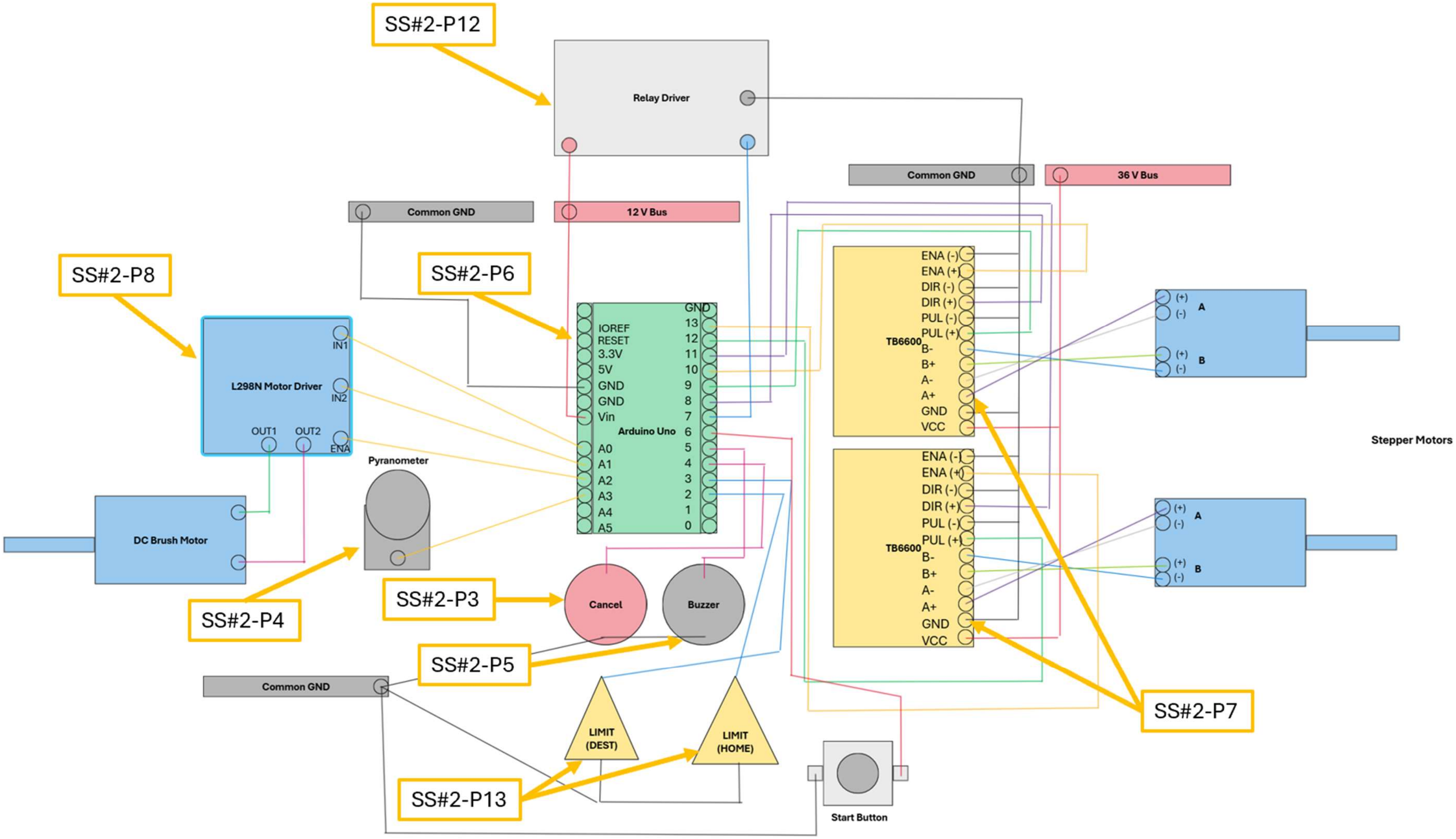

SS#2-P12
Relay Driver
Common GND
36 V Bus
Common GND
12 V Bus
SS#2-P8
SS#2-P6
L298N Motor Driver
IN1
IN2
OUT1
OUT2
ENA
IOREF
RESET
3.3V
5V
GND
GND
Vin
Arduino Uno
A0
A1
A2
A3
A4
A5
GND
13
12
11
10
9
8
7
6
5
4
3
2
1
0
TB6600
ENA (-)
ENA (+)
DIR (-)
DIR (+)
PUL (-)
PUL (+)
B-
B+
A-
A+
GND
VCC
TB6600
ENA (-)
ENA (+)
DIR (-)
DIR (+)
PUL (-)
PUL (+)
B-
B+
A-
A+
GND
VCC
(+)
(-)
A
(+)
(-)
B
Stepper Motors
(+)
(-)
A
(+)
(-)
B
DC Brush Motor
Pyranometer
SS#2-P4
SS#2-P3
Cancel
Buzzer
SS#2-P5
Common GND
LIMIT (DEST)
LIMIT (HOME)
Start Button
SS#2-P13
SS#2-P7

Figure XIII-6: Exploded View Assembly Drawing of Sub-System SS#2 – Cleaning System

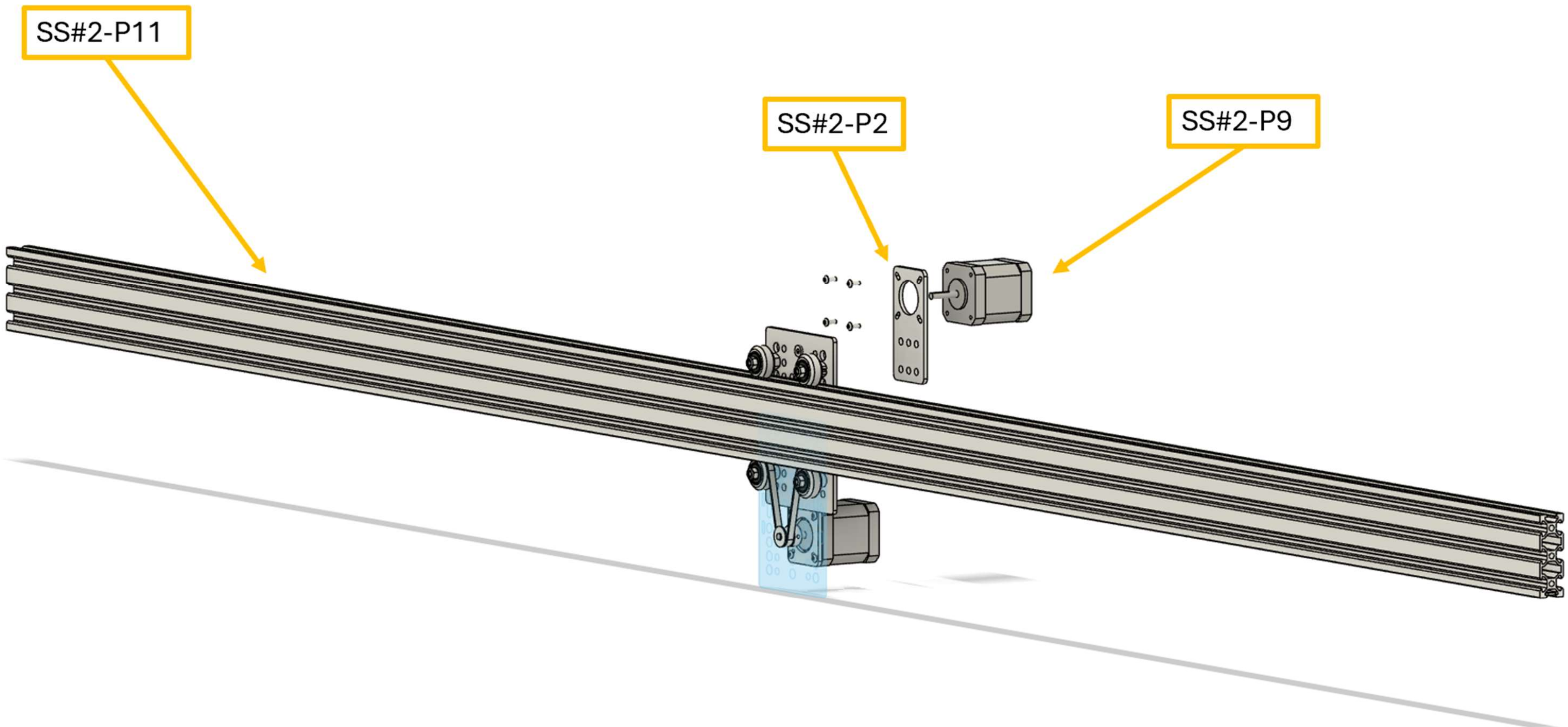


Figure XIII-7: Exploded View Assembly Drawing of Sub-System SS#2 – Cleaning System

XIII.C.2.a. Comprehensive Parts List for SS#2 – Cleaning System

Table XIII-2: List of Parts for Sub-System SS#2

Part #	Quantity	Material	Name
SS#2-P1	2	Aluminum	Stepper Motor
SS#2-P2	2	Aluminum	Motor Gantry Plate
SS#2-P3	1	Aluminum	Cancel Button
SS#2-P4	1	Plastic	Pyranometer
SS#2-P5	1	Plastic	Buzzer
SS#2-P6	1	Metal/Plastic	Arduino Uno
SS#2-P7	2	Metal/Plastic	TB6600 Stepper Motor Driver
SS#2-P8	1	Metal/Plastic	L298N Motor Driver
SS#2-P9	1	Metal	DC Geared Motor
SS#2-P10	1	Plastic	Solar Panel Safe Radial Brush
SS#2-P11	2	Aluminum	Linear Actuator Assembly
SS#2-P12	1	Plastic	Relay Driver
SS#2-P13	2	Plastic	Limit Switch
SS#2-P14	2	Aluminum	Flange Couplings

~~Off-The-Shelf Parts and Component Specifications for SS#2 – Cleaning System~~

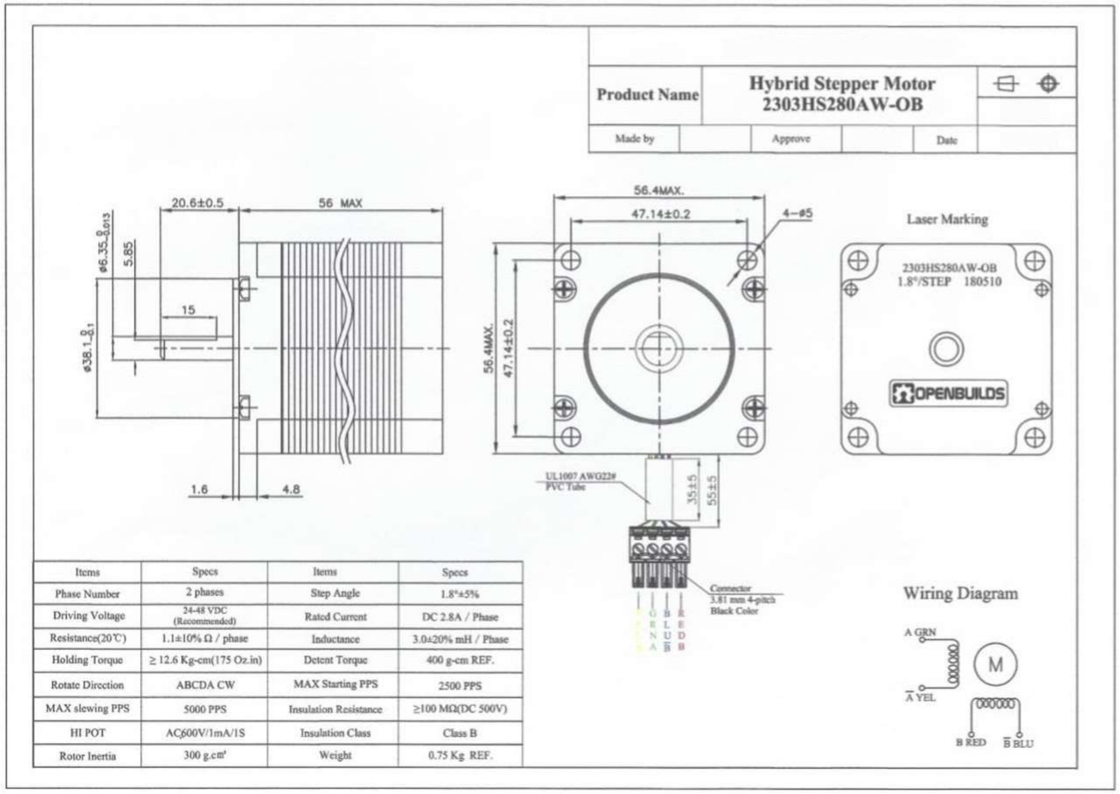


Figure XIII-8: Manufacturing Drawing of part SS#2-P#1 – NEMA 23 Stepper Motor

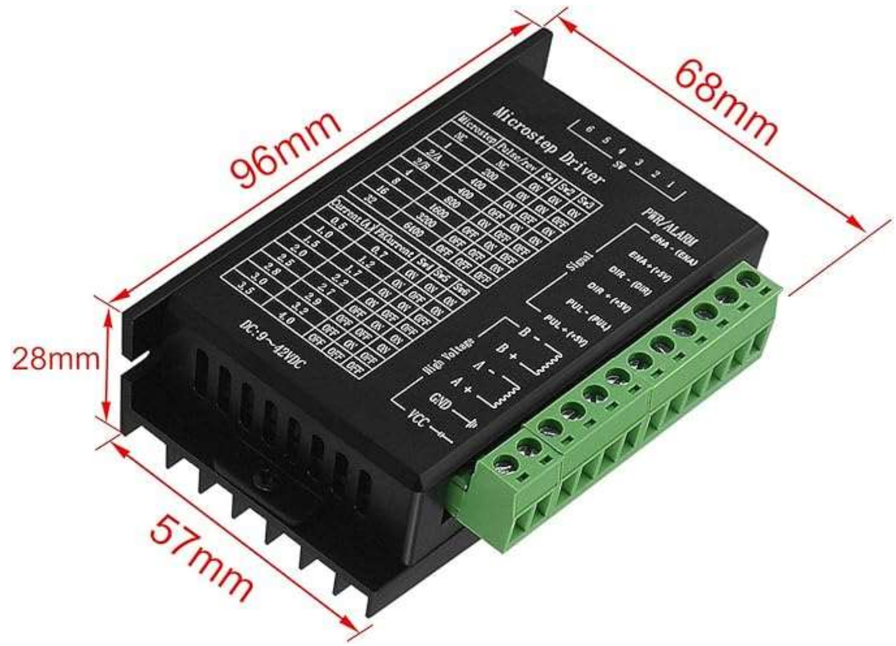


Figure XIII-9: Manufacturing Drawing of part SS#2-P#7– TB6600 Stepper Motor Driver

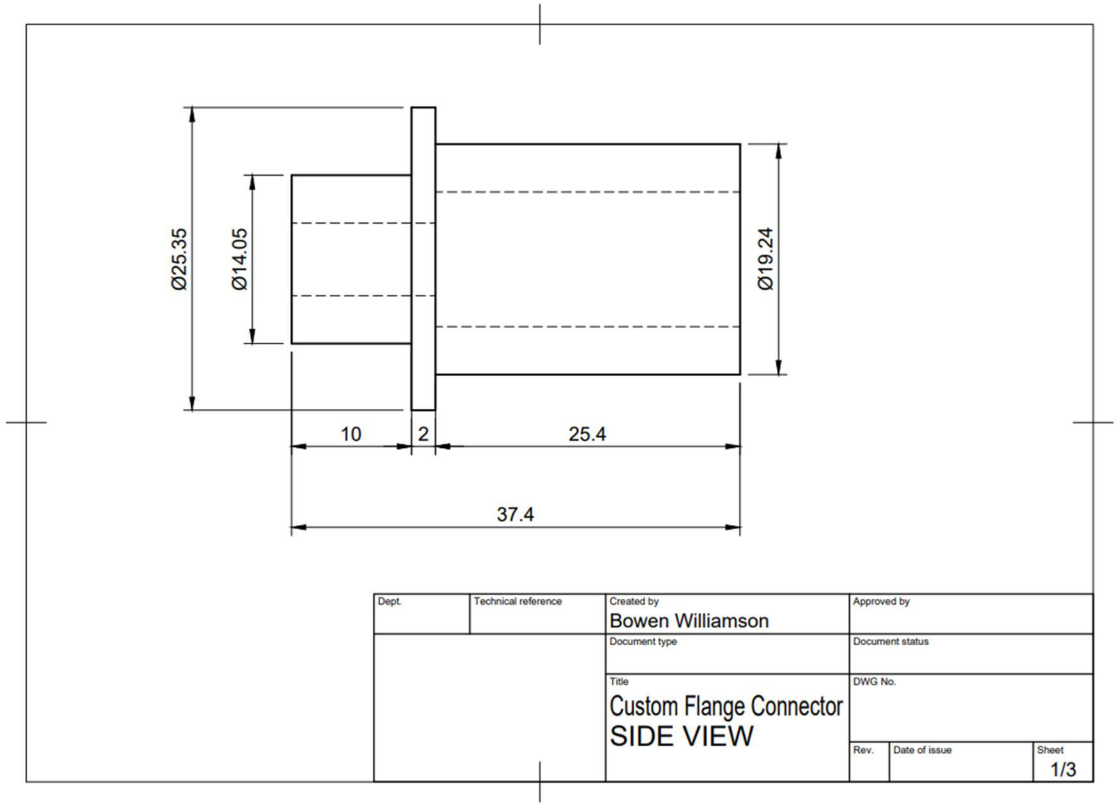


Figure XIII-10: Manufacturing Drawing of part SS#2-P#14- Flange Coupling

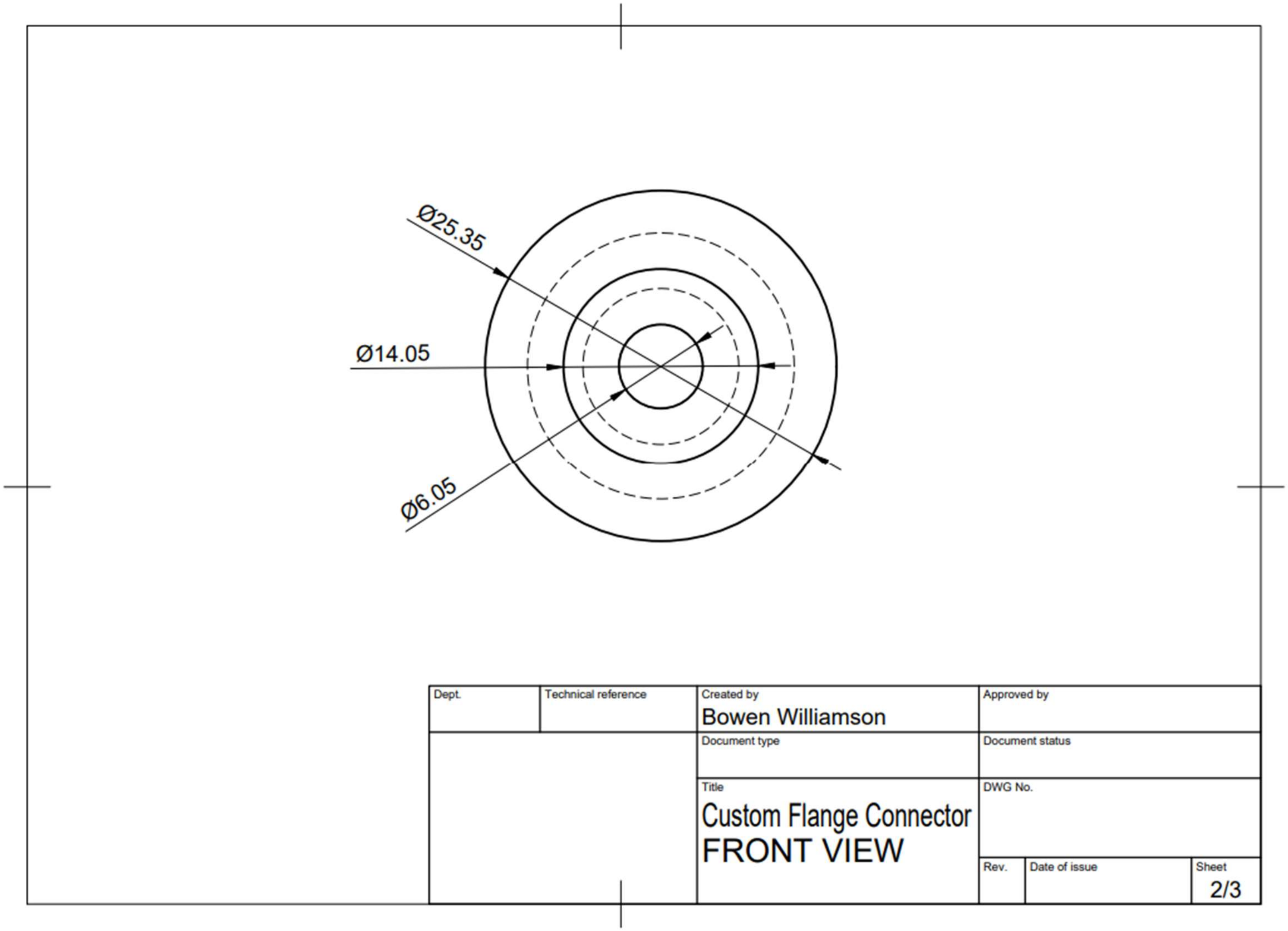


Figure XIII-11: Manufacturing Drawing of SS#2-P#14 - Flange Coupling

Figure XIII-12: Manufacturing Drawing of part SS#2-P#3- Cancel Button

Specification	SP-215-SS
Compliance	ISO 9060:2018 – Class C (fast response)
Power Supply	5.5 to 24 V DC
Current Draw	300 µA
Output (Sensitivity)	2.5 mV per W/m²
Calibration Factor	0.4 W/m² per mV
Calibration Uncertainty	Less than 3%
Measurement Repeatability	Less than 1%
Long-term Drift	Less than 2% per year
Non-linearity	Less than 1% up to 2000 W/m²
Response Time	Less than 1 ms
Field of View	180°
Spectral Range	360 to 1120 nm
Directional (Cosine) Response	±5% at 75° zenith angle
Temperature Response	0.04 ± 0.04% per °C
Operating Environment	–40 to 70 °C; 0–100% RH; submersible up to 30 m
Dimensions	24 mm diameter; 33 mm height
Mass (with 5 m cable)	90 g
Cable	5 m shielded, twisted-pair; TPR jacket (high UV/water resistance)
Warranty	4 years against defects in materials and workmanship

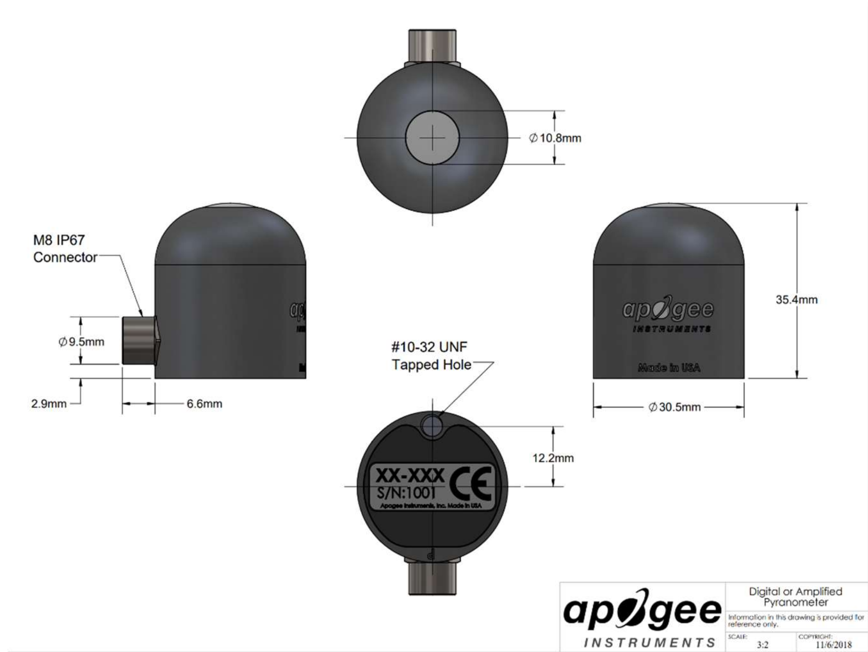


Figure XIII-13: Manufacturing Drawing of part SS#2-P#4- Pyranometer

Type	SFM-27-W
Rated Voltage	12VDC
Opcration Voltage	3～24VDC
Rated Current	15mA at 12VDC
Sound Output	100dB at 12VDC/30cm
Resonant Freq	3.0±0.2
Operating Temp	-20-+45
Storage Temp	-20-+60

Figure XIII-14: Manufacturing Drawing of part SS#2-P#5- Buzzer

Microcontroller	ATmega328
Clock Speed	16MHz
Operating Voltage	5V
Maximum supply Voltage (not recommended)	20V
Supply Voltage (recommended)	7-12V
Analog Input Pins	6
Digital Input/Output Pins	14
DC Current per Input/Output Pin	40mA
DC Current in 3.3V Pin	50mA
SRAM	2KB
EEPROM	1KB
Flash Memory	32KB of which 0.5KB used by boot loader

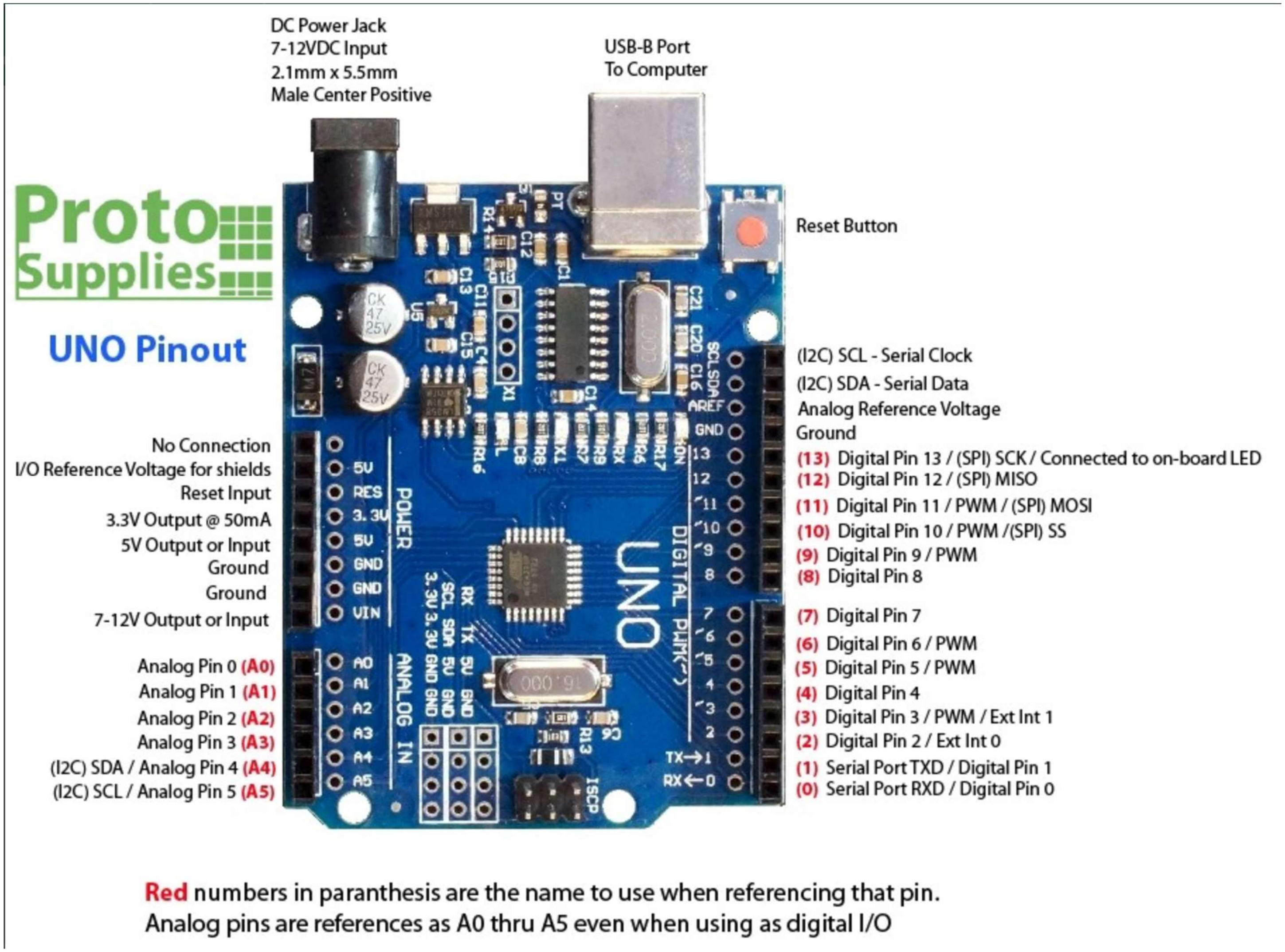
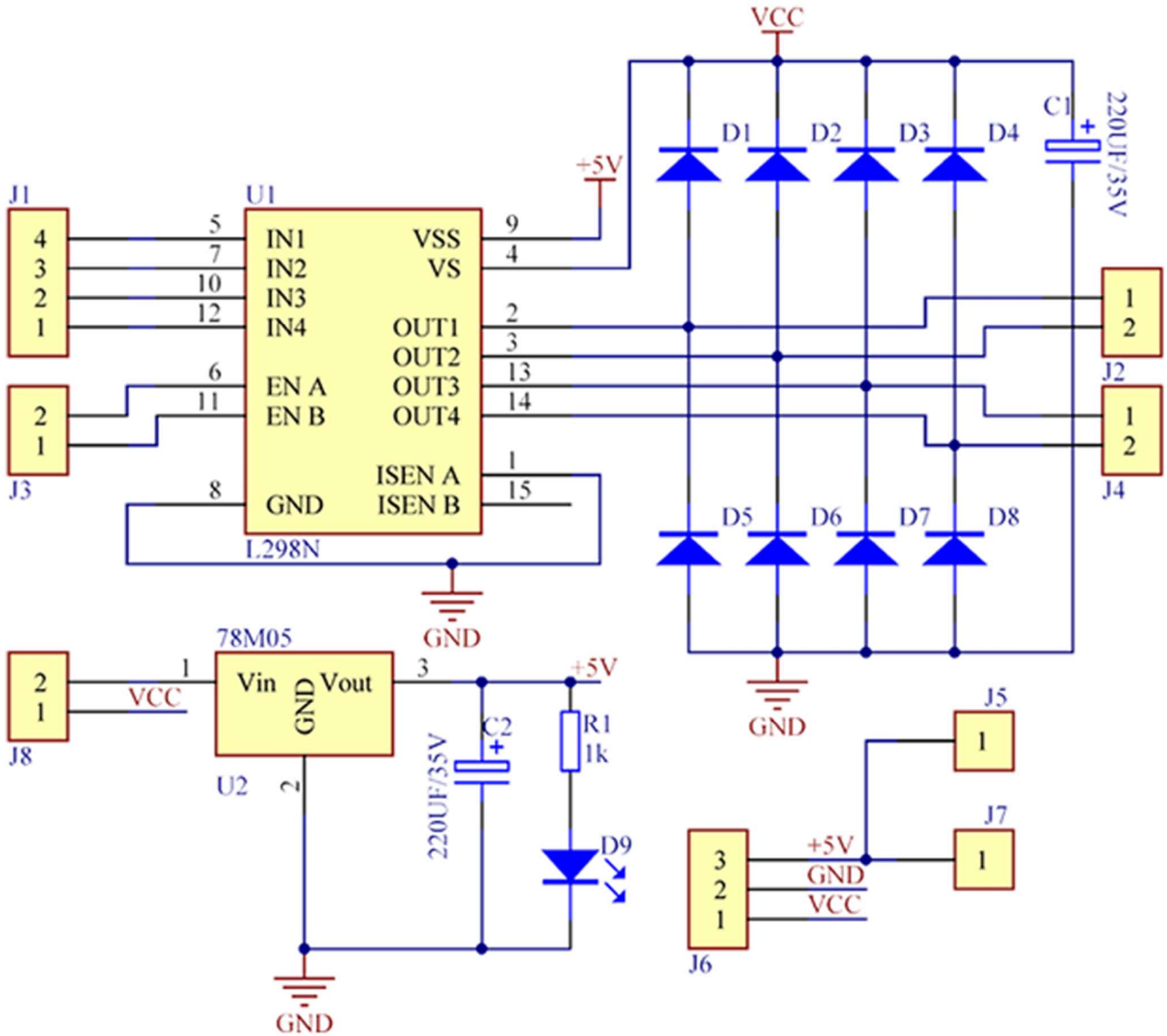


Figure XIII-15: Manufacturing Drawing of part SS#2-P#6- Arduino Uno


VCC
D1
D2
D3
D4
C1
220UF/35V
+5V
J1
U1
IN1
IN2
IN3
IN4
VSS
VS
OUT1
OUT2
OUT3
OUT4
EN A
EN B
ISEN A
ISEN B
GND
L298N
J2
J3
J4
D5
D6
D7
D8
GND
78M05
Vin
Vout
U2
VCC
J8
C2
220UF/35V
R1
1k
D9
J5
J6
J7
+5V
GND
VCC

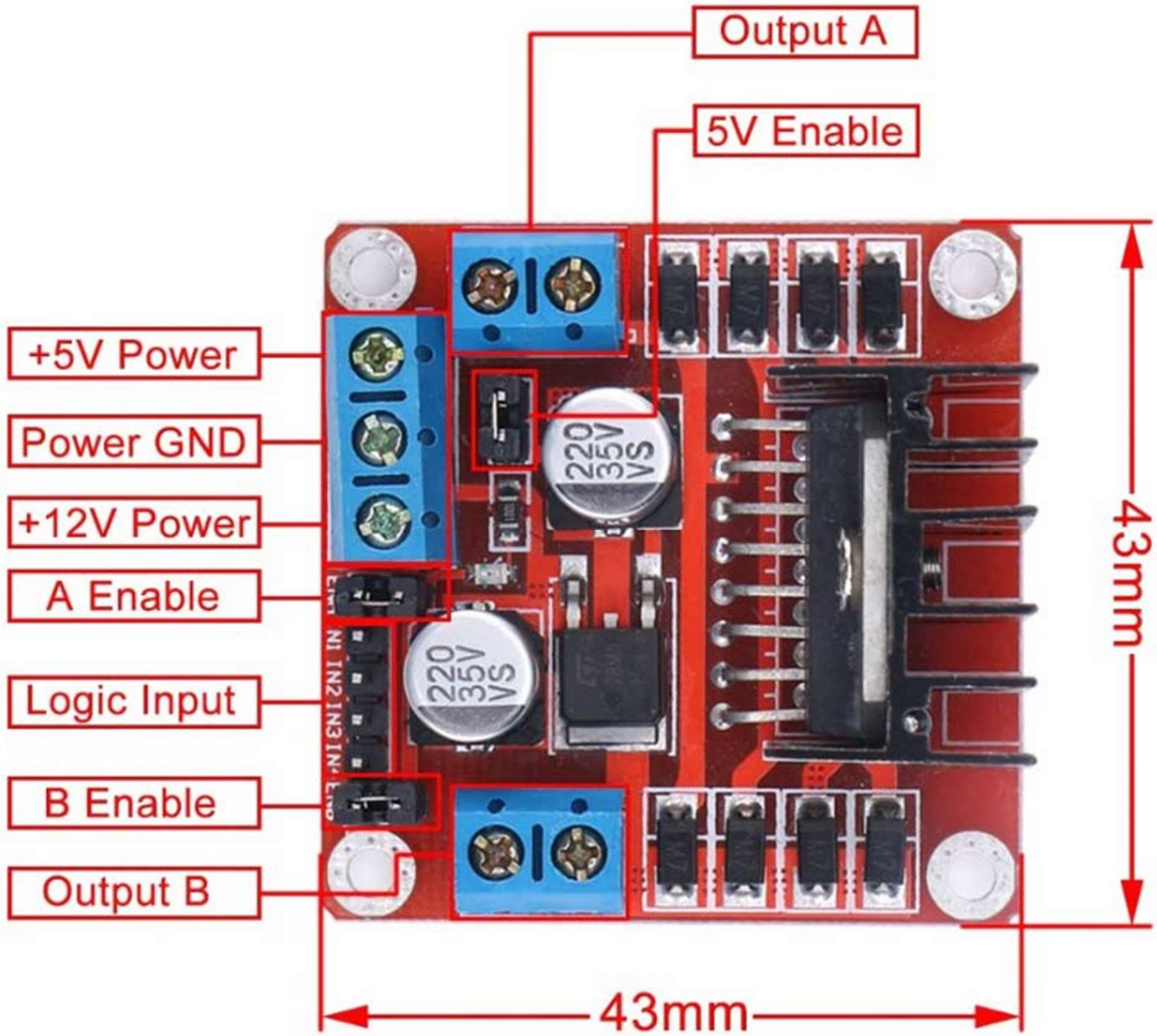


Figure XIII-16: Manufacturing Drawing of part SS#2-P#8- L298N Motor Driver

Operating Voltage	24V DC
No-Load current	60mA
Max Current draw	2A
Max power consumption	48W

Figure XIII-17: Manufacturing Drawing of part SS#2-P#9- DC Geared Motor

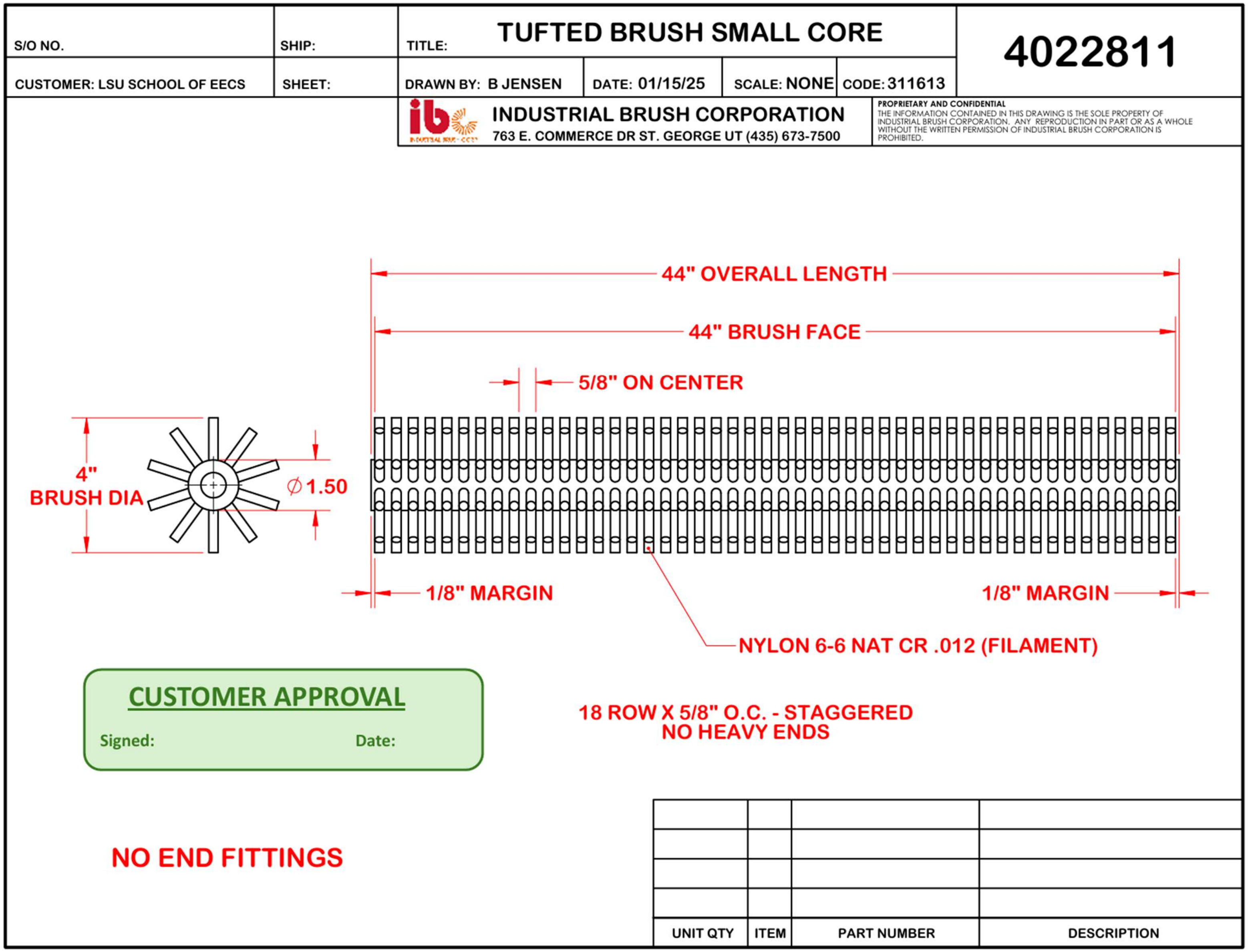


Figure XIII-18: Manufacturing Drawing of part SS#2-P#10- Solar Panel Safe Radial Brush

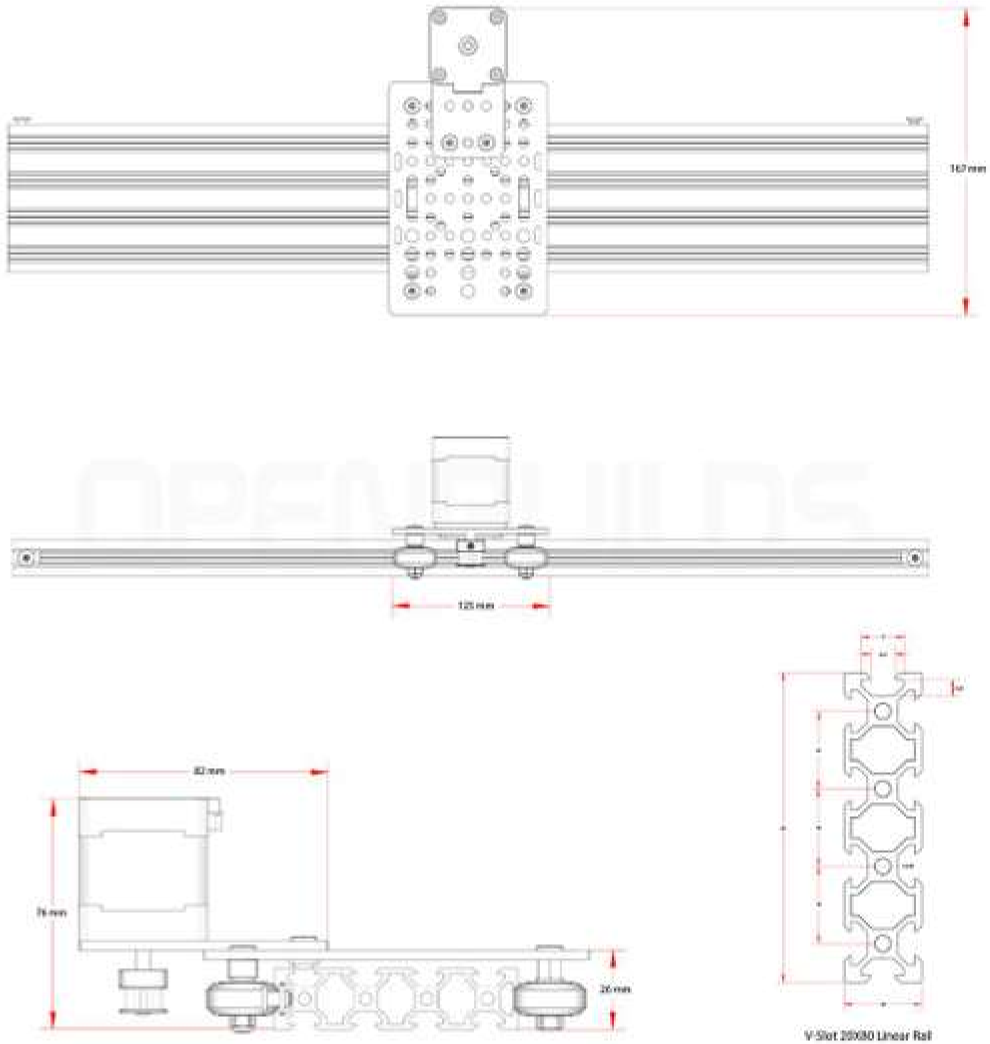


Figure XIII-19: Manufacturing Drawing of part SS#2-P#11- Linear Actuator Assembly

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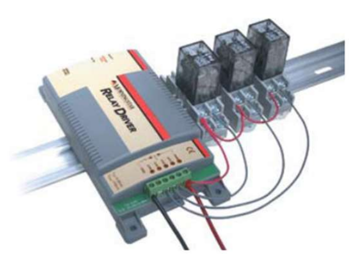
Technical Specifications

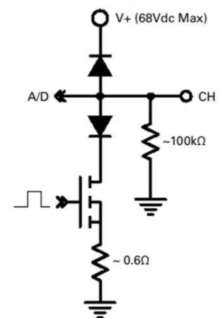
Version	RD-1
Electrical	
System Voltage	12 to 48 volts*
Max. Channel Current	750 mA
Max. Channel Input Voltage	< Battery Voltage
Voltage Accuracy	2% ± 50 mV
Temperature Accuracy	± 2°C
Min. Operating Voltage	8 volts
Max. Operating Voltage	68 volts
Self Consumption	< 20 mA
Temperature Sensor Range	−40°C to +85°C
Transient Surge Protection	1500 W / channel
Comm. Ports (opto-isolated)	(2) RJ-11 meter bus connections (1) 9-pin serial RS-232
Environmental	
Operating Temperature	−40°C to +45°C
Storage Temperature	−55°C to +85°C
Humidity	100% (NC)
Tropicalization	Conformal coating on both sides of printed circuit board
Mechanical	
Dimensions	Length: 16.3 cm / 6.4 inch Width: 8.1 cm / 3.2 inch Depth: 3.3 cm / 1.3 inch
Weight	0.2 kg / 0.4 lb
Largest Wire	1.0 to 0.25 mm² 16 to 24 AWG
Torque Terminals	0.4 Nm / 3.5 in-lb
Enclosure	Type 1, indoor rated
DIN Rail Attachment	35 mm standard

* Voltage of user selected relays must be same as battery voltage

DIN Rail Mounted Relay Driver with Relays



General Channel Schematic



Electronic Protections

- Reverse Polarity Protection
- Short-Circuit Protection
- Overcurrent Protection
- Lightning and Transient Surge Protection

Compatible with:

- TriStar MPPT 600V Controller
- TriStar MPPT Controller
- SunSaver MPPT Controller
- Additional System Relay Drivers
- TriStar (PWM) Controller
- SunSaver Duo Controller
- SureSine (Classic) Inverter
- MeterHub

Warranty:
Five years warranty period. Contact Morningstar or your authorized distributor for complete terms.

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Figure XIII-20: Manufacturing Drawing of part SS#2-P#12- Relay Driver

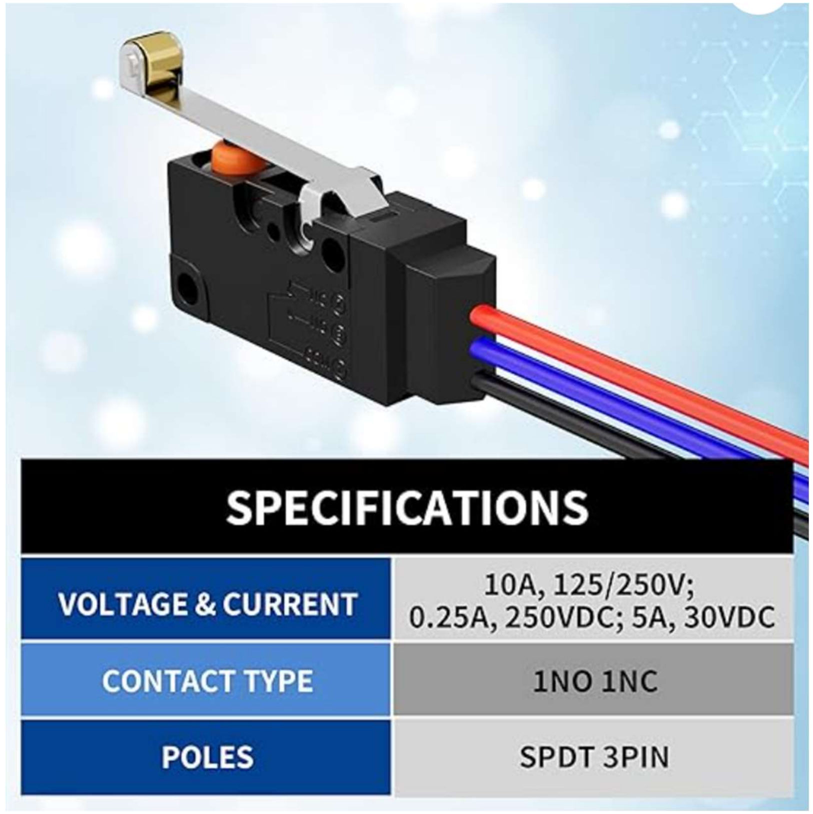


Figure XIII-21: Manufacturing Drawing of part SS#2-P#13- Limit Switches

XIII.D. Engineering Analysis and Materials Selection Supplement

XIII.D.1. Eng. Analysis Details for SS#1 – Solar Panel and Charge Controller

XIII.D.1.a. Eng. Analysis and Materials Selection Details for SS#1-P#1&2 – Solar Panel and Charge Controller

Bowen Williamson and William Mancuso

The solar charge controller selected for this project is the Morningstar TriStar MPPT 30 (TS-MPPT-30), a high-performance maximum power point tracking (MPPT) device widely used in professional off-grid solar energy systems. Its primary function in our design is to efficiently regulate power from the solar panel array to the battery bank while ensuring optimal energy harvesting under a range of environmental conditions. The controller constantly adjusts its operating point to track the maximum power point of the connected PV array, which is particularly valuable when irradiance or panel temperature fluctuates. This functionality allows it to achieve efficiency gains of up to 30% compared to traditional pulse-width modulation (PWM) charge controllers.

The TS-MPPT-30 also offers robust battery management features, supporting various battery chemistries including flooded, sealed, and gel types with customizable charging profiles. To protect both the controller and the system components, it incorporates multiple layers of electrical protection, including overload, short-circuit, and reverse polarity safeguards. Furthermore, it includes communication capabilities through RS-232, Modbus (EIA-485), and Ethernet (with optional accessories), which enables system monitoring, data logging, and remote configuration. Designed for long-term reliability in demanding outdoor environments, the controller is passively cooled and rated for operation between -40°C and +60°C without performance derating. Its enclosure is made from anodized aluminum, which was selected for its excellent thermal conductivity, corrosion resistance, and structural durability. The anodized finish also enhances the electrical insulation and protects the housing from oxidation. Internally, the charge controller uses high-temperature-rated capacitors and power transistors (MOSFETs) mounted on heatsinks that rely on the external case for heat dissipation. This passive thermal design avoids the use of fans or moving parts, reducing maintenance needs and increasing system longevity.

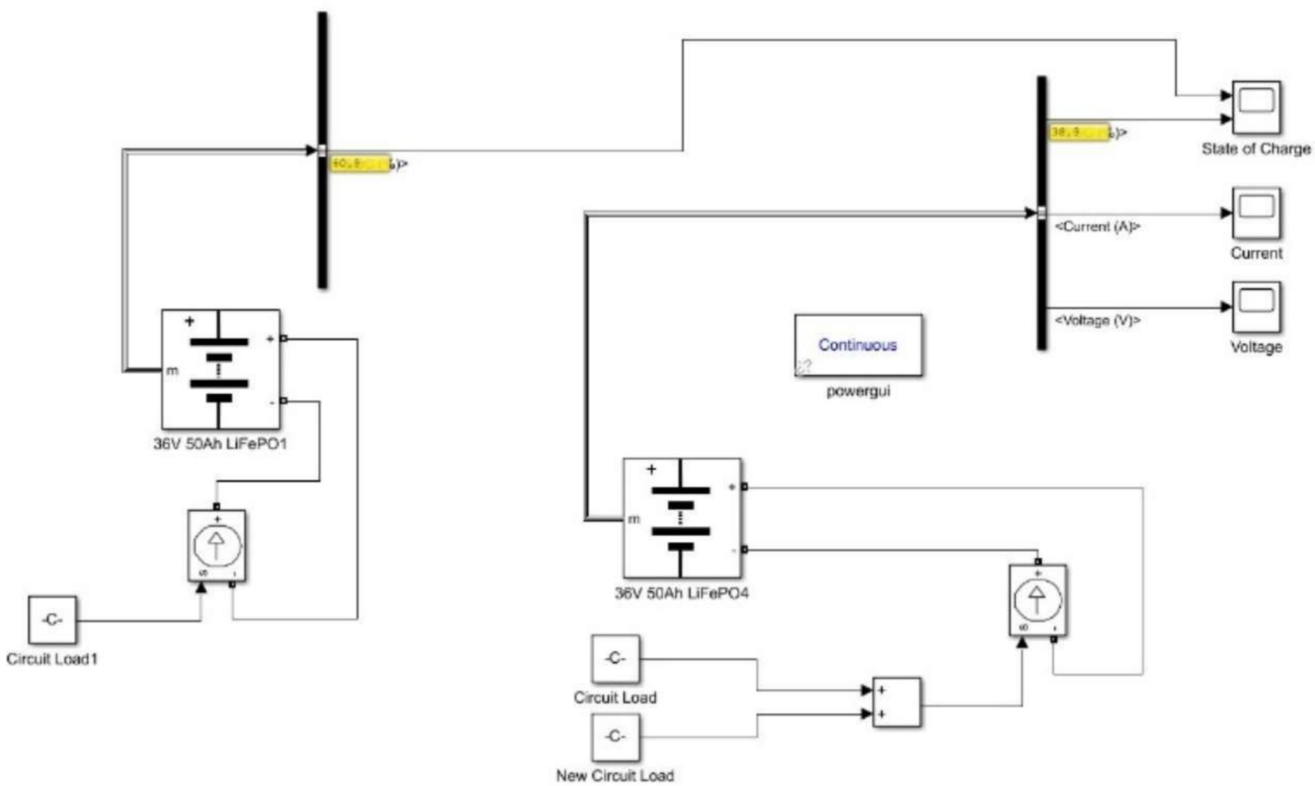
The electrical terminals are typically composed of brass or copper alloy, often with a nickel or tin plating to improve corrosion resistance and minimize electrical resistance at connection points. To protect the internal electronics from environmental contaminants, the printed circuit board is coated with a conformal layer—typically an acrylic or urethane-based compound—that guards against moisture, dust, and chemical exposure.

The decision to use the Morningstar TriStar MPPT 30 was based on its proven track record in industrial and remote solar applications, its high efficiency under varying solar conditions, and its rugged build quality. The combination of intelligent MPPT tracking, comprehensive protection features, passive cooling, and durable material choices ensures that the controller meets the technical and environmental demands of our project.

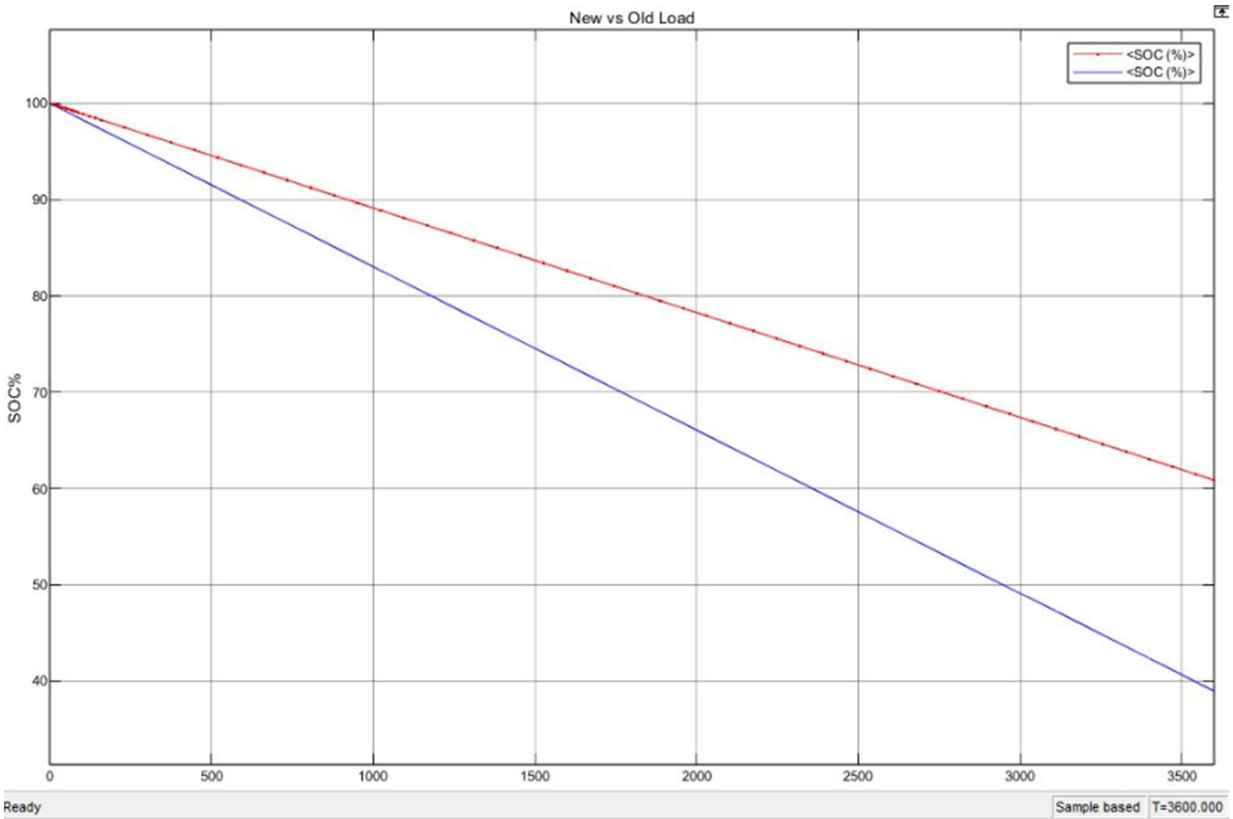
XIII.D.1.b. Eng. Analysis and Materials Selection Details for SS#1 - Overall Circuit

Kenny Bui

This analysis supported the material selection for our overall circuit. Using the load analysis of the previous team along with our new load analysis we were able to analyze the capabilities of our battery. Using these load analysis values, I was able to input these load consumption values into a Simulink model that represents our overall circuit. After running the Simulink model, I received a graph with two lines of state of charge, one for the original circuit and one for the new circuit. In this graph, we can see that the original circuit's state of charge is at 60.9% after an elapsed time of one hour and the new circuit has a state of charge of 38.9% after one hour. These state of charge results indicate that there is a dip in state of charge due to the increase in load consumption but confirms that the battery can still handle this additional load. Analysis images can be found below.



Overall Circuit Simulink Model



Red: Original Load (60.9%) Blue: New Load (38.9%)

Overall Circuit Analysis Results

XIII.D.2. Eng. Analysis Details for SS#2 – The Cleaning System

XIII.D.2.a. Eng. Analysis and Materials Selection Details for SS#2-P#6- Arduino Uno

Kenny Bui

In selecting the appropriate electronic intelligence for our rover system, we conducted extensive research on various options, including integrated circuits, microcontrollers, and microcomputers. Each option presented distinct advantages and trade-offs in terms of complexity, cost, and functionality. Integrated circuits offered a cost-effective and straightforward solution. However, they would require numerous individual components and could be vulnerable to electromagnetic interference within the rover's chassis, making system integration more challenging. On the other hand, microcomputers, while powerful and highly versatile, proved to be overly complex and costly for our project's scope, better suited for high-level computing tasks. Microcontrollers emerged as the ideal middle ground. They offer sufficient processing capabilities and input/output control for embedded applications without unnecessary complication. After evaluating several microcontrollers, we selected the Arduino Uno due to its affordability, widespread availability, and suitability for our needs. The Arduino Uno features a well-balanced set of digital and analog I/O pins, making it easy to interface with various sensors and actuators. Its onboard memory and processing speed are more than adequate for our system's requirements, and its extensive community support enhances development and troubleshooting. Overall, the Arduino Uno provides the perfect balance of simplicity, functionality, and cost for our rover's electronic intelligence system.

XIII.D.2.b. Eng. Analysis and Materials Selection Details for SS#2-P#1&7 – NEMA 23 Stepper Motor and TB6600 Stepper Motor Driver

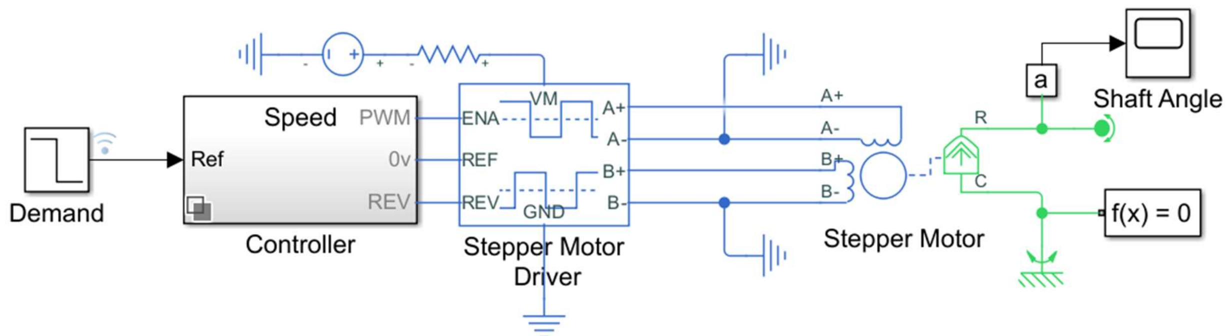
Margaret Burkes

To determine the current draw, I used the Simulink file entitled "Stepper Motor with Control" on MATLAB's website. It works as follows:

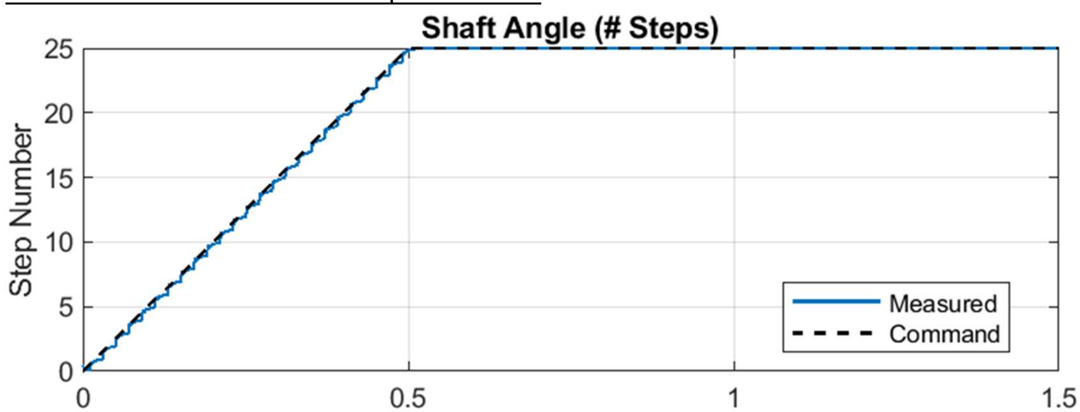
1. Demand: is the demand speed in revolutions per second. Ours was 2.5 RPS due to the following equations:

$$\omega = \frac{v}{r} = \frac{\frac{150mm}{s}}{9.5mm} = 15.71\frac{rad}{s}$$
$$RPS = \frac{\omega}{2\pi} = \frac{15.71\,rad/s}{2\pi} = 2.5\,RPS$$

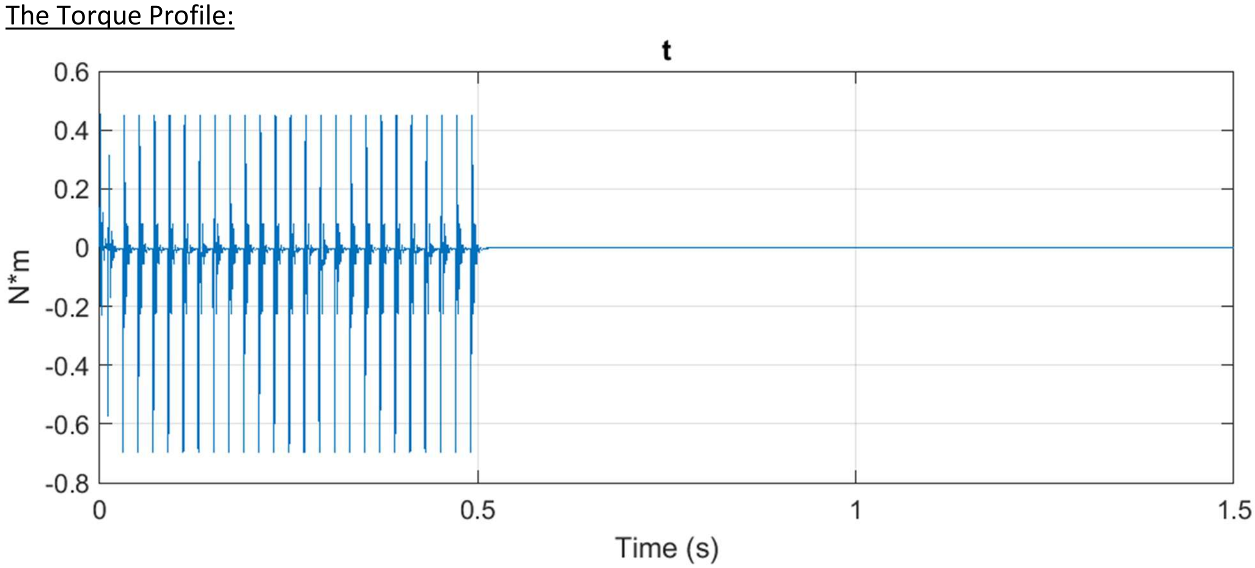
2. Controller: is set in speed mode. This means it uses the speed demand signal and converts the command into a pulse train that controls the stepper motor driver. This is representing our Arduino Uno.
3. Stepper Motor Driver: is set in microstepping mode at 4 steps per full step. This is to be able to control the motor. The motor is rated for 2.8A/phase, and the TB6600 can be set to 2.8A/phase. Therefore, the maximum continuous current per phase on the driver is set to 2.8A. In general, the driver is responsible for converting the low-power input signals from the controller into high-power electrical pulses to drive the motor.
4. Stepper Motor: is the NEMA 23 stepper motor that the team will be using. The information used in the block is directly from the datasheet.



The Commanded vs Measured Speed Profile:

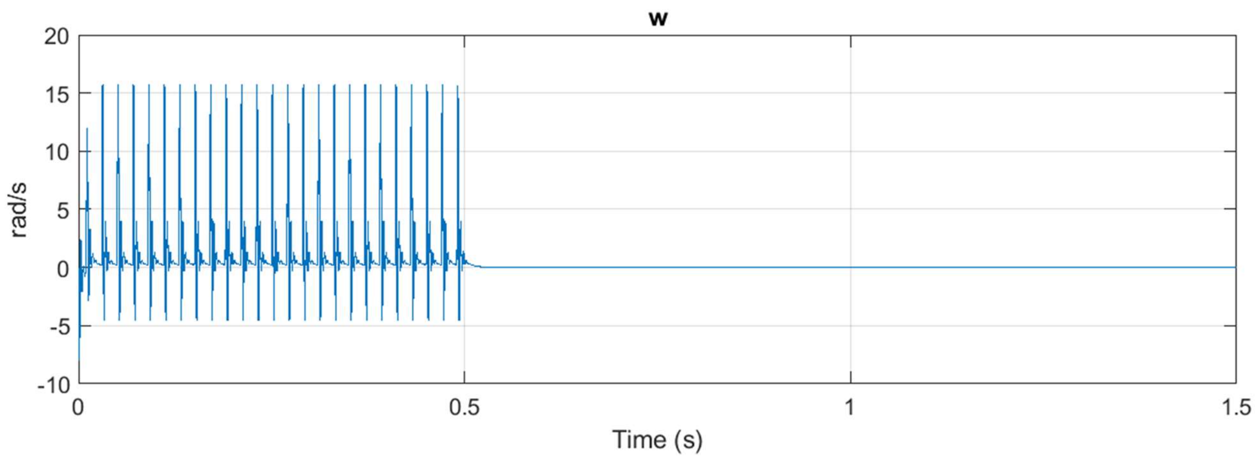


The Torque Profile:

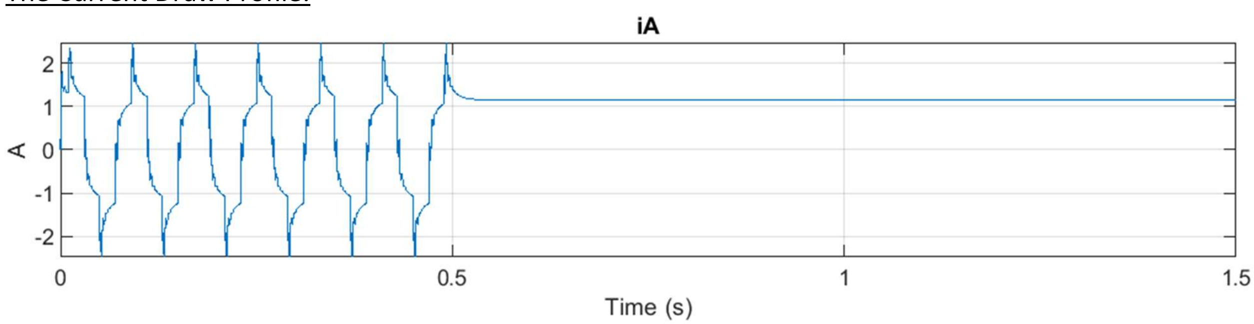


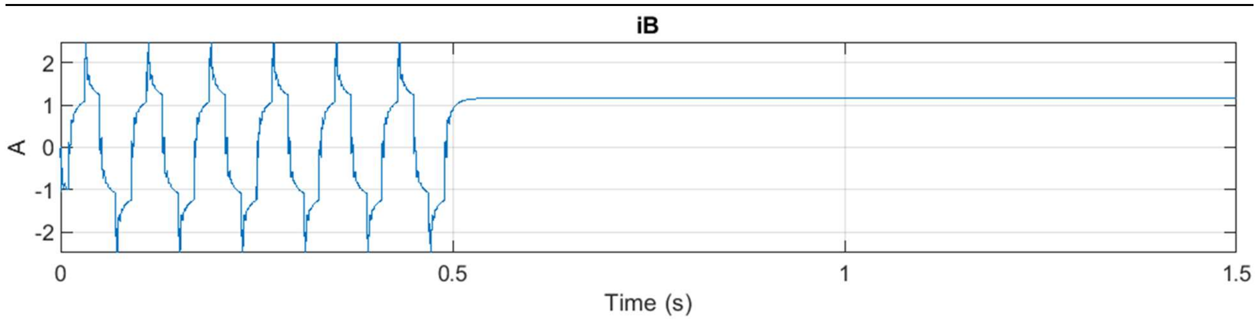
The maximum torque required shown in the graph above is 0.456 Nm, which the NEMA 23 stepper motor can more than handle.

The Speed Profile:



<u>The Current Draw Profile:</u>



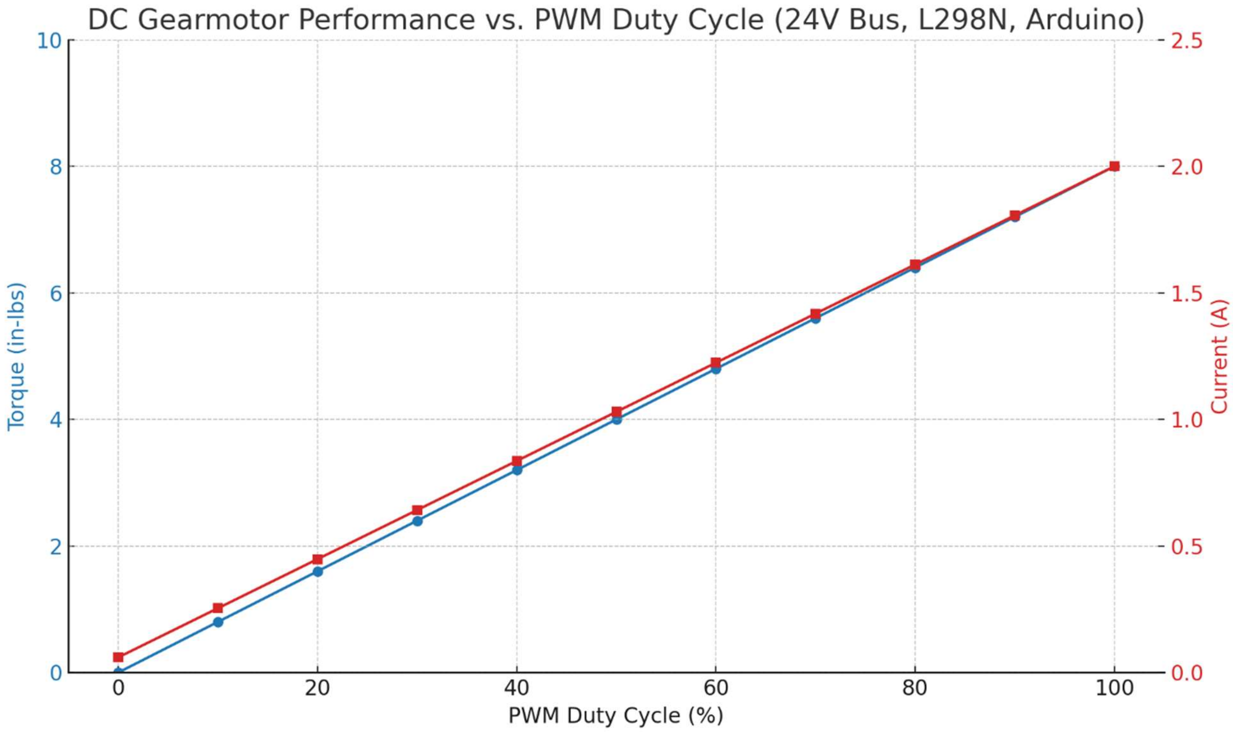


As seen here, the current being drawn is a maximum of 2.484A/phase.

XIII.D.2.c. Eng. Analysis and Materials Selection Details for SS#2-P#9 - 25 RPM 24V DC Inline Gearmotor (Model SEI 37JB270G/32ZYT30-2468

Keldon Ngo

Operating Voltage	24V DC
No-Load current	60mA
Max Current draw	2A
Max power consumption	48W



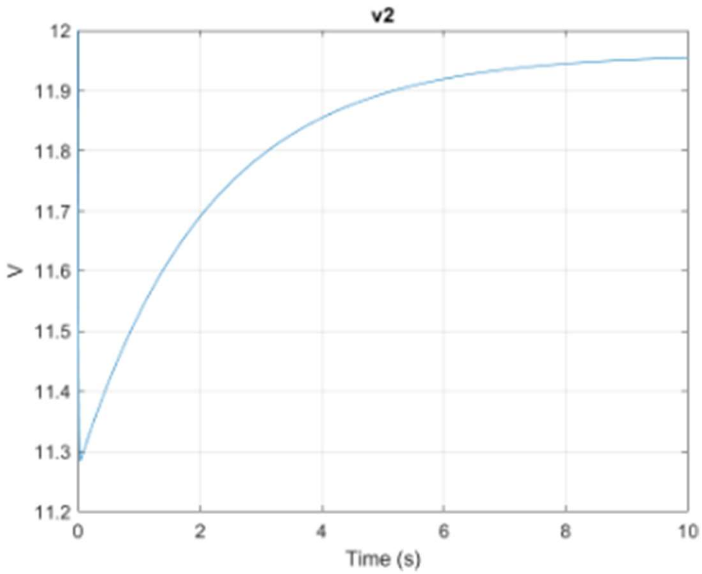
XIII.D.2.d. Eng. Analysis and Materials Selection Details for SS#2-P#8 - L298N dual H-bridge motor driver

Keldon Ngo

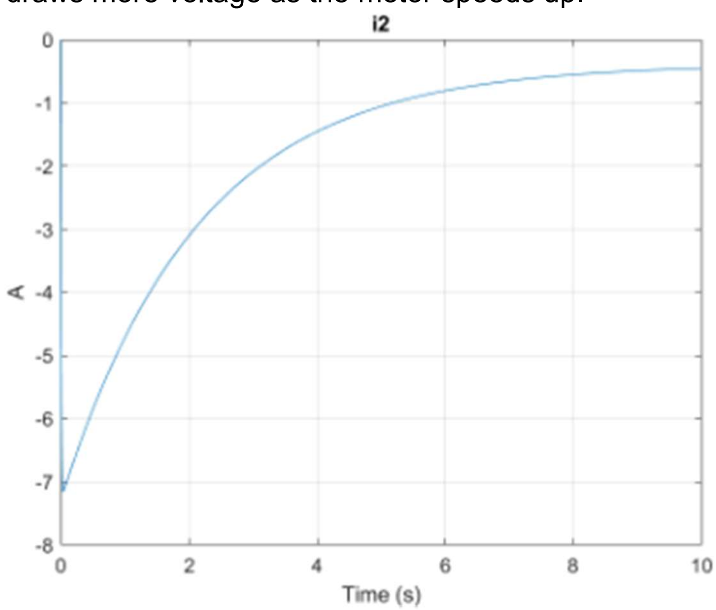
<u>L298N Dual H-bridge Motor Driver Specs:</u>

Operating Voltage	3.2-40V
Input Current	0-36mA
Max Output Current	2A
Input Control Voltage	3.2-24V
Max Power Consumption	20W

The above chart shows how the operating voltage was within the operating voltage of the L298N Dual H-Bridge Motor Driver.



Shown above is the voltage to time of the motor controller. As we can see, the controller slowly draws more voltage as the motor speeds up.



Shown above is a current to time of the motor controller. As we can see, the controller slowly draws more current as the motor speeds up.

XIII.D.2.e. Eng. Analysis and Materials Selection Details for SS#2-P#4- Pyranometer

Bowen Williamson

The pyranometer selected for our project is the Apogee Instruments SP-215-SS, a silicon-cell pyranometer designed to measure global horizontal irradiance, which is essential for evaluating the performance of our solar-powered rover. Although it is an off-the-shelf component, its selection was based on specific engineering criteria that align with our system's requirements.

The SP-215-SS provides a spectral response from 360 to 1120 nm, closely matching the operational spectrum of most photovoltaic panels, making it highly suitable for solar energy studies. Its compact size and rugged construction also made it ideal for integration on a mobile outdoor platform.

This sensor offers a fast response time of less than 1 millisecond and comes factory-calibrated with an accuracy of ±5% under clear sky conditions. Its output is a 0–5 V voltage signal, which integrates easily with standard data acquisition systems and microcontrollers. The internal circuitry includes temperature compensation to ensure stable performance under varying environmental conditions, which is particularly important for outdoor use. This allowed us to log solar irradiance data accurately throughout our testing without additional signal conditioning or filtering.

While the SP-215-SS's internal sensing element and circuitry were not customized, we evaluated its external materials for durability and environmental resilience. The sensor features a robust anodized aluminum housing and a UV-stabilized acrylic diffuser, both designed to resist degradation from prolonged exposure to sunlight, moisture, and temperature extremes. The model's IP68 rating confirmed its suitability for outdoor deployment, giving us confidence in its long-term reliability during extended field tests with the rover.

XIII.E. Manufacturing & Assembly Supplement

XIII.E.1. Manufacturing Processes

Manufacturing started by cutting 20x20 1500mm rails to about 1240mm in length. These rails were then drilled with a size 19 bit and then threaded them with a M5x0.8 tapping bit. We then used a size 11 bit to drill through the linear actuator rails to secure the rails to each other. The holes were drilled such that the lined up to the part of the frame of the solar panel we were welding too. We then laid the panel on its face and set the frame on top lining it up even and welded all 8 points of contact, to ensure the welds held up under some potential pressure. We also had to remove some knockout conduit from the solar charge controller to add in the wire grommets. For wiring the solar panel IP68 cable connecters were removed and replaced with a waterproof switch and another switch was added between the battery and charge controller connection too.

Table XIII-3: Manufacturing Processes for Parts of Sub-System SS#1

Part #	Manufacturing Process
SS#-P1	Purchased/ Welded
SS#-P2	Purchased/ Drilled
SS#-P3	Purchased/ Cut to length

For the manufacturing of sub-system 2, most of the components were purchased. However the gantry plate was milled. We first measured the holes on the base plate to be 20mm apart, with that, we 3d printed a gantry plate with 5mm of space between them as a stencil. After drawing where the motor and ball bearing would be best fit on the stencil we then used a drill press to to create holes large enough for the motor shaft and the attaching screws. We then secured the motor to the stencil and placed the entire assembly on the base plate. After attaching the assembly, we realized we needed the slots to be about 11mm longer to have enough adjustable room for the brush to sit comfortably on the panel. We then took an aluminum plate and cut the appropriate-sized piece from it, sanding the material to make it uniform and remove jagged edges. After manufacturing the plate, we milled in the appropriate slots, leaving a 5mm gap between the next slot. We also extended the existing 25mm slots to 36mm. After ensuring the plate fit we then used the previously created stencil to drill the appropriate holes for the bearing and motor. During assembly of the entire system, we realized that the previously used T-track rails were too small for the newly milled slots. We then took a smaller aluminum plate and cut it into a rectangle of 25mm and tapped it to created a mounting bar for the screws to secure the gantry plate to the base plate. After mounting the plates, we made sure the flanges are lined up and at the appropriate height to ensure they sit evenly on the panel. This concludes my manufacturing and assembly of the custom gantry plates.

The team originally planned to mount the radial cleaning brush to the gear motor using a flange coupling screwed into the end of the brush. However, feedback during the final presentation raised concerns about centering the coupling on the brush core. To address this, the team designed a custom flange coupling that extends into the hollow brush core, improving both alignment and structural integrity by shifting load-bearing from screws to the coupling itself. Due to its axial symmetry, the part was machined on a lathe. With no prior lathe experience among the five electrical engineering students, machine shop staff assistance was essential. The coupling was made from aluminum for its light weight and machinability, as it only needed to support the brush's weight and minimal frictional force. Precise caliper measurements were taken of the motor shaft and brush core. The coupling's outer diameter was gradually turned down on the lathe to fit the brush, while a slightly undersized drill bit was used to bore the inner diameter for the motor shaft. Because one end of the brush is motor-driven and the other is bearing-supported, the couplings were made to slightly different lengths and adjusted after partial assembly for proper alignment. The motor-side coupling was drilled and tapped with an M5 set screw to secure it to the D-shaft, using a #20 bit and an M5 tap. The bearing-side coupling fit snugly and required no fastening. Since the motor-side fit was looser, a more permanent fix involved drilling through the brush and one wall of the coupling, then tapping for an M5 screw. This prevents slipping during rotation and poses no risk to the panel, as the screw head sits nearly flush and is shielded by the bristles.

Aside from the machining all of the electrical components had to wired together, in this case, the stepper motor driver, L298N motor driver, pyranometer and relay driver were all connected to the arduino for sending and receiving signals. The stepper motors were wired to the stepper motor drivers, and the DC geared motor was connected to the L298N motor driver. Finishing the signal wiring, the charge controller was connect to the relay driver. For power, the stepper motor drivers were powered by the 36 V bus bar, the L298N was power by the 24V bus, the pyranometer and relay driver were on the 12V bus, and the arudino was on the 5 V bus.

Table XIII-4: Manufacturing Processes for Parts of Sub-System SS#2

Part #	**Manufacturing Process**
SS#2-P1	Purchased
SS#2-P2	Milled
SS#2-P3	Purchased
SS#2-P4	Purchased
SS#2-P5	Purchased
SS#2-P6	Purchased
SS#2-P7	Purchased
SS#2-P8	Purchased
SS#2-P9	Purchased
SS#2-P10	Purchased
SS#2-P11	Purchased
SS#2-P12	Purchased
SS#2-P13	Purchased
SS#2-P14	Lathe

XIII.E.2. Assembly Processes

XIII.E.2.a. Assembly Process for Sub-System SS#1 – Solar Panel and Charge Controller

For subsystem 1, assembly began with wiring the solar panel (16 awg) to the solar charge controller as well as the battery (4 awg) to the charge controller, and the charge controller to the relay driver via a meter-bus cable. After that, it is important that the cleaning system linear actuator rails are screwed into the mounting t track rails that were welded to the frame. Following this the solar panel can be mounted to the frame by adding the t track nuts down the t track rails to secure the electrical conduit straps around the frame of the rover and then screwing those straps through the t track nuts.

XIII.E.2.b. Assembly Process for Sub-System SS#2-The Cleaning system.

For subsystem2, assembly began with attaching the solar panel to the frame manufactured by the team. After the solar panel was attached the base plates with both the rotational and linear actuating components can be attached to the rails. A teethed belt was then attached to the stepper motors responsible for linear actuation. The custom flanges were then attached to both the geared motor, responsible for rotation of the brush, and the ball bearing was attached. The flanges were then attached to the brush via a combination of setscrews and glue. Thus, completing the assembly of the cleaning system.

XIII.F. Design Phase Testing Supplement

XIII.F.1. Design Phase Testing Methods and Details

Team 66 conducted no design-phase testing.

XIII.F.2. Design Phase Testing Results

Team 66 conducted no design-phase testing.

XIII.G. Testing & Validation Supplement

XIII.G.1. Testing & Validation of Function F# 1– Convert Solar Energy to Power the Rover

Keldon Ngo

XIII.G.1.a. Test Protocol Description - F#1

The initial test is an ideal conditions test in which the panel is placed outside during a sunny day with no cloud cover, and the power output is measured via a solar panel tester. The same will be done on a day with light to heavy cloud cover, as the reflection of sunlight through the clouds hinders solar irradiation. The final test will be an indoor test, with standard room lights, to see if any power is produced while in storage.

XIII.G.1.b. Equipment and Instrumentation - F#1

- Solar panel
- Solar panel multimeter

XIII.G.1.c. Data Acquisition & Analysis – F#1

The panel was placed in the same spot for all the aforementioned tests except for the indoor test, to ensure consistency of environmental factors. We then tested the current and voltage output of the panel using a current clamp and a voltage meter.

XIII.G.1.d. Results Details – F#2

The results were higher than expected, apart from our ideal conditions test, which came in lower than expected. However, after retesting post-assembly, the output was much closer to expected rates.

Table XIII-5: Test Result Data Table

Condition Tested	Average Voltage	Average Current	Average Power
Sunny	51.9V	7.4A	384W

Partly Cloudy	50.5V	8.05A	406.5W
Cloudy	48.2V	7.8A	376W
Indoor	47.25V	6.34A	300W

XIII.G.2. Testing & Validation of Function F#2 – Autonomous Power Switching Between Power Sources

Bowen Williamson

XIII.G.2.a. Test Protocol Description - F#2

In order to test the autonomous power flow, current measurements were taken of both power sources, the battery and the solar panel. The comparison of these measurements confirms the autonomous power flow, as the solar panel should carry more or less of the electrical load as the weather permits.

XIII.G.2.b. Equipment and Instrumentation - F#2

- Enduro battery mobile application
- Clamp-style multimeter
- iPhone

XIII.G.2.c. Data Acquisition & Analysis – F#2

All the driving motors of the rover were turned on at full speed. Testing was conducted with no solar power, with solar power on a cloudy day, and with solar power on a sunny day. The mobile app for the battery was used to monitor the current draw of the battery, and the multimeter was used for measuring the solar power current. An iPhone was used for monitoring the battery as well as controlling the rover.

XIII.G.2.d. Results Details – F#2

The results of this test were as expected. For the baseline, 11 amps were being pulled from the battery with no solar. For the full speed test with the solar power turned on during a cloudy day, the battery only had to supply 8.8 amps. Finally, the last test was on a sunny day when the panel would be producing max power, and the battery only had a current draw of about 3.4 amps, explaining the dramatic increase battery life expectancy when compared to the original load of 11 amps. This was confirmation that the solar charge controller and battery were functioning as they should.

Table XIII-6: Test Result Data Table

	No Solar	Cloudy	Full Sun
PV Current		2.5A	7.7A
Battery Current	11.0A	8.8A	3.4A

Figure XIII-22: Test Results

XIII.G.3. Testing & Validation of Function F#3 – Clean the Solar Panel

Margaret Burkes and Kenny Bui

XIII.G.3.a. Test Protocol Description - F#3

As stated earlier, there were two tests to validate this functional requirement:

1. Activation Test: To conduct this test, we set the rover outdoors and uploaded the code onto the Arduino. We then simulated dust and debris by laying jackets onto the rover. Our team would then record the activation take place to determine if the activation system embedded onto the Arduino works. We would conduct this test a few more times under different code conditions to see which code edition would most benefit the rover.
2. Effectiveness Test: To conduct this test, a baseline power generation was first measured for a clean solar panel. Then, 350mg of flour was placed in a bowl and put on top of a scale to ensure an accurate weight. Following that, the flour was placed evenly on the solar panel to simulate a thin layer of dust or debris. Power generation was remeasured. Next, the cleaning system was run to clean the panel. The power generation was remeasured for a third time.

XIII.G.3.b. Equipment and Instrumentation – F#3

Activation Test: We needed an iPhone as a recording device.
Effectiveness Test: We used flour, a multimeter, a scale, a bowl, and our cleaning system.

XIII.G.3.c. Data Acquisition & Analysis – F#3

Activation Test: We needed a recording device/iPhone. Using the recordings we would be able to see the time it would take for the activation to take place along with if the pyranometer threshold was set adequately.
Effectiveness Test: We used a multimeter to measure power generation and recorded it. To analyze our results, we wanted to see how much the solar panel generation improved from dirty to clean and how much the power generation was able to return to its original value. The following equations were used:

$$\eta_{cleaning} = \frac{P_{after} - P_{during}}{P_{during}} \times 100\%$$

$$Restoration = \frac{P_{after}}{P_{before}}$$

P_{before} referring to the original, clean panel's performance, P_{during} referring to the "dirty" panel's performance, and P_{after} referring to the "cleaned" panel's performance.

XIII.G.3.d. Results Details – F#3

Activation Test: This test confirms that our activation system does work. We were also able to fine tune the parameters of our activation. This included a signal hold for 1 minute and a pyranometer threshold of about 300. These two requirements must be met before the Arduino Uno considers activating the cleaning system.

Effectiveness Test:

Table XIII-7: Test Result Data Table

	Current (A)	Power (W)
Before	8.00 A	450 W
During	2.25 A	125 W
After	6.5 A	350 W

Consequently, the cleaning system improved the power generation by 180% from dirty to clean and was able to return it to 77.8% of its original power generation. As stated earlier, there was variable cloud coverage during the day of testing. Likely, these results are even higher. According to Keldon's tests, the power generation can change relatively quickly, especially in cloudy conditions, therefore these results give a good idea of its effectiveness but not a perfect assessment.

Figure XIII-23: Test Results

XIII.G.4. Testing & Validation of Qualitative Constraint Q#1 – Compatibility with Existing Design

Margaret Burkes

XIII.G.4.a. Test Protocol Description - Q#1

To verify that the rover could function as it did prior to our additions, we conducted a maximum speed test. The first step was going to an empty lot to maintain the safety of all. The location we chose was the Alex Box Stadium lot. The next step was measuring 15 meters to set the standard distance using a tape measurer. The next step was setting the starting and stopping points for LOUIS. Following this, the team ran LOUIS as fast as they could, recorded a video of this to ensure accuracy, and recorded the time it took for him to cross the line.

XIII.G.4.b. Equipment and Instrumentation - Q#1

The team needed a tape measurer, an iPhone to record the run, a timer via the phone app, and an open lot.

XIII.G.4.c. Data Acquisition & Analysis – Q#1

To acquire this information, the runs were recorded via video. To analyze the run, we recorded the distance and time it took. Then, we divided the distance by the time to get the speed.

XIII.G.4.d. Results Details – Q#1

Consequently, the rover can run up to 1.24m/s with an average of 1.22m/s being its top speed.

Table XIII-8: Test Result Data Table

Trial Number	Distance	Time	Speed	Average:
1	15 meters	12.14 seconds	1.24 m/s	1.22 m/s
2	15 meters	12.21 seconds	1.23 m/s	
3	15 meters	12.60 seconds	1.19 m/s	

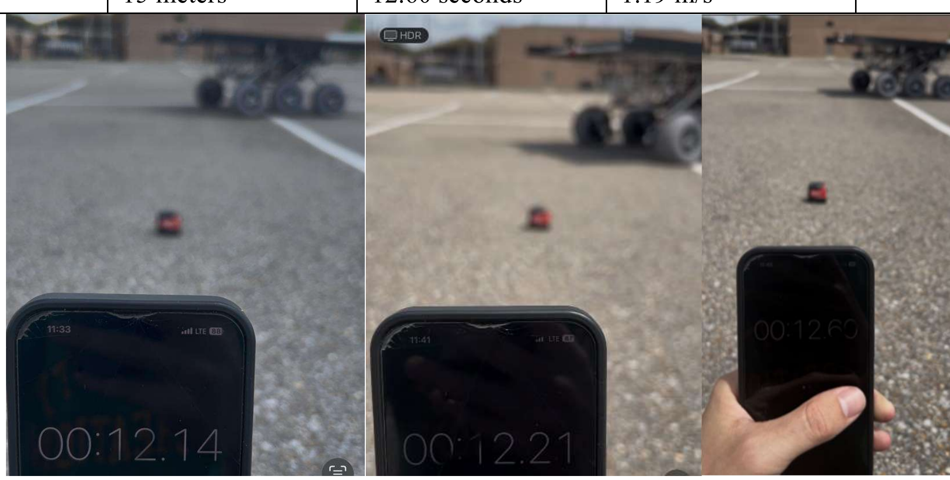


Figure XIII-24: Test Results

XIII.G.5. Testing & Validation of Qualitative Constraint Q#2 – Weatherproof Design

Keldon Ngo

XIII.G.5.a. Test Protocol Description – Q#2

The rover is to be driven and transported through typical Louisiana weather conditions. The conditions that were focused on were high heat with direct sunlight, high humidity, and light to moderate rain showers.

XIII.G.5.b. Equipment and Instrumentation - Q#2

The equipment needed was the rover itself and an area with suitable weather conditions. A trailer, although not required, was very useful in transporting the rover to and from those areas.

XIII.G.5.c. Data Acquisition & Analysis – Q#2

The rover was able to operate under all weather conditions tested, including an outside temperature of 85 F with no cloud cover, so the sun was warming up the rover quite a bit. High humidity, the humidity on the days LOUIS was tested were in excess of 70%

XIII.G.5.d. Results Details – Q#2

The rover sustained no noticeable degradation in performance in these conditions. The rover was then also able to complete several different tests later that day with little to no maintenance in between tests.

XIII.G.6. Testing & Validation of Qualitative Constraint Q#3 - Portability

William Mancuso

XIII.G.6.a. Test Protocol Description - Q#3

The point of this constraint was to make sure the additions didn't completely inhibit the portability of the rover. It was expected that the additions would decrease the overall ease of portability, so to combat that we made the entire design removeable. The goal of this test was to ensure that if necessary, the rover could be brought back to its more portable state, and to see how long that takes. I timed myself multiple times assembling and disassembling the additions to see how time consuming the process could be.

XIII.G.6.b. Equipment and Instrumentation - Q#3

The equipment needed for this test was a screwdriver with multiple bits of different heads and sizes.
Particularly, flat head, phillips and, star bits. We also used a stopwatch to time the entire process.

XIII.G.6.c. Data Acquisition & Analysis – Q#3

The data gathered from the this test was the time it took to complete both disassembly and assembly of the rover, two times for each.

XIII.G.6.d. Results Details – Q#3

Looking at these results it took a little over 10 minutes to assemble the additions to the rover and about 6.5 to disassemble. This is expected to be due to having to line up components when assembling which can sometimes take longer depending on the day.

Table XIII-9: Test Result Data Table

	Time 1	Time 2
Disassembly	6 minutes 39 seconds	6 minutes 24 seconds
Assembly	10 minutes 8 seconds	10 minutes 27 seconds

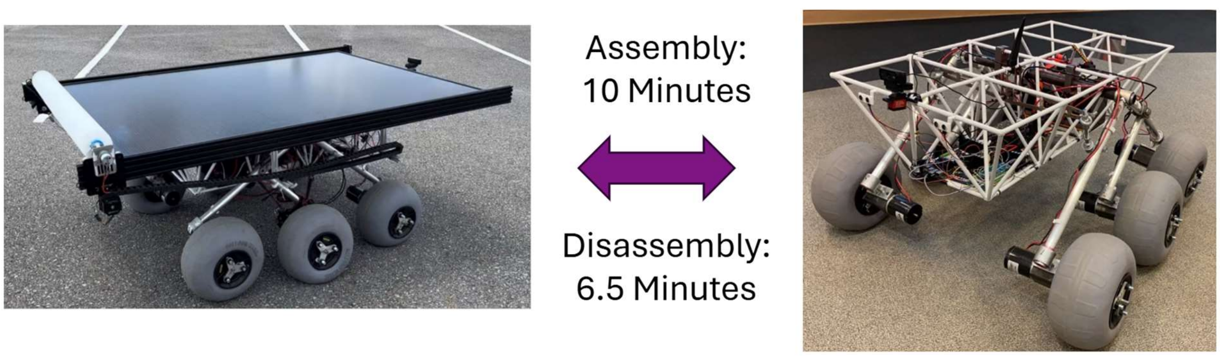


Figure XIII-25: Test Results

XIII.G.7. Testing & Validation of Quantitative Constraint M#1 - Budget

Bowen Williamson

XIII.G.7.a. Test Protocol Description - M#1

Although there were not any actual tests conducted for this constraint, a detailed expense list was maintained throughout the entire senior design process to ensure that the team had adequate funds to purchase anything necessary to see the project to completion. This project had a budget of $4000 that we could not exceed.

XIII.G.7.b. Equipment and Instrumentation - M#1

- Microsoft Excel
- Laptop

XIII.G.7.c. Data Acquisition & Analysis – M#1

Every expense was kept track of in a Microsoft Excel spreadsheet. This master list included prices, quantities, and links to find each item online.

XIII.G.7.d. Results Details – M#1

The project was completed with a total cost of $3525, well under the $4000 maximum set in place by the sponsor.

Table XIII-10: Test Result Data Table

Mounting Hardware	$333.76
Electronics/Wiring	$921.86
iPad	$1387.61
Mechanical Cleaning System Parts	$657.52
Testing	$79.11
Misc.	$145.52
Total	**$3525.38**

Part	Quantity	Price per	Cost	Link
flange coupling connectors	1	7.99	7.99	https://a.co/d/4aaC36U
threaded plastic inserts	1	14.99	14.99	https://a.co/d/5S0wAls
electro-seal	1	26.95	26.95	https://a.co/d/dKIXVPQ
4awg battery cable	1	34.88	34.88	https://a.co/d/6gZXvLq
on/off breaker	2	19.09	38.18	https://a.co/d/9JgP1Cp
stepper motor drivers	1	16.99	16.99	https://a.co/d/evWXyUx
arduino uno	1	27.60	27.60	https://a.co/d/6r3D8bH
l298n motor drivers	1	14.99	14.99	https://a.co/d/cxUyKq9
16awg red/black wire	1	16.98	16.98	https://a.co/d/dbyQNZE
conduit strap	25	4.74	118.50	https://bit.ly/3DzckEd
1000mm rail	2	10.69	21.38	https://us.openbuilds.com/v-slot-20x20-linear-rail
2000mm linear actuator	2	200.43	400.86	https://us.openbuilds.com/nema-23-belt-pinion-actuator-bundle/
motor mount plate	2	7.99	15.98	https://us.openbuilds.com/motor-mount-plate-nema-23-stepper-motor/
t-track nuts	1	3.90	3.90	https://us.openbuilds.com/drop-in-tee-nuts/
OpenBuilds Shipping (1)	1	24.91	24.91	
gear motor	1	34.58	34.58	https://www.robotshop.com/products/e-s-motor-22mm-planetary-gear-motor-w-12v-encoder-185rpm
RobotShop Shipping	1	16.50	16.50	
cleaning brush	1	196.75	196.75	Industrial Brush Corp.
limit switches	1	15.99	15.99	https://a.co/d/11pT2lY
microfiber cloths	1	7.99	7.99	https://a.co/d/6Lo63Pz
USB serial adapter	1	12.99	12.99	https://a.co/d/iX4SWSE
t-track nuts	1	3.90	3.9	https://us.openbuilds.com/drop-in-tee-nuts/
hidden corner bracket	12	1.99	23.88	https://us.openbuilds.com/inside-hidden-corner-bracket/
1500mm rail	4	15.99	63.96	https://us.openbuilds.com/v-slot-20x20-linear-rail/
OpenBuilds Shipping (2)	1	14.95	14.95	
pyranometer	1	284.12	284.12	https://www.apogeeinstruments.com/sp-215-ss-amplified-0-5-volt-pyranometer/

Figure XIII-26: Test Results

XIII.G.8. Testing & Validation of Quantitative Constraint M#2 – Interior Space

Kenny Bui

XIII.G.8.a. Test Protocol Description - M#2

To conduct this test, we utilized the previous teams' measurement of the interior space of the rover before our additions. We then determined the volumes of the components that we plan on adding within the chassis of the rover. To determine the new interior space measurement, we subtracted the total interior space by the volumes of our new components. This resulted in the new interior space measurement.

XIII.G.8.b. Equipment and Instrumentation - M#2

Measuring tape and component technical documents.

XIII.G.8.c. Data Acquisition & Analysis – M#2

I utilized an Excel spreadsheet to document all the new components along with their total volumes. I then combined the total volume of the new components and then subtracted them from the existing interior space measurement.

XIII.G.8.d. Results Details – M#2

After the appropriate calculations, we determined that the new interior space measurement was 0.971 cubic meters as opposed to the old 0.977 cubic meters. This means that our additions only consumed an interior space of 0.006 cubic meters leaving an adequate amount for future additions to the rover interior chassis.

Table XIII-11: Test Result Data Table

Components	Volume (M^3)		
L298N	0.000049851		
TB6600 (2)	0.000354816		
Arduino Uno	0.000054909		
TriStar	0.005378		
RD-1	0.0004346		
Bus Bar (3)	0.0000648		
	0.006336976		
Original Volume	Occupied Space		
0.977	0.006336976	0.970663024	

Figure XIII-27: Test Results

XIII.G.9. **Testing & Validation of Quantitative Constraint M#3 - Weight**

William Mancuso

XIII.G.9.a. Test Protocol Description - M#3

The goal of this test was to make sure that after the project was complete, we were under the 36.3 kg max tested load. The group was confident that we could go over the weight if necessary which is why this constraint was weighted so low, however the goal was to try to be under that weight, without decreasing solar efficiency and cleaning effectiveness. To complete this test we weighed the rover as a whole on a weigh station, and also calculated the weight of the rover, using the weight of the parts we ordered. We did this because weigh station are meant for large trucks and have a hard time distinguishing small weight changes.

XIII.G.9.b. Equipment and Instrumentation - M#3

Truck, trailer, and ratchet straps for transportation, and a calculator to add all of the components separately

XIII.G.9.c. Data Acquisition & Analysis – M#3

To acquire our data, we compared a weigh station, with our calculated results. The way we did this was by weighing my truck and trailer with the rover on and with the rover off and calculating the difference. The weigh stations are calibrated to weigh 10s of thousands of pounds, so a small 250-pound difference is difficult to weigh especially since it only gives numbers by 20s. We did this to see if the number we calculated was close to right with about 20-40 pounds of error since the scale isn't accurate for our application.

We compared this weight to the weight calculated by hand, since we considered that to be a more accurate portrayal of the weight of the additions, this is because, we also weren't 100% positive how accurate the previous weight of 77.25 kg was before our addition.

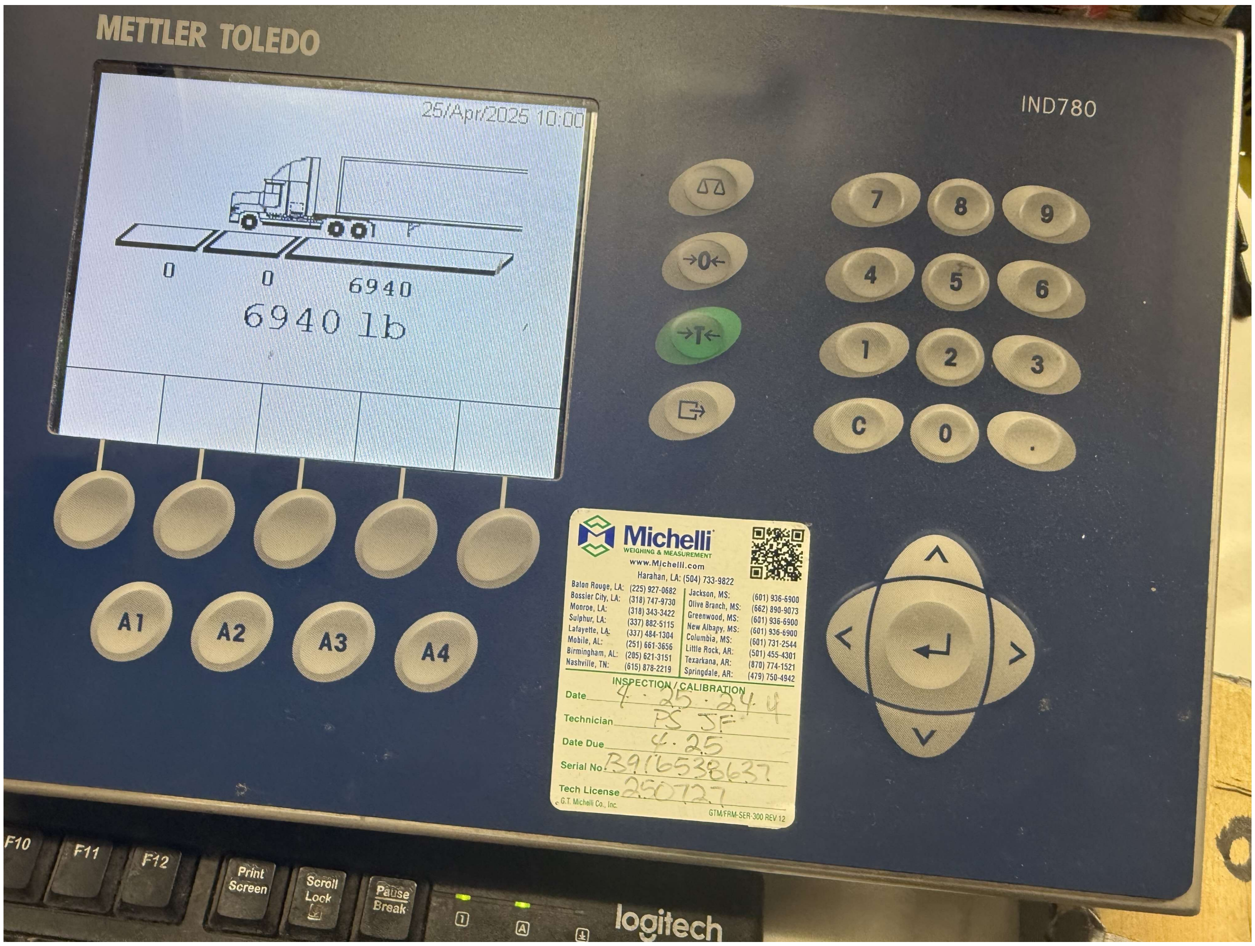
Brush: 6 lbs
Rails from openbuild: 18lbs
Solar Panel: 22.7 kg or 49.94 lbs
Electronics: about 1 lbs
Charge controller: 9lbs
Rover: 77.25 kgs
Rails cut: -4 lbs
Total: 249.89 or 113.45 kg

XIII.G.9.d. Results Details – M#3

Table XIII-12: Test Result Data Table

Weight with rover ±20 lbs	Weight without rover ±20 lbs	Rover scale weight	Rover calculated weight
6940 lbs	6,660 lbs	280 lbs	249.5 lbs
			113.45 kgs

This using the calculated weight, this puts us at a weight of 113.45 kgs or 249.5 lbs for the rover. Comparing this with the scale which approximated the weight to be about 280 lbs, this falls in our margin of error, between 20-40 pounds. Which means we can accurately say the rover is about 250 lbs, or 113.5 kgs, making the additions we made only 36.2 kgs, just under the weight constraint we set.



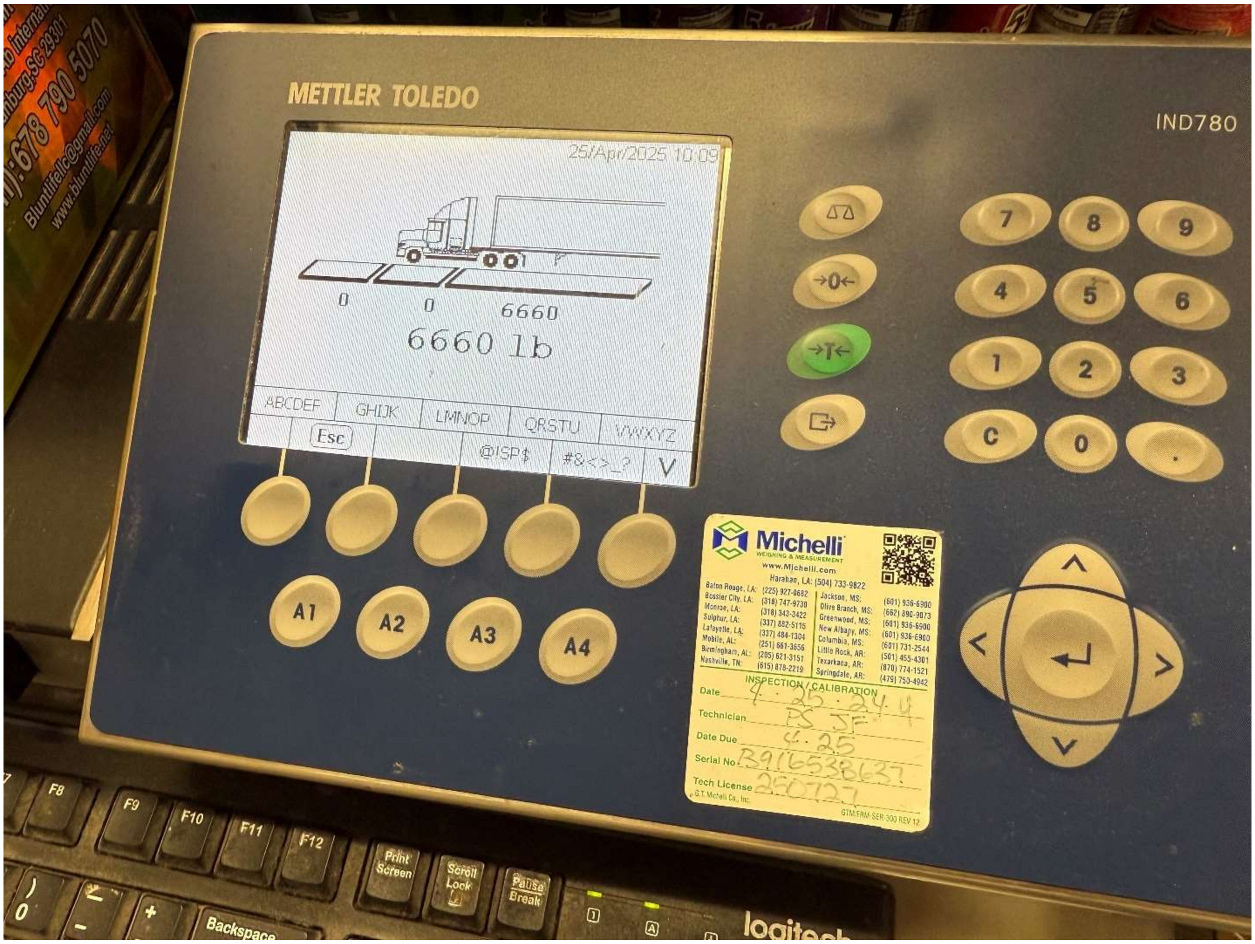


Figure XIII-28: Test Results

XIII.H. Project Management Supplement

XIII.H.1. Schedule and Milestones Details

During our project analysis phase, our group did have some realizations. These involved running simulations and calculations for our stepper motors, DC motor, battery state of charge with additional load, a code for brush cleaning system activation, and solar panel power generation. These were all crucial realizations because these analyses were additional factors that our group did not realize would play a major part in the complete picture of our individual function objectives. For manufacturing, we also realized that we needed to figure out how to connect motors to the rails for the brush cleaning system, mount the solar panel to the rover frame, and program the solar charge controller for use. For the testing phase, our group realized that we need to make sure that our brush cleaning system motors actuate properly, test the power generation of the solar panel in different weather types, and test the solar charge controller switching with different battery state of charge. All these realizations are very beneficial as they play a role in determining the overall success of our project. They also help our group make the proper adjustments needed in terms of decisions to get us closer to the success of our project. Now that our team is finished with our manufacturing and testing phases, we did have a couple of realizations. Our first one occurred when one of our rover axles broke during a test. The test required the rover to be mobile, so since the axle broke, we had to repair it. Repairing this axle took a few days resulting in some delay for testing. The second realizations are that some fine tuning and adjustments would be needed to perfect our Arduino coding. Through a series of different codes, we finally reached an Arduino code that was to our liking. This also required a couple of days and create some time delays. These time delays did cause our team to rush a little bit, but we were able to complete the senior design project completely and within a timely manner.

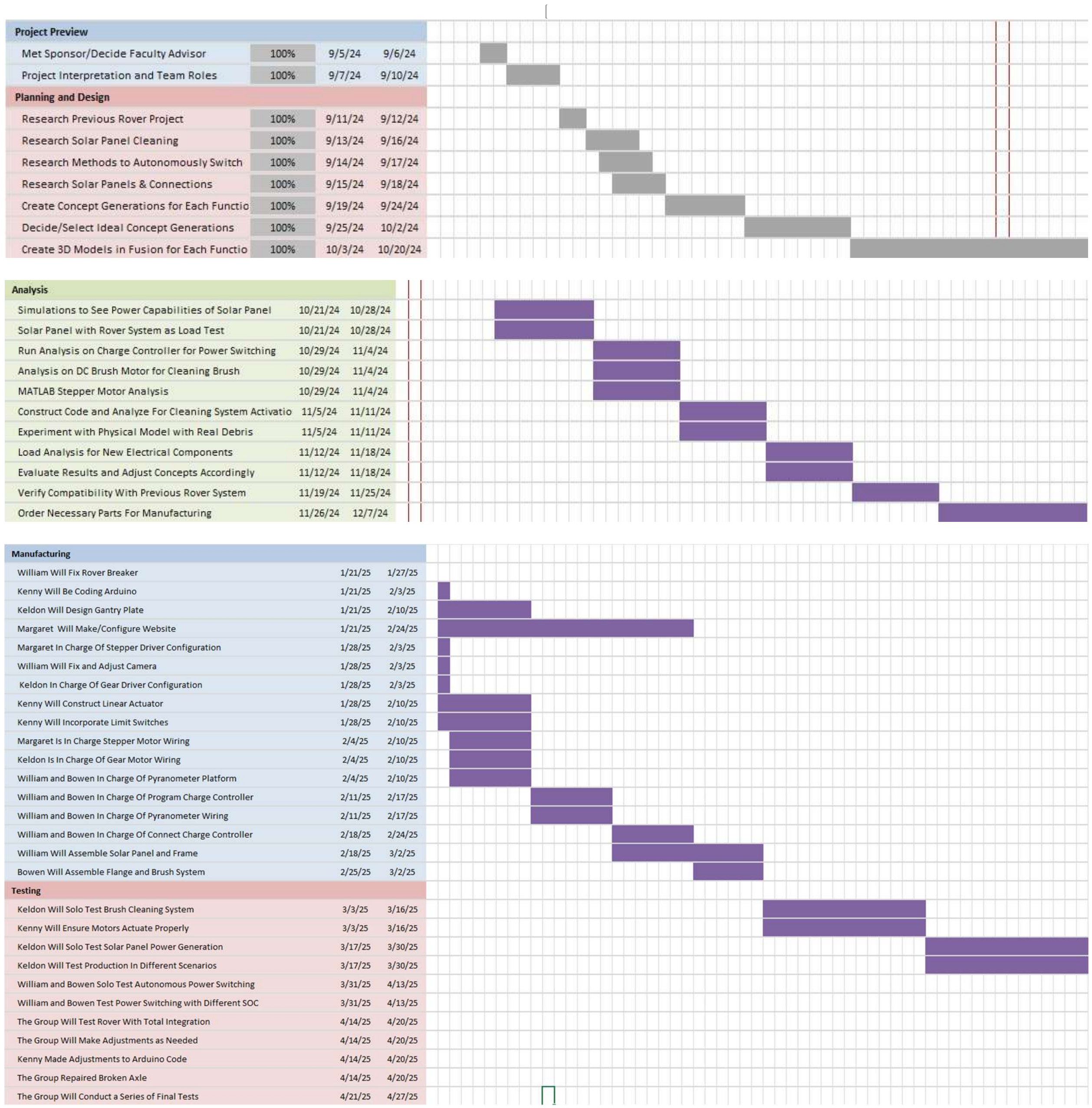


Figure XIII-29: Gantt Chart with Detailed Project Time-Line

XIII.H.2. **Budget Details**

The total spending for the project was $3525.38. This includes all components ordered, shipping costs, the costs of manufacturing, and testing costs. The most expensive part of this project turned out to be the cleaning system, as the solar panel and solar charge controller were both donated to the team. A full breakdown of all the costs for each major category can be seen in Table XIII10 below.

Table 0-1: Bill of Materials/Parts

Category	Components	Quantity	Cost/Item	Total Cost	Manufacturer
Mounting Hardware	Conduit strap	25	4.74	118.50	Grainger
	1000mm rail	2	10.69	21.38	OpenBuilds
	T-track nuts (1)	1	3.90	3.90	OpenBuilds
	OpenBuilds Shipping (1)	1	24.91	24.91	OpenBuilds
	T-track nuts (2)	1	3.90	3.90	OpenBuilds
	Hidden corner bracket	12	1.99	23.88	OpenBuilds
	1500mm rail	4	15.99	63.96	OpenBuilds
	OpenBuilds Shipping (2)	1	14.95	14.95	OpenBuilds
	Rubber strip	1	19.98	19.98	Amazon
	Adhesive tape	1	15.29	15.29	Amazon
	Flathead rail screws	1	8.99	8.99	Amazon
	Hex rail screws	1	8.29	8.29	Amazon
	ACE fasteners	1	5.83	5.83	ACE
Electronics/Wiring	4awg battery cable	1	34.88	34.88	Amazon
	On/off breaker	2	19.09	38.18	Amazon
	Stepper motor drivers	1	16.99	16.99	Amazon
	Arduino Uno	1	27.60	27.60	Arduino
	L298N motor drivers	1	14.99	14.99	Amazon
	16awg red/black wire	1	16.98	16.98	Amazon
	Gear motor	1	34.58	34.58	RobotShop
	RobotShop Shipping	1	16.50	16.50	RobotShop
	Limit switches	1	15.99	15.99	Amazon
	Pyranometer	1	284.12	284.12	Apogee Instruments
	Cable carriers	4	13.49	53.96	Amazon
	Temp sensor	1	11.95	11.95	Amazon
	Solar panel in-line fuse	1	4.99	4.99	Amazon
	Battery in-line fuse	1	12.38	12.38	Amazon
	Buzzer	1	5.49	5.49	Amazon
	Cancel pushbutton	1	9.99	9.99	Amazon
	DC gear motor	1	14.17	14.17	Surplus Center
	Gear motor shipping	1	9.17	9.17	Surplus Center
	Terminal connectors	1	6.99	6.99	Amazon
	Battery disconnect switch	1	15.97	15.97	Amazon
	PV disconnect switch	1	12.49	12.49	Amazon
	RS-485 Arduino shield	1	79.37	79.37	Zihatec
	RS-485 to RS-232 adapter	1	10.80	10.80	Amazon
	Bus bars	2	27.11	54.22	McMaster-Carr
	VGA cable	1	19.88	19.88	Best Buy
	DB9 male/female adapter	1	5.07	5.07	Southern Electronics
	Spare Arduinos	2	27.99	55.98	Arduino
	Spare circuit breakers	2	19.09	38.18	Amazon

Mechanical Cleaning System Parts	Flange coupling connectors	1	7.99	7.99	Amazon
	Threaded plastic inserts	1	14.99	14.99	Amazon
	2000mm linear actuator	2	200.43	400.86	OpenBuilds
	Motor mount plate	2	7.99	15.98	OpenBuilds
	Cleaning brush	1	196.75	196.75	Industrial Brush Corp.
	Set screws	1	4.96	4.96	Amazon
	Bearings	1	15.99	15.99	Amazon
Testing	Flour	2	2.58	5.16	Amazon
	Soap	1	19.99	19.99	Amazon
	Battery pack	1	7.99	7.99	Amazon
	Multimeter	1	39.99	39.99	Amazon
	Flour	2	2.99	5.98	Amazon
Misc.	Electro-seal	1	26.95	26.95	Amazon
	Microfiber cloths	1	7.99	7.99	Amazon
	USB serial adapter	1	12.99	12.99	Amazon
	QR code keychain	5	11.99	59.95	Amazon
	Rubber grommets	1	14.79	14.79	Amazon
	Large grommets	1	7.95	7.95	Amazon
	Hole saw	1	14.90	14.90	Lowe's
	iPad	1	1000.00	1000.00	Apple
	Apple pen	1	110.00	110.00	Apple
	iPad case	1	277.61	277.61	Apple
TOTAL				3525.38	

XIII.I. Reference Materials

XIII.I.1. Codes and Standards

The following codes were considered in the design and execution process:

1. NEC: This lays out the proper sizing of wiring and overcurrent protection, grounding, disconnects, proper labeling of equipment, and solar photovoltaic systems designs and considerations.
2. UL 9540A: This highlights different ways to test to protect against thermal runaway risk.
3. UL 1703/UL 61730: This describes PV module safety standards, including heat, cold, and humidity tolerance.
4. UL 1742: This details how to safely integrate charge controllers into solar systems.
5. EPA Guidelines: This details how to dispose of electrical components and restricts the use of lead, mercury, or cadmium for environmental safety.
6. IEC 61508: This standard ensures that a given control system can detect faults and act appropriately.
7. ISO 13849: This standard assesses the mechanical safety of the system with regards to emergency stops or other means of protecting the user and/or the rover.
8. ISO 14120: This code shows how to ensure safeguards, like barriers, within mechanical systems for the safety of the user.
9. IEC 61850: This code lays out communication protocols between intelligent electronic devices (IEDs) to ensure synchronization and reliability.

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